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A Coupled Component GARCH Model for Intraday and Overnight Volatility*

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Abstract

We propose a semi-parametric coupled component GARCH model for intraday and overnight volatility that allows the two intraday periods to have different properties. To capture the very heavy tails of overnight returns, a dynamic conditional score model with t innovations is adopted. We propose a several step estimation procedure that captures the nonparametric slowly moving components by kernel estimation and the dynamic parameters by estimated maximum likelihood. We establish the consistency and asymptotic normality of our semiparametric estimation procedures. We extend the modelling to the multivariate case where we allow time varying correlation between stocks. We apply our model to the study of the Dow Jones industrial average component stocks over the period 1991-2016 and the CRSP capitalization-based portfolios over the period 1992-2015. We show that the ratio of overnight to intraday volatility has actually increased in importance for Dow Jones stocks during the last two decades. This ratio has also increased for large stocks in CRSP, but decreased for small stocks in CRSP. Notably, the slope increases monotonically from the smallest cap decile to the largest cap decile. The multivariate model shows that overnight and intraday correlations have both increased considerably during this period.

KEYWORDS: DCS, GAS, GARCH, size portfolios, Testing

JEL CLASSIFICATION: C12, C13

1 Introduction

The balance between intraday and overnight returns is of considerable interest as it potentially sheds light on many issues in finance: the efficient markets hypothesis, the calendar time versus trading time models, the process by which information is impacted into stock prices, the relative merits

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of auction versus continuous trading, the effect of high frequency trading on market quality, and the globalization and connectedness of international markets. We propose a bivariate time series model for intraday and overnight returns that respects their temporal ordering and permits the two processes to have different marginal properties, and to feedback into each other, and allows for both short run and long components. In particular, our volatility model for each return series has a long run component that slowly evolves over time, and is treated nonparametrically, and a parametric dynamic volatility component that allows for short run deviations from the long run process, where those deviations depend on previous intraday and overnight shocks. We adopt a dynamic conditional score (DCS) model, Harvey (2013) and Harvey and Luati (2014), that links the news impact curves of the innovations to the shock distributions, which we assume to be t-distributions with unknown degrees of freedom (which may differ between day and night). In practice, the overnight return distribution is more heavy tailed than the intraday return, and in fact very heavily tailed. Our model allows for a difference in tail thickness in the conditional distributions. The short run dynamic process allows for leverage effects and separates the overnight shock from the intraday shock. We also introduce a multivariate model that allows for time varying correlations between stocks and between overnight and intraday returns.

We apply our model to the study of 28 Dow Jones industrial average component stocks over the period 1991-2016, a period that saw several substantial institutional changes. There are several purposes for our application. First, many authors have argued that the introduction of computerized trading and the increased prevalence of High Frequency Trading (HFT) strategies in the period post 2005 has lead to an increase in volatility, see Boehmer, Fong, and Wu (2015) and Linton, O'Hara, and Zigrand (2013). A direct comparison of volatility before and after would be problematic here because of the Global Financial Crisis (GFC), which raised volatility during the same period that HFT was becoming more prevalent. There are a number of studies that have investigated this question with natural experiments methodology (Brogaard, 2011), but the conclusions one can draw from such work are event specific. We model the volatility process with a view to addressing this hypothesis in a more general way. One implication of this hypothesis is that ceteris paribus the ratio of intraday to overnight volatility should have increased during this period because trading is not taking place during the market close period. We would like to evaluate whether this has occurred. One could just compare the daily return volatility from the intraday segment with the daily return volatility from the overnight segment, as many studies such as French and Roll (1986) have done. However, this would ignore both fast and slow variation in volatility through business cycle and other causal factors. Also, overnight raw returns are very heavy tailed and so sample (unconditional) variances are not very reliable. We use our dynamic two component model that allows for both fast and slow dynamic components to volatility, as is now common practice (Engle and Lee, 1999; Engle and Rangel, 2008; Hafner and Linton, 2010; Rangel and Engle, 2012; and Han and Kristensen, 2015). Our model also allows dynamic feedback between overnight and intraday volatility, which is of interest in itself. Our model generates heavy tails in observed returns, but the parameter estimates we employ are robust to this phenomenon. Our methodology therefore allows us to compare the long run components of volatility over this period without over reliance on

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Gaussian-type theory. We show that for the Dow Jones stocks the long run component of overnight volatility has actually increased in importance during this period relative to the long run component of intraday volatility. We provide a formal test statistic that quantifies the strength of this effect; our test can be interpreted as carrying out a difference in difference analysis but in ratio form. This finding seems to be hard to reconcile with the view that trading has increased volatility. We also document the short run dynamic processes. Notably, we find, unlike Blanc, Chicheportiche, and Bouchaud (2014), that overnight returns significantly affect future intraday volatility. We also find overnight return shocks to have t-distributions with degrees of freedom roughly equal to three, which emphasizes the potential fragility of Gaussian-based estimation routines that earlier work has been based on. We also estimate a multivariate model and document that there has been an upward trend in the long run component of contemporary overnight correlation between stocks as well as in the long run component of contemporary intraday correlation between stocks. However, the trend development for the overnight correlations started later than for intraday, and started happening only after 2005, whereas the intraday correlations appear to have slowly increased more or less from the beginning of the period.

We also apply our model to 10 market capitalization-based portfolios of CRSP stocks over the period 1992-2015. We find that the ratio of overnight to intraday volatility has indeed increased for large stocks, but has decreased for small stocks. Notably, the slope increases monotonically from the smallest to the largest cap decile. From the multivariate model, we find small stocks had rather weak co-movement with the market in the early 90s. However the co-movement has increased considerably during the recent period, although it still remains smaller than that of big stocks.

A second practical purpose for our model is to improve forecasts of intraday volatility or close to close volatility. Our model allows us to condition on the opening price to forecast intraday volatility or to update the close to close volatility forecast and also to take into account the full dynamic consequences of the overnight shock and previous ones. We compare forecast performance of our model with a procedure based only on close to close returns and find in most cases superior performance.

We work only with the return series, although for some stocks intraday transaction and quote records are available for the duration of our study, which would permit the computation of realized volatility measures, which are for some the preferred measure of intraday volatility. This however would pose some additional questions in terms of the joint modelling of discrete time returns and realized volatility, and puts an imbalance between the measurement of intraday and overnight volatility.[1] Furthermore, it would be problematic to implement some of those techniques on the small CRSP stocks in the early part of the sample period, so it is not a silver bullet. Instead we do make use of alternative measures of market (SP500) volatility - the VIX (which includes overnight volatility) and the Rogers and Satchell (1991) intraday volatility measure - to conduct a robustness check. We find that these measures confirm the main finding regarding the rise of overnight volatility relative to intraday.

[1]Our main findings involve averages of the daily volatility series and so the efficiency gain of realized volatility may not be so large in this context.

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Related Literature. Overnight returns have attracted much attention during recent years. Specifically, Cooper, Cliff, and Gulen (2008) suggest that the US equity premium over the period 1993-2006 is solely due to overnight returns. Berkman, Koch, Tuttle, and Zhang (2012) find positive overnight returns tend to be followed by reversals in the subsequent trading day, because retail investors who dominate open prices tend to buy high-attention stocks, driving up the open price. Following this, Aboody, Even-Tov, Lehavy, and Trueman (forthcoming) suggest the use of overnight returns as a firm-specific sentiment. On the other hand, Aretz and Bartram (2015) find evidence that intraday returns outperform overnight returns from a sample with stocks in 35 countries, and suggest this also holds in the US market. In contrast with the mixed evidence for mean values, it is generally agreed that overnight returns are less volatile at least in the US market (French and Roll, 1986;Lockwood and Linn, 1990; Aretz and Bartram, 2015), but more leptokurtic (Ng and Masulis, 1995; Blanc, Chicheportiche, and Bouchaud, 2014).

Tsiakas, 2008 proposed a stochastic volatility model for daytime returns with feedback from night to day and leverage effects built in. He assumed Gaussian innovations; he did not model the overnight returns. In the literature on realized volatility, many authors have considered how to incorporate overnight returns into daily variance modeling and forecasting, by scaling the intraday measure (e.g., Martens, 2002 and Fleming, Kirby, and Ostdiek, 2003), or by combining daytime realized volatility and the squared overnight return with optimally chosen weight parameters (e.g., Hansen and Lunde, 2005). However, these authors also did not model the overnight returns either. Andersen, Bollerslev, and Huang (2011) decomposed the total daily return variability into the continuous sample path variance, the discontinuous intraday jumps, and the overnight variance. For this overnight variance, they used an augmented GARCH-t type structure with the immediately preceding daytime realized volatility as an additional explanatory variable. Blanc, Chicheportiche, and Bouchaud (2014) employ a quadratic ARCH model with flexible dynamics for both intraday and overnight returns; they also allow for feedback from overnight to intraday returns and leverage effects. They use a t distributed shock to drive each process and to define an estimation algorithm. They impose a pooling assumption on the model parameters across 280 S&P500 stocks that are continually in the index over 2000-2009, and assume stationarity over the period in question. Creal, Koopman, and Lucas (2013) introduced a general class of time series models called Generalized Autoregressive Score models (GAS). Simultaneously, Harvey, Chakravarty et al. (2008)developed a score driven model specifically for volatilities, called the Beta-t-(E)GARCH model, built on exactly the same philosophy. Harvey (2013) settles on the dynamic conditional score model terminology, and we shall follow that nomenclature. Engle and Rangel (2008) introduced nonparametric slowly varying trends into GARCH models; Hafner and Linton (2010) propose a multivariate extension and develop the distribution theory for inference.

2 Model and Properties

We let r_t^D denote intraday returns and r_t^N denote overnight returns on day t. We take the ordering that night precedes day so that $r_t^D = \ln(P_t^C/P_t^O)$ and $r_t^N = \ln(P_t^O/P_{t-1}^C)$, where P_t^O denotes the

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opening price on day t and P_t^C denotes the closing price on day t. Daily close to close returns satisfy $r_t = r_t^D + r_t^N$. The timeline is illustrated below

$$\cdots \longrightarrow P_{t-1}^C \quad \Big| \underbrace{\overbrace{\longrightarrow}^{Night\ \mathrm{t}}}_{r_t^N} \quad P_t^O \quad \underbrace{\overbrace{\longrightarrow}^{Day\ \mathrm{t}}}_{r_t^D} \quad P_t^C \Big| \overbrace{\longrightarrow}^{Night\ \mathrm{t+1}} \quad P_{t+1}^O \longrightarrow \cdots$$

r_t^N r_t^D r_t r_{t+1}^N

We do not distinguish between weekend, holiday weekends, and ordinary midweek over night periods, although we comment on this issue in the concluding section below.

Our model allows intraday returns to depend on overnight returns with the same t, but overnight returns just depend on lagged variables. Suppose that

$$\begin{pmatrix} 1 & \delta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r_t^D \\ r_t^N \end{pmatrix} = \begin{pmatrix} \mu_D \\ \mu_N \end{pmatrix} + \Pi \begin{pmatrix} r_{t-1}^D \\ r_{t-1}^N \end{pmatrix} + \begin{pmatrix} u_t^D \\ u_t^N \end{pmatrix}, \tag{1}$$

where u_t^D and u_t^N are conditional mean zero shocks. Under the EMH, $\delta = 0$ and $\Pi = 0$, but we allow these coefficients to be nonzero to pick up what could be misspricing effects or short run effects such as might arise from the market microstructure, Scholes and Williams (1977).

We further suppose that the error process has conditional heteroskedasticity, with both long run and short run effects. Specifically, we suppose that

$$u_t = \begin{pmatrix} u_t^D \\ u_t^N \end{pmatrix} = \begin{pmatrix} \exp(\lambda_t^D)\exp(\sigma^D(t/T))\varepsilon_t^D \\ \exp(\lambda_t^N)\exp(\sigma^N(t/T))\varepsilon_t^N \end{pmatrix}, \tag{2}$$

where: ε_t^D and ε_t^N are i.i.d. mean zero shocks from t distributions with v_D and v_N degrees of freedom, respectively, while $\sigma^D(\cdot)$ and $\sigma^N(\cdot)$ are unknown but smooth functions that will represent the slowly varying (long-run) scale of the process, and T is the number of observations. Suppose that for $j = D, N$:

$$\sigma^j(s) = \sum\nolimits_{i=1}^{\infty} \theta_i^j \psi_i^j(s), \quad s \in [0,1] \tag{3}$$

for some orthonormal basis $\{\psi_i^j(s)\}_{i=1}^{\infty}$ with $\int_0^1 \psi_i^j(s)\, ds = 0$ and

$$\int \psi_i^j(s)\, \psi_k^j(s)\, ds = \begin{cases} 1 & \text{if } i = k \\ 0 & \text{if } i \neq k. \end{cases}$$

Different from Hafner and Linton (2010), we suppose $\sigma^D(\cdot)$ and $\sigma^N(\cdot)$ integrate to zero to achieve identification. So we do not have to restrict the parameters of the short run dynamic processes. In the following, j is always used to denote D, N without further mentioning.

Regarding the short run dynamic part of (2), we adopt a dynamic conditional score approach, Creal, Koopman, and Lucas (2011) and Harvey and Luati (2014). Let $e_t^j = \exp(-\sigma^j(t/T))u_t^j$, and the conditional score function associated with the t shocks is

$$m_t^j = \frac{(1+v_j)(e_t^j)^2}{v_j \exp(2\lambda_t^j) + (e_t^j)^2} - 1, \quad v_j > 0.$$

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We suppose that λ_t^D and λ_t^N are linear combinations of past values of the shocks determined by the conditional score function

$$\begin{aligned}\lambda_t^D &= \omega_D(1-\beta_D) + \beta_D\lambda_{t-1}^D + \gamma_D m_{t-1}^D + \rho_D m_t^N \\ &\quad + \gamma_D^*(m_{t-1}^D+1)\text{sign}(e_{t-1}^D) + \rho_D^*(m_t^N+1)\text{sign}(e_t^N) \qquad (4)\\ \lambda_t^N &= \omega_N(1-\beta_N) + \beta_N\lambda_{t-1}^N + \gamma_N m_{t-1}^N + \rho_N m_{t-1}^D \\ &\quad + \rho_N^*(m_{t-1}^D+1)\text{sign}(e_{t-1}^D) + \gamma_N^*(m_{t-1}^N+1)\text{sign}(e_{t-1}^N). \qquad (5)\end{aligned}$$

This gives two dynamic processes for the short run scale of the overnight and intraday return. We allow the overnight shock to affect the intraday scale through the parameter ρ_D, and we allow for leverage effects through the parameters $\gamma_D^*, \rho_D^*, \rho_N^*$, and γ_N^*.[2] We set the intercepts this way so that ω_D is the unconditional mean of λ_t^D and ω_N is the unconditional mean of λ_t^N; we may consider $\exp(\omega_D - \omega_N)$ as measuring the relative mean contribution of the daily process and the overnight process. As $v_j \to \infty$, the shocks converge to the quadratic Gaussian innovation. Let

$$\phi = (\omega_D, \beta_D, \gamma_D, \gamma_D^*, \rho_D, \rho_D^*, v_D, \omega_N, \beta_N, \gamma_N, \gamma_N^*, \rho_N, \rho_N^*, v_N)^\intercal \in \mathbb{R}^{14}$$

be the finite dimensional parameters of interest.

Harvey (2013) argues that the quadratic innovations that feature in GARCH models naturally fit with the Gaussian distribution for the shock, but once one allows heavier tail distributions like the t-distribution, it is anomalous to or not obvious why to focus on quadratic innovations, and indeed this focus leads to a lack of robustness because large shocks are fed substantially into the volatility update. He argues that it is more natural to link the shock to volatility to the distribution of the rescaled return shock, which in the case of the t distribution has the advantage that large shocks are automatically down weighted, and in such a way driven by the shape of the error distribution.[3] The DCS model has the incidental advantage that there are analytic expressions for moments, autocorrelation functions, multi-step forecasts, and the mean squared forecast errors.

Before introducing our estimation procedure we comment on some properties of our model that are useful in applications. In the appendix we prove that if $|\beta_j| < 1$, $j = D, N$, then e_t^j and λ_t^j are strongly stationary and β-mixing with exponential decay. We note that although the conditional distribution of returns is symmetric about the mean, the unconditional distribution implied by our model may be asymmetric because of the conditional mean process and the asymmetric news impact curve that we allow for, He, Silvennoinen, and Teräsvirta (2008). We may obtain the dynamic intraday value at risk conditional on overnight returns and past information as follows

$$VaR_t^D(\alpha) = \mu_t^D + s_t^D t_\alpha(v_D),$$

$$\mu_t^D = E(r_t^D|\mathcal{F}_{t-1}, r_t^N) = \mu_D - \delta r_t^N - \Pi_{11} r_{t-1}^D - \Pi_{12} r_{t-1}^N$$

[2]The shock variable m_t^j can be expressed as $m_t^j = (v_j+1)b_t^j - 1$, where b_t^j has a beta distribution, $beta\,(1/2, v_j/2)$.

[3]This type of argument is similar to the argument in limited dependent variable models such as binary choice where a linear function of covariates is connected to the observed outcome by a link function determined by the distributional assumption.

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$$s_t^D = \mathrm{sd}(r_t^D|\mathcal{F}_{t-1}, r_t^N) = \sqrt{\frac{v_D}{v_D-2}}\exp(\lambda_t^D)\exp(\sigma^D(t/T)),$$

where $t_\alpha(v)$ is the α quantile of the t-distribution with degrees of freedom v. Here, $\mathcal{F}_{t-1}$ is the sigma field generated by $\{r_{t-1}^D, r_{t-1}^N, r_{t-2}^D, r_{t-2}^N, \ldots\}$ and $\mathcal{F}_{t-1} \cup \{r_t^N\}$ is the sigma field generated by $\{r_t^N, r_{t-1}^D, r_{t-1}^N, r_{t-2}^D, r_{t-2}^N, \ldots\}$. This process depends on all components of the model, the dynamic mean, the dynamic variance, the trend variance, and the tail thickness of the shock distribution. To obtain the value at risk given only past intraday returns say, requires some further arguments and this is presented in the appendix.

3 Estimation

Suppose that we know δ, μ, Π (in practice these can be replaced by root-T consistent estimators), we next describe how we estimate the unknown quantities ϕ and $\sigma^j(.)$. For any $\alpha > 0$, we have for $j = N, D$,

$$E\left(\left|u_t^j\right|^\alpha\right) = E\left(\left|\varepsilon_t^j\right|^\alpha\right) E\left(\exp\left(\alpha\lambda_t^j\right)\right)\exp(\alpha\sigma^j(t/T)) = c^j(\phi;\alpha) \times \exp(\alpha\sigma^j(t/T)),$$

where c^j is a constant that depends in a complicated way on the parameter vector ϕ and on the choice of α. Therefore, we can estimate $\sigma^N(s), \sigma^D(s)$ as follows. We use kernel technology to estimate the nonparametric part. Let $K(u)$ be a kernel with support $[-1,1]$ and h a bandwidth, and let $K_h(.) = K(./h)/h$. Then let

$$\widetilde{\sigma}^j(s) = \frac{1}{\alpha}\log\left(\frac{1}{T}\sum_{t=1}^{T} K_h(s - t/T)\left|u_t^j\right|^\alpha\right) \tag{6}$$

for any $s \in (0,1)$. In fact, we employ a boundary modification for $s \in [0,h] \cup [1-h,1]$, whereby K is replaced by a boundary kernel, which is a function of two arguments $K(u,c)$, where the parameter c controls the support of the kernel; thus left boundary kernel $K(u,c)$ with $c = s/h$ has support $[-1,c]$ and satisfies $\int_{-1}^{c} K(u,c)du = 1$, $\int_{-1}^{c} uK(u,c)du = 0$, and $\int_{-1}^{c} u^2K(u,c)du < \infty$. Similarly for the right boundary. The purpose of the boundary modification is to ensure that the bias property holds throughout $[0,1]$,Härdle and Linton (1994) . For identification, we rescale $\widetilde{\sigma}^j(t/T)$ as

$$\widetilde{\sigma}^j(t/T) = \widetilde{\sigma}^j(t/T) - \frac{1}{T}\sum_{t=1}^{T}\widetilde{\sigma}^j(t/T). \tag{7}$$

Note that $\widetilde{\sigma}^j(u)$ can be written as $\widetilde{\sigma}^j(u) = \sum_{i=1}^{\infty}\widetilde{\theta}_i^j\psi_i^j(s)$ for some coefficients $\widetilde{\theta}_i^j$ determined uniquely by the sample value. For notational convenience we will represent $\widetilde{\sigma}^j$ in terms of $\{\widetilde{\theta}_i^j\}_{i=1}^{\infty}$ or just $\widetilde{\theta}$ for shorthand.

Let $\widetilde{e}_t^N = \exp(-\widetilde{\sigma}^N(t/T))u_t^N$ and $\widetilde{e}_t^D = \exp(-\widetilde{\sigma}^D(t/T))u_t^D$, and let $\widetilde{\theta}$ denote $\{\widetilde{\sigma}^j(s),\ s \in [0,1],\ j = N, D\}$. Define the global log-likelihood function for ϕ (apart from an unnecessary constant and

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conditional on the estimated values of θ)

$$l_T(\phi;\widetilde{\theta}) = \frac{1}{T}\sum_{t=1}^{T}\left(l_t^N(\phi;\widetilde{\theta}) + l_t^D(\phi;\widetilde{\theta})\right),$$
$$l_t^j(\phi;\widetilde{\theta}) = -\lambda_t^j(\phi;\widetilde{\theta}) - \frac{v_j+1}{2}\ln\left(1+\frac{(\widetilde{e}_t^j)^2}{v_j\exp(2\lambda_t^j(\phi;\widetilde{\theta}))}\right) + \ln\Gamma\left(\frac{v_j+1}{2}\right) - \frac{1}{2}\ln v_j - \ln\Gamma\left(\frac{v_j}{2}\right), \tag{8}$$

where $\lambda_t^j(\phi;\widetilde{\theta})$ are defined in (4) and (5). For practical purposes, $\lambda_{1|0}^j$ may be set equal to the unconditional mean, $\lambda_{1|0}^j = \omega_j$. We estimate ϕ by maximizing $l_T(\phi;\widetilde{\theta})$ with respect to ϕ. Let $\widetilde{\phi}$ denote these estimates.

Given estimates of ϕ and the preliminary estimates of $\sigma^D(\cdot), \sigma^N(\cdot)$, we calculate

$$\widetilde{\eta}_t^N = \exp(-\widetilde{\lambda}_t^N)u_t^N \ ; \ \widetilde{\eta}_t^D = \exp(-\widetilde{\lambda}_t^D)u_t^D,$$

where $\widetilde{\lambda}_t^j = \lambda_t^j(\widetilde{\phi};\widetilde{\theta})$. We then update the estimates of $\sigma^D(\cdot), \sigma^N(\cdot)$ using the local likelihood function in Severini and Wong (1992) given $\widetilde{\eta}_t^j$ and $\widetilde{v}_j$, i.e., we minimize the objective function

$$\widetilde{L}_T^j(\gamma;\widetilde{\lambda}^j,s) = \frac{1}{T}\sum_{t=1}^{T}K_h(s-t/T)\left[\gamma + \frac{\widetilde{v}_j+1}{2}\ln\left(1+\frac{(\widetilde{\eta}_t^j\exp(-\gamma))^2}{\widetilde{v}_j}\right)\right] \tag{9}$$

with respect to $\gamma \in \mathbb{R}$, for $j = D, N$ separately, where $\widetilde{\lambda}^j = (\widetilde{\lambda}_1^j, \ldots, \widetilde{\lambda}_T^j)^\intercal$. Likewise here we use a boundary kernel for $s \in [0,h] \cup [1-h,1]$. In practice we use Newton-Raphson iterations making use of the derivatives of the objective functions, which are given in (21) in Appendix A.

To summarize, the estimation algorithm is as follows.

Algorithm

STEP 1. Estimate δ, μ^j, Π by least squares and $\widetilde{\sigma}^j(u)$, $u \in [0,1]$, $j = N, D$ from (6) and (7)

STEP 2. Estimate ϕ by optimizing $l_T(\phi;\widetilde{\theta})$with respect to ϕ (by Newton-Raphson) to give $\widetilde{\phi}$.

STEP 3. Given the initial estimates $\widetilde{\theta}$ and $\widetilde{\phi}$, we replace λ_t^j with $\widetilde{\lambda}_t^j = \lambda_t^j(\widetilde{\phi};\widetilde{\theta})$. Then let $\widehat{\sigma}^j(s)$ optimize $\widetilde{L}_T^j(\sigma^j(s);\widetilde{\lambda},s)$ with respect to $\sigma^j(s)$. For identification, we rescale $\widehat{\sigma}^j(t/T) = \widehat{\sigma}^j(t/T) - \frac{1}{T}\sum_{t=1}^{T}\widehat{\sigma}^j(t/T)$.

STEP 4. Repeat Steps 2-3 to update $\widehat{\theta}$ and $\widehat{\phi}$ until convergence. We define convergence in terms of the distance measure

$$\Delta_r = \sum_{j=D,N}\int\left[\widehat{\sigma}^{j,[r]}(u) - \widehat{\sigma}^{j,[r-1]}(u)\right]^2 du + \left(\widehat{\phi}^{[r]} - \widehat{\phi}^{[r-1]}\right)^\intercal\left(\widehat{\phi}^{[r]} - \widehat{\phi}^{[r-1]}\right),$$

that is, we stop when $\Delta_r \leq \epsilon$ for some prespecified small ϵ.

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4 Large Sample Properties of Estimators

In this section we give the asymptotic distribution theory of the estimators considered above. The proofs of all results are given in the Appendix. Let $h_t^j = \lambda_t^j + \sigma^j(t/T)$, and let:

$$A_t = \begin{bmatrix} 1 & a_t^{DN} \\ 0 & 1 \end{bmatrix}, \quad B_{t-1} = \begin{bmatrix} \left(\beta_D + a_{t-1}^{DD}\right) & 0 \\ a_{t-1}^{ND} & \left(\beta_N + a_{t-1}^{NN}\right) \end{bmatrix},$$

$$\begin{aligned} a_{t-1}^{DD} &= -2\left(\gamma_D + \gamma_D^* \text{sign}(u_{t-1}^D)\right)(v_D+1)\, b_{t-1}^D\left(1-b_{t-1}^D\right) \\ a_t^{DN} &= -2\left(\rho_D + \rho_D^* \text{sign}(u_t^N)\right)(v_N+1)\, b_t^N\left(1-b_t^N\right) \\ a_{t-1}^{NN} &= -2\left(\gamma_N + \gamma_N^* \text{sign}(u_{t-1}^N)\right)(v_N+1)\, b_{t-1}^N\left(1-b_{t-1}^N\right) \\ a_{t-1}^{ND} &= -2\left(\rho_N + \rho_N^* \text{sign}(u_{t-1}^D)\right)(v_D+1)\, b_{t-1}^D\left(1-b_{t-1}^D\right) \end{aligned}$$

$$b_t^D = \frac{(e_t^D)^2}{v_D \exp(2\lambda_t^D) + (e_t^D)^2} \quad ; \quad b_t^N = \frac{(e_t^N)^2}{v_D \exp(2\lambda_t^N) + (e_t^N)^2}.$$

We use the maximum row sum matrix norm, $\|\cdot\|_\infty$, defined by

$$\|A\|_\infty = \max_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}|.$$

Assumptions A

1. $\|E(A_t \otimes A_t)\|_\infty < \infty, \|EB_tEA_t\|_\infty < 1, \|E(B_{t-1}A_{t-1} \otimes B_{t-1}A_{t-1})\|_\infty < \|EB_tEA_t\|_\infty$, *and the top-Lyapunov exponent of the sequence of* A_tB_{t-1} *is strictly negative. The top Lyapunov exponent is defined as Theorem 4.26 of* Douc, Moulines, and Stoffer (2014).

2. $0 \le |\beta_j| < 1$.

3. h_t^j *starts from infinite past. The parameter* ϕ_0 *is an interior point of* $\Phi \subset \mathbb{R}^{14}$, *where* Φ *is the parameter space of* ϕ_0.

4. *The functions* σ^j *are twice continuously differentiable on* $[0,\,1], j = D, N$.

5. $E|u_t^j|^{2+\delta} < \infty$ *for some* $\delta > 0$, $j = D, N$.

6. *The kernel function* K *is bounded, symmetric about zero with compact support, that is* $K(v) = 0$ for all $|v| > C_1$ with some $C_1 < \infty$. *Moreover, it is Lipschitz, that is* $|K(v) - K(v')| \le L|v - v'|$ *for some* $L < \infty$ *and all* $v, v' \in \mathbb{R}$. *Denote* $||K||_2^2 = \int K(s)^2 ds$.

7. $h(T) \to 0$,*as* $T \to \infty$ *such that* $T^{1/2-\delta}h \to \infty$ *for some small* $\delta > 0$.

Assumptions A3-A7 are used to derive the properties of $\widetilde{\sigma}^j(s)$, in line with Vogt and Linton (2014) and Vogt (2012). But we only require that $E|u_t^j|^{2+\delta} < \infty$, since we use $\widetilde{\sigma}^j(s) = \log\left(\frac{1}{T}\sum_{t=1}^T K_h(s - t/T)\left|u_t^j\right|\right)$. This is in line with the fact that the fourth-order moment of overnight returns may not exist for some datasets. The mixing condition in Vogt and Linton (2014) is replaced

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by Assumption A2, because of our tight model structure. Assumption A1 is required to derive the stationarity of score functions, where $\|E(A_t \otimes A_t)\|_\infty < \infty$ can be verified easily, since b_t^N in A_t follows a beta distribution.

The first result gives the uniform convergence rate of the initial estimator $\widetilde{\sigma}^j(s)$. The proof mainly follows Theorem 3 in Vogt and Linton (2014).

Lemma 1 *Suppose that Assumptions A2-A7 hold. Then,*

$$\sup_{u\in[0,1]} \left|\widetilde{\sigma}^j(u) - \sigma_0^j(u)\right| = O_p\left(h^2 + \sqrt{\frac{\log T}{Th}}\right).$$

We note that this estimator is robust to the specification of the short run dynamic process in the sense that Lemma 1 continues to hold under the weak dependence assumptions for whatever stationary process is assumed for λ_t^j. A CLT is also available for this estimator.

We next present an important orthogonality condition that allows us to establish a simple theory for the parametric component.

Theorem 1 *Suppose that Assumptions A1-A4 hold. Then, for each k and i, for $k \in \{1,\ldots,\infty\}$ and $i \in \{1,\ldots,14\}$, we have*

$$\frac{1}{T}\sum_{t=1}^{T} E\left[\frac{\partial l_t(\phi_0;\theta_0)}{\partial \theta_k}\frac{\partial l_t(\phi_0;\theta_0)}{\partial \phi_i}\right] = O(\frac{1}{T}).$$

The proof of Theorem 1 is provided in Appendix A. Theorem 1 implies that the score functions with respect to θ and ϕ are asymptotic orthogonal. The intuition behind this is that σ^j is a function of a deterministic variable, t/T, while $\lambda_t^j$is a stationary process independent of time t. The cross product of their score functions can be somehow separated, see Linton (1993) for a similar result. The asymptotic orthogonality implies that the particular asymptotic property of $\tilde{\phi}$ and $\hat{\phi}$ in Theorem 2 follows.

Let

$$I(\phi_0) = \lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T} E\left[\frac{\partial l_t(\phi_0;\theta_0)}{\partial \phi}\frac{\partial l_t(\phi_0;\theta_0)}{\partial \phi^\intercal}\right].$$

Theorem 2 *Suppose that Assumptions A1-A7 hold. Then*

$$\sqrt{T}\left(\tilde{\phi} - \phi_0\right) = \sqrt{T}\left(\widehat{\phi} - \phi_0\right) + o_P(1) \Longrightarrow N\left(0, I(\phi_0)^{-1}\right)$$

Theorem 3 *Suppose that Assumptions A1-A7 hold. Then for $s \in (0,1)$*

$$\sqrt{Th}\begin{pmatrix} \widehat{\sigma}^D(s) \\ \widehat{\sigma}^N(s) \end{pmatrix} - \begin{pmatrix} \sigma_0^D(s) \\ \sigma_0^N(s) \end{pmatrix} \Longrightarrow N\left(0, ||K||_2^2 \begin{pmatrix} \frac{(v_D+3)}{2v_D} & 0 \\ 0 & \frac{(v_N+3)}{2v_N} \end{pmatrix}\right). \tag{10}$$

Theorem 2 shows that $\tilde{\phi}$ and $\hat{\phi}$ have the same asymptotic property. Together with Theorem 3, we obtain the consistency and asymptotic normality of $\tilde{\phi}$, $\widehat{\phi}$ and $\widehat{\sigma}^j(s)$. The form of the limiting variance in (10) is consistent with the known Fisher information for the estimation of scale parameters of a

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t-distribution with known location and degrees of freedom (these quantities are estimated at a faster rate), which makes this part of the procedure also efficient in the sense considered in Tibshirani (1984).

The proofs of Theorem 2 and 3 are provided in Appendix A. The information matrix, $I(\phi_0)$, can be computed explicitly. We can conduct inference with Theorem 2 and Theorem 3 using plug-in estimates of the unknown quantities.

The kernel estimator $\tilde{\sigma}(s)$ in the initial step can be replaced by the sieve estimator of θ. It is easy to show that $\tilde{\theta}$ converges to θ, paralleling Lemma 1, and Theorem 1, 2 and 3 remain the same.

4.0.1 Test of Constancy of the Ratio of Long Run components

We provide a test of the constancy of the ratio of long run overnight to intraday volatility. We may consider the null hypothesis

$$H_0 : \exp\left(\sigma_0^N(s)\right) = \rho \exp\left(\sigma_0^D(s)\right), \quad s \in (0,1) \tag{11}$$

for some $\rho \in \mathbb{R}_+$, versus the general alternative. By Theorem 3 and the delta method, $\exp(\widehat{\sigma}^D(s))$ and $\exp(\widehat{\sigma}^N(s))$ converge jointly to a normal distribution, and are asymptotically mutually independent. Therefore, consider the t-ratio

$$\widehat{t}(s) = \frac{\sqrt{Th}\left(\widehat{\rho}(s) - \widehat{\rho}\right)}{\sqrt{\widehat{\omega}(s)}},$$

$$\widehat{\rho}(s) = \frac{\exp(\widehat{\sigma}^N(s))}{\exp(\widehat{\sigma}^D(s))}, \qquad \widehat{\rho} = \int_0^1 \frac{\exp(\widehat{\sigma}^N(s))}{\exp(\widehat{\sigma}^D(s))} ds$$

$$\widehat{\omega}(s) = \widehat{\rho}^2 ||K||_2^2 \left(\frac{\widehat{v}_N + 3}{2\widehat{v}_N} + \frac{\widehat{v}_D + 3}{2\widehat{v}_D} \right).$$

Large values of $|\widehat{t}(s)|$ are inconsistent with the null hypothesis. It follows from the CLT that for $s \in (0,1)$, $\widehat{t}(s) \Longrightarrow N(0,1)$ under the null hypothesis. We may carry out the pointwise test statistic or confidence interval based on this. We also consider an integrated version of this, specifically

$$\tau = \frac{\int \widehat{t}(s)^2 dW_T(s) - a_T}{b_T},$$

where W_T is some weighting function, for example Lebesgue measure on $[0,1]$. Under the null hypothesis, $E\widehat{t}(s)^2 \simeq 1$ so we take $a_T = 1$. Under the null hypothesis

$$\text{var}\left(\int \widehat{t}(s)^2 dW_T(s) \right) = E\left(\int \int \widehat{t}(s)^2 \widehat{t}(r)^2 dW_T(s) dW_T(r) \right) - 1.$$

In the special case that W_T is the measure that puts equal mass on the points $s_1, \ldots, s_M$ with $M = O(Th)$ so that $\widehat{t}(s_l)$ and $\widehat{t}(s_k)$ are asymptotically independent for $l \neq k$, we may take $b_T = \sqrt{2}$, because $E(\widehat{t}(s)^4) \simeq 3$. Under the null hypothesis, $\tau \Longrightarrow N(0,1)$. This testing strategy is well suite to detect general alternatives to the null hypothesis of constancy of the volatility ratio.

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5 A Multivariate model

We next consider an extension to a multivariate model. We keep a similar structure to the univariate model except that we allow the slowly moving component to be matrix valued. Suppose that

$$r_t = \begin{pmatrix} r_t^D \\ r_t^N \end{pmatrix}; \ \mu = \begin{pmatrix} \mu_D \\ \mu_N \end{pmatrix},$$

where r_t^D, r_t^D are $n \times 1$ vectors containing all the intraday and overnight returns respectively, and let

$$\mathbf{D}r_t = \mu + \Pi r_{t-1} + u_t,$$

where u_t^D and u_t^N are mean zero shocks, while

$$\mathbf{D} = \begin{pmatrix} I_n & \text{diag}\,(\Delta) \\ 0 & I_n \end{pmatrix}; \quad \Pi = \begin{pmatrix} \text{diag}(\Pi_{11}) & \text{diag}(\Pi_{12}) \\ \text{diag}(\Pi_{21}) & \text{diag}(\Pi_{22}) \end{pmatrix},$$

and $\Delta, \Pi_{11}, \Pi_{12}, \Pi_{21}$,and Π_{22} are $n \times 1$ vectors.

We also consider an alternative modelling approach when we have also market returns. In this case, we can specify r_{it}^j using a microstructure adjusted Market Model

$$r_{it}^D = a_i^D + \beta_i^{DD} r_{mt}^D + \beta_i^{DN} r_{mt}^N + u_{it}^D \tag{12}$$

$$r_{it}^N = a_i^N + \beta_i^{NN} r_{mt}^N + \beta_i^{ND} r_{mt-1}^D + u_{it}^N, \tag{13}$$

where r_{mt}^D and r_{mt}^N are the market returns and r_{it}^D and r_{it}^N are the returns of stock i. Including a lagged return in the market model to account for microstructure goes back to Scholes and Williams (1977).

We now consider the specification of the variance equation. We suppose that

$$u_t = \begin{pmatrix} \Sigma^D(\frac{t}{T})^{\frac{1}{2}} \text{diag}\left(\exp(\lambda_t^D)\right) & 0 \\ 0 & \Sigma^N(\frac{t}{T})^{\frac{1}{2}} \text{diag}\left(\exp(\lambda_t^N)\right) \end{pmatrix} \begin{pmatrix} \varepsilon_t^D \\ \varepsilon_t^N \end{pmatrix},$$

where: ε_{it}^j are i.i.d. shocks (mutually independent across i, j, and t, identically distributed over t) from univariate t distributions with v_{ij} degrees of freedom, while λ_t^j are $n \times 1$ vectors. We assume that $\Sigma^D(.)$ and $\Sigma^N(.)$ are smooth matrix functions but are otherwise unknown. They allow slowly evolving correlation between stocks in the day or night, and for those correlations to vary by stock and over time.

We can write the covariance matrices in terms of the correlation matrices and the variances as follows

$$\Sigma^j(s) = \text{diag}\left(\exp(\sigma^j(s))\right) R^j(s) \text{diag}\left(\exp(\sigma^j(s))\right), \quad j = D, N, \tag{14}$$

with $\text{diag}\,(\exp(\sigma^j(s)))$ being the volatility matrix and $R^j(s)$ being the correlation matrix with unit diagonal elements and off-diagonal elements $R_{il}^j(s)$ with $-1 \leq R_{il}^j(s) \leq 1$. For identification, we still assume $\int_0^1 \sigma_i^j(s)ds = 0$, for $i \in \{1, \ldots n\}$ and $j = D, N$.

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As with the univariate model, define $e_t^j = \text{diag}\left(\exp(\lambda_t^j)\right)\varepsilon_t^j$, and suppose that

$$m_{it}^j = \frac{(1+v_{ij})(e_{it}^j)^2}{v_{ij}\exp(2\lambda_{it}^j) + (e_{it}^j)^2} - 1,$$

$$\begin{aligned}
\lambda_{it}^D &= \omega_{iD}(1-\beta_{iD}) + \beta_{iD}\lambda_{it-1}^D + \gamma_{iD}m_{it-1}^D + \rho_{iD}m_{it}^N \\
&\quad + \gamma_{iD}^*(m_{it-1}^D+1)\text{sign}(u_{it-1}^D) + \rho_{iD}^*(m_{it}^N+1)\text{sign}(u_{it}^N), \\
\lambda_{it}^N &= \omega_{iN}(1-\beta_{iN}) + \beta_{iN}\lambda_{it-1}^N + \gamma_{iN}m_{it-1}^N + \rho_{iN}m_{it-1}^D \\
&\quad + \rho_{iN}^*(m_{it-1}^D+1)\text{sign}(u_{it-1}^D) + \gamma_{iN}^*(m_{it-1}^N+1)\text{sign}(u_{it-1}^N).
\end{aligned}$$

For each i define the parameter vector $\phi_i = (\omega_{iD}, \beta_{iD}, \gamma_{iD}, \gamma_{iD}^*, \rho_{iD}, \rho_{iD}^*, v_{iD}, \omega_{iN}, \beta_{iN}, \gamma_{iN}, \gamma_{iN}^*, \rho_{iN}, \rho_{iN}^*, v_{iN})^\intercal \in \mathbb{R}^{14}$ and let $\phi = (\phi_1^\intercal, \ldots, \phi_n^\intercal)^\intercal$ denote all the dynamic parameters.

Define ι_i the vector with the ith element 1 and all others 0, so that $\varepsilon_{it}^j = \iota_i^\intercal \text{diag}\left(\exp(-\lambda_t^j)\right)\left(\Sigma^j(\frac{t}{T})\right)^{-1/2} u_t^j$. The normalized global log-likelihood function is

$$l_T(\phi, \Sigma(\cdot)) = \frac{1}{T}\sum_{t=1}^T \left(l_t^N + l_t^D\right)$$

$$\begin{aligned}
l_t^j(\phi, \Sigma(\cdot)) = &\sum_{i=1}^n \left(-\prod_{i=1}^n \lambda_{it}^j - \frac{v_{ij}+1}{2}\ln\left(1 + \frac{(\iota_i^\intercal \text{diag}\left(\exp(-\lambda_t^j - \sigma^j(t/T))\right)\left(\Sigma^j\left(\frac{t}{T}\right)\right)^{-1/2} u_t^j)^2}{v_{ij}}\right)\right) \\
&- \frac{1}{2}\log\left|\Sigma^j\left(\frac{t}{T}\right)\right| + \sum_{i=1}^n \left(\ln\Gamma\left(\frac{v_{ij}+1}{2}\right) - \frac{1}{2}\ln v_{ij} - \ln\Gamma\left(\frac{v_{ij}}{2}\right)\right).
\end{aligned}$$

We first define an initial estimator for $\Sigma^j(\frac{t}{T})$ and then obtain an estimator of ϕ, and then we update them. Suppose that we know Δ, Π and μ. To give an estimator of $\Sigma^j(\frac{t}{T})$ robust to heavy tails, we estimate the volatility parameter

$$\widetilde{\sigma}_i^j(s) = \log\left(\frac{1}{T}\sum_{t=1}^T K_h(s - t/T)\left|u_{it}^j\right|\right). \tag{15}$$

For identification, we rescale $\widetilde{\sigma}^j(t/T)$ as

$$\widetilde{\sigma}_i^j(t/T) = \widetilde{\sigma}_i^j(t/T) - \frac{1}{T}\sum_{t=1}^T \widetilde{\sigma}_i^j(t/T). \tag{16}$$

Supposing that the heavy tails issue is less severe in the estimation of correlation, which seems reasonable; we estimate the correlation parameter by standard procedures

$$\widetilde{R}_{ik}^j(s) = \frac{\sum_{t=1}^T K_h(s - t/T)u_{ik}^j u_{ik}^j}{\sqrt{\sum_{t=1}^T K_h(s - \frac{t}{T})u_{it}^j u_{it}^j \sum_{t=1}^T K_h(s - \frac{t}{T})u_{kt}^j u_{kt}^j}} \tag{17}$$

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for $s \in (0,1)$, and boundary modification as previously detailed. Alternatively, we can use a robust correlation estimator. Omitting the superscript $j = D, N$ here, we first compute the pairwise Kendall tau

$$\hat{\tau}_{k,l}(s) = \frac{\sum_{i=1}^{T}\sum_{j=i}^{T-1} K_h(s-\frac{i}{T})K_h(s-\frac{j}{T})\left(I\left\{(u_{i,k}-u_{j,k})(u_{i,l}-u_{j,l})>0\right\} - I\left\{(u_{i,k}-u_{j,k})(u_{i,l}-u_{j,l})<0\right\}\right)}{\sum_{i=1}^{T}\sum_{j=i}^{T-1} K_h(s-\frac{i}{T})K_h(s-\frac{j}{T})\left(I\left\{(u_{i,k}-u_{j,k})(u_{i,l}-u_{j,l})>0\right\} + I\left\{(u_{i,k}-u_{j,k})(u_{i,l}-u_{j,l})<0\right\}\right)}.$$

Then applying the relation between Kendall tau and the linear correlation coefficient for the elliptical distribution suggested by Lindskog, Mcneil, and Schmock (2003) and Battey and Linton (2014), we obtain the robust linear correlation estimator, $\hat{\rho}_{k,l}(s) = \sin(\frac{\pi}{2}\hat{\tau}_{k,l}(s))$. In some cases, the matrix of pairwise correlations must be adjusted to ensure that the resulting matrix is positive definite.

We have

$$\widetilde{\Sigma}^j(s) = \text{diag}\left(\exp(\widetilde{\sigma}^j(s))\right)\widetilde{R}^j(s)\text{diag}\left(\exp(\widetilde{\sigma}^j(s))\right),\ j = D, N. \tag{18}$$

Letting $\widetilde{e}_t^j = \widetilde{\Sigma}^j(\frac{t}{T})^{-1/2}u_t^j$, we obtain $\widetilde{\phi}_i$ by maximizing the univariate log-likelihood function of $\widetilde{e}_{it}^j$ in (8) for each $i = 1, \ldots, n$. To update the estimator for each $\Sigma^j(\frac{t}{T})$, denote $\Theta = (\Sigma^j)^{-1/2}$. We first obtain $\widehat{\Theta}$ with the local likelihood function given $\widetilde{\lambda}_t^j$ and $\widetilde{v}_j$, i.e., maximize the local objective function

$$L_T^j(\Theta; \widetilde{\lambda}, s) = \frac{1}{T}\sum_{t=1}^{T} K_h(s-t/T)\left[\log|\Theta| - \sum_{i=1}^{n}\left(\frac{\widetilde{v}_{ij}+1}{2}\ln\left(1+\frac{(\iota_i^\intercal \text{diag}\left(\exp(-\widetilde{\lambda}_t^j)\right)\Theta u_t^j)^2}{\widetilde{v}_{ij}}\right)\right)\right]$$

with respect to vech(Θ), and let $\widehat{\Sigma}^j(t/T) = \widehat{\Theta}^{-2}$. The derivatives of the objective function are given in (31) and (32) in Appendix B.

To summarize, the estimation algorithm is as follows.

Algorithm

STEP 1. Estimate $\mathbb{D}, \mu, \Pi$ (or the CAPM structure)by least squares, $\widetilde{\sigma}^j(u)$, $u \in [0,1]$, $j = N, D$ from (15) and (16), and $\widetilde{R}_{ik}^j$ from (17).

STEP 2. Let $\widetilde{e}_t^j = \widetilde{\Sigma}^j(\frac{t}{T})^{-1/2}u_t^j$. We obtain $\widetilde{\phi}_i$ by maximizing the univariate log-likelihood function of $\widetilde{e}_{it}^j$ in (8) for each $i = 1, \ldots, n$.

STEP 3. We replace λ_t^j with $\widetilde{\lambda}_t^j = \lambda_t^j(\widetilde{\phi}; \widetilde{\theta})$. For each $\Sigma_j(\frac{t}{T})$, denote $\Theta^j = (\Sigma^j)^{-1/2}$. Then obtain $\widehat{\Theta}$ with the local likelihood function given $\widetilde{\lambda}_t^j$ and $\widetilde{v}_j$, with the Newton-Raphson iterations making use of the derivatives of the objective functions.

STEP 4. Repeat Steps 2-3 to update $\widehat{\phi}$ and $\widehat{\Sigma}^j(\frac{t}{T})$ until convergence.

Our multivariate model can be considered as a diagonal DCS EGARCH model with a slowly moving correlation matrix. Assuming diagonality on the short run component λ_t^j enables us to estimate the model easily and rapidly. In particular, the computation time of the initial estimator is only of order n, with n being the number of assets considered; it is thus feasible even with quite large

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n. The extension to models with non-diagonal short run components is possible, but only feasible with small n.

Blanc, Chicheportiche, and Bouchaud (2014) impose a pooling assumption in their modelling, which translates here to the restriction that $\phi_i = \phi_1$ for all $i = 1, \ldots, n$. This improves efficiency when the restriction is true. We can test the restriction by a standard Wald procedure or Likelihood ratio statistic. In the application we find these pooling restrictions are strongly rejected by the data.

6 Application to Dow Jones stocks

6.1 Data and preliminary analysis

We investigate 28 components of the Dow Jones industrial average index over the period 1991-11-12 to 2016-04-13. The 28 stocks are: MMM, AXP, AAPL.O, BA, CAT, CVX, CSCO.O, KO, DD, XOM, GE, HD, IBM, INTC.O, JNJ, JPM, MCD, MRK, MSFT.O, NKE, PFE, PG, TRV, UNH, UTX, VZ, WMT, DIS. The stocks GS and V are excluded since they did not officially go public until 1999 and 2008, respectively. The data is obtained from Thompson Reuters Eikon, and has been adjusted for corporate actions such as dividends and stock splits. We define overnight returns as the log price change between the close of one trading day to the opening of the next trading day. We do not incorporate weekend and holiday effects into our model, since they are not the focus of this paper, and our model is already rather complicated in terms of both model specification and estimation. In addition, although the weekend effect is documented by studies such as French (1980) and Rogalski (1984), and further supported by Cho, Linton, and Whang (2007) with a stochastic dominance approach, many studies suggest the disappearance of the weekend effect, including Mehdian and Perry (2001) and Steeley (2001). In particular, Sullivan, Timmermann, and White (2001) claim that calendar effects are the result of data-snooping.

Many studies find significantly higher overnight returns, for example, Cooper, Cliff, and Gulen (2008) and Berkman, Koch, Tuttle, and Zhang (2012). Cooper, Cliff, and Gulen (2008) even suggest the US equity premium is solely due to overnight returns during the research period from 1993 to 2006. To investigate this overnight anomaly, Figure B.1 in the online appendix plots the cumulative returns for these 28 stocks. For AAPL, CAT, CSCO, HD, INTC, JPM, PFE, UTX and WMT, positive cumulative returns mainly stem from overnight periods, while for MMM, KO, DD, XOM, JNJ, MCD, MRK, NKE, PG, TRV, VZ and DIS, positive cumulative returns mainly come from intraday periods. There is no clear dominance of positive overnight returns from these Dow Jones stocks.

Berkman, Koch, Tuttle, and Zhang (2012) find significant positive mean overnight returns of +10 basis points (bp) per day, along with -7 bp for intraday returns from the 3000 largest U.S. stocks. Following Berkman, Koch, Tuttle, and Zhang (2012), we first compute the cross-sectional mean (or median) returns for each day, then compute the time series mean and the standard deviation of these cross-sectional mean (or median) values. The mean intraday return is 2.18 bp with standard error 1.22 bp, while the mean overnight return is 1.60 bp with standard error 0.70 bp. The difference

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between overnight and intraday mean is not statistically significant.

Table B.1 in the online appendix provides summary statistics for intraday and overnight returns. Compared with intraday returns, overnight returns exhibit more negative skewness and leptokurtosis. More specifically, 10 of these 28 stocks exhibit negative intraday skewness, while 26 of these 28 stocks have negative overnight skewness. The largest sample kurtosis for overnight returns is extremely high, suggesting non-existence of the population kurtosis. We find that the per hour variance of intraday returns is roughly 11 times the per hour variance of overnight returns, which is somewhat less than the range of 13-100 times larger that French and Roll (1986) found.[4]

6.2 Results of the univariate model

Table B.2 in the online appendix reports the estimates and their robust standard errors in the mean equations. We multiply returns by 100 to give more readable coefficients. Π_{ij} refers to the element of the ith row jth column in the coefficient matrix Π. For the prediction of intraday returns, 12 of the 28 stocks have significant Π_{11} that are all negative, and 7 of 28 stocks have significant δ which are all positive. This suggests that both overnight and intraday returns tend to have a negative effect in their subsequent intraday return. However, we do not find clear patterns for predicting overnight returns. The constant terms, μ_D and μ_N, are positive for most Dow Jones stocks.

Table B.3 in the online appendix gives the estimates of dynamic parameters. Parameters β_D and β_N are significantly different from 1, and ρ_D, γ_D, ρ_N and γ_N are positive and significant. In addition, we find significant leverage effects, with negative and significant ρ_D^*, γ_D^*, ρ_N^* and γ_N^*, suggesting higher volatility after negative returns.

We are also concerned about the difference between overnight and intraday parameters. Table B.4 in the online appendix reports Wald tests with the null hypothesis that the intraday and overnight parameters are equal within each stock. The parameter ω_D, determining the unconditional short-run scale, is significantly larger than ω_N. The overnight degree-of-freedom parameter is around 3, significantly smaller than the intraday counterpart at approximately 8. Both are in line with the descriptive statistics in Table B.1 in the online appendix and previous studies that suggest overnight returns are more leptokurtic but less volatile. With other pairs of intraday and overnight parameters, $\beta_j, \gamma, \rho_j, \gamma_j^*, \rho_j^*$, the null hypothesis is seldom rejected. However, the joint null hypothesis, $(\beta_D, \gamma, \rho_D, \gamma_D^*, \rho_D^*) = (\beta_N, \gamma, \rho_N, \gamma_N^*, \rho_N^*)$, is rejected by many stocks. It is noteworthy that the null hypothesis $H_0 : \gamma_N = \rho_D$ is not rejected by our data, which is inconsistent with Blanc, Chicheportiche, and Bouchaud (2014). They suggest that past overnight returns weakly affect future

[4]Suppose that hourly stock returns satisfy

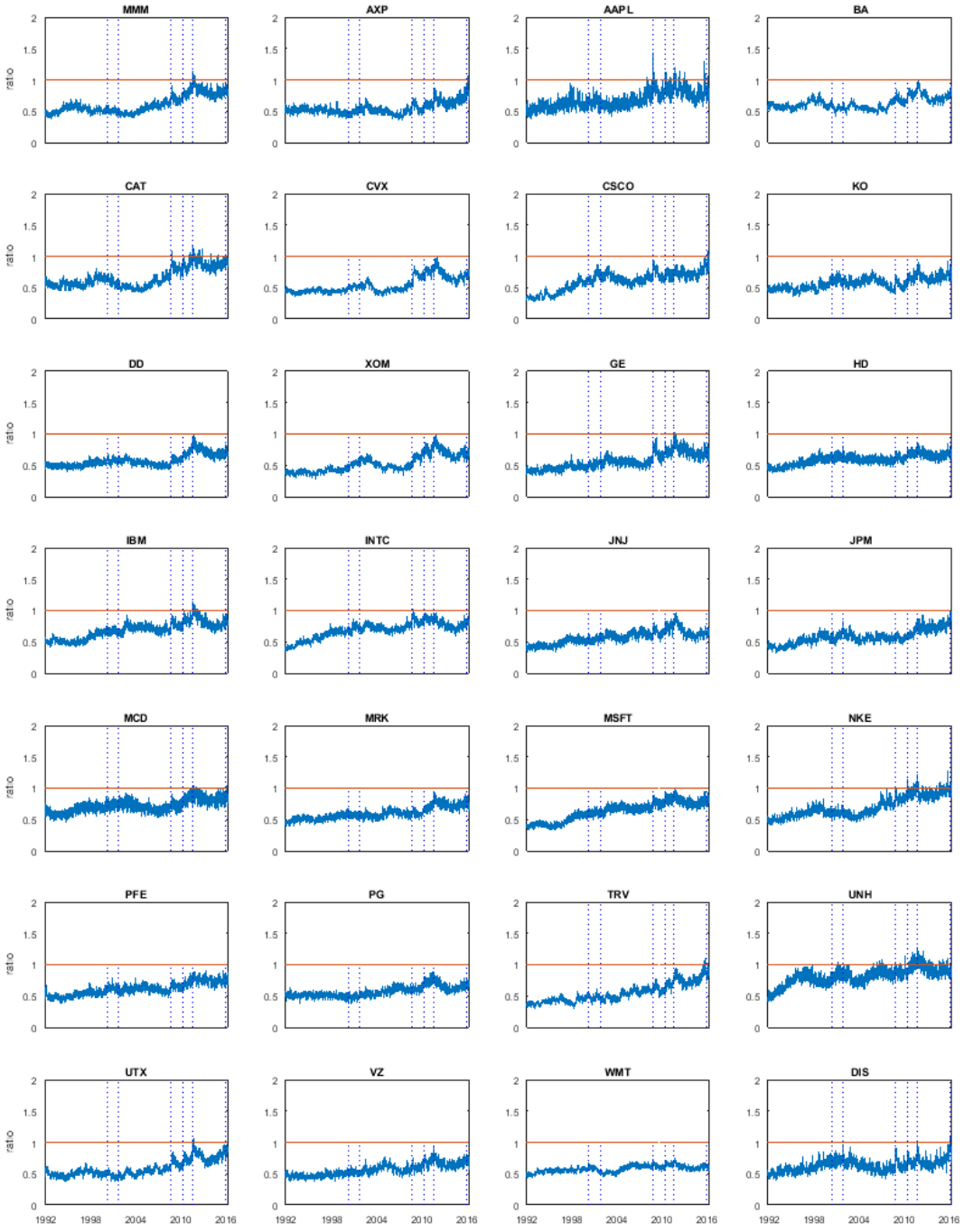
$$r_{ht} \sim \mu_h, \sigma_h^2, \kappa_{3h}, \kappa_{4h},$$

which is consistent with French and Roll (1986). Then daily (based on 8 hour trading day) and weekend (64 hours from Friday close to Monday open) returns should satisfy

$$r_{Dt} \sim 8\mu_h, 8\sigma_h^2, \frac{\kappa_{3h}}{\sqrt{8}}, \frac{\kappa_{4h}}{8} \quad ; \quad r_{Wt} \sim 64\mu_h, 64\sigma_h^2, \frac{\kappa_{3h}}{8}, \frac{\kappa_{4h}}{64}.$$

In fact, overnight returns including weekend returns are very leptokurtic

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This figure shows the dynamic ratio of overnight to intraday volatility, based on the univariate coupled component model with one subplot for each stock. The five dashed vertical lines from left to right represent the dates: 10 March 2000(dot-com bubble), 11 September 2001(the September 11 attacks), 16 September 2008(financial crisis), 6 May 2010(flash crash) and 1 August 2011(August 2011 stock markets fall), respectively. Intraday and overnight volatiles are defined as $\sqrt{\frac{\nu_j}{\nu_j-2}exp(2\lambda_t^j + 2\sigma^j(\frac{t}{T}))}$, for $j = D, N$.

Figure 1: Ratios of overnight to intraday volatility: univariate model

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intraday volatilities, except for the very next one, but substantially impact future overnight volatilities. This inconsistency is probably because the dynamic conditional score model shrinks the impact of extreme overnight observations. After this shrinkage, overnight innovations become closer to the intraday innovations.

Many papers have argued that the introduction of high frequency trading in the period post 2005 has led to an increase in volatility. Figure A.1 in Appendix C plots the intraday and overnight volatilities, $\sqrt{\frac{\nu_j}{\nu_j-2}\exp(2\lambda_t^j + 2\sigma^j(\frac{t}{T}))}$, for $j = D, N$. The five dashed vertical lines from left to right indicate the dates: 10 March 2000 (dot-com bubble), 11 September 2001 (the September 11 attacks), 16 September 2008 (financial crisis), 6 May 2010 (flash crash) and 1 August 2011 (August 2011 stock markets fall). The intraday volatility (red lines) significantly dominated the overnight volatility (black lines) in the first half of the study period, but this domination has disappeared gradually, especially after the 2008 financial crisis. We can also notice that the intraday volatilities after 2005 are in general smaller than that before 2005, except for the financial crisis period. This is contrary to they typical argument that high frequency trading increasing volatilities. To further investigate this, we plot the ratios of overnight to intraday volatility in Figure 1. The stocks all exhibit upward trends over the 24-year period considered here, and many of them experienced peaks around August 2011, corresponding to the August 2011 stock markets fall.

6.2.1 Constancy of the Ratio of overnight to intraday volatility

Figure B.2 in the online appendix depicts the long-run intraday and overnight components, $\sigma^D(t/T)$ and $\sigma^N(t/T)$, and their 95% point-wise confidence intervals. Most stocks arrived at their first peaks around 10 March 2000, corresponding to the Dot-com bubble event, while some arrived at around September 2011, right after the 9-11 attacks. The intraday components reached their second peaks during the financial crisis in September 2008, while overnight components continued to rise, reaching their highest points during the 2010 Flash crash. Roughly speaking, the intraday components were larger than the overnight ones before the first peaks, but smaller after the financial crisis of September 2008. However, it is imperative to remember that the long-run components are constructed with rescaling $\int_0^1 \sigma(s)ds = 0$. In general, the intraday volatility is still larger.

We test the constancy of the ratio of long run overnight to intraday volatility. Figure B.3 in the online appendix displays the test statistics $\widehat{\tau}(s)$ and the 95 % point-wise confidence intervals for $s \in [0, 1]$. Consistent with the results above, the equal ratio null hypothesis is mostly rejected before the first peaks occurred (in 2000) and after the second peaks (in 2010).

Cumulatively, this evidence indicates that the overnight volatility has increased in importance during the 24-year period considered here, relative to the intraday volatility for the Dow Jones stocks.

6.2.2 Volatility Forecast Comparison

We also want to compare our coupled component GARCH model with its one component version for the open to close return to assess the improvement in volatility forecast from using overnight returns. We construct 10 rolling windows, each containing 5652 in-sample and 50 out-of-sample observations.

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In each rolling window, the parameters in the short-run variances are estimated with the in-sample data once and stay the same during the one-step out-of-sample forecast. In the one-step ahead forecast of the long-run covariance matrices, the single-side weight function is used. For instance, to forecast the long-run covariance matrix of period τ ($s = \tau/T$), we set the two-side weight function $K_h(s - t/T) = 0$, for $t >= \tau$, and then rescale $K_h(s - t/T)$ to obtain a sum of 1. Table B.5 in the online appendix reports Giacomini and White (2006) model pair-wise comparison tests with the out-of-sample quasi Gaussian and student t log-likelihood loss functions. For most stocks, the coupled component GARCH model dominates the one component model. Some dominances are statistically significant. We omit the comparison for overnight variance forecast between the one component and the coupled component model, since it is not plausible to estimate a GARCH model with overnight returns alone.

6.2.3 Diagnostic Tests

Ljung-Box tests on the absolute and squared standardized residuals are used to verify whether the coupled component GARCH model is adequate to capture the heteroskedasticity, shown in Table B.6 in the online appendix . With the absolute form, strong heteroskedasticity exists in both intraday and overnight returns, but disappears in the standardized residuals, implying that our model captures the heteroskedasticity well. On the other hand, we are sometimes unable to detect the heteroskedasticity in overnight returns with squared values. In general, the use of the absolute form is more robust when the distribution is heavy tailed.

Figure B.6 in the online appendix displays the quantile-quantile(Q-Q) plots of the intraday innovations, comparing these with the student t distribution with $\hat{\nu}_D$ degrees of freedom. The points in the Q-Q plots approximately lie on a line, showing that the intraday innovations closely approximate the t distribution. Figure B.7 in the online appendix displays the Q-Q plots of the overnight innovations. Many stocks have several outliers in the lower left corners. Our model only partly captures the negative skewness and leptokurtosis of overnight innovations.

6.3 Results of the multivariate model

Figure A.2 and Figure A.3 in Appendix C present the long run correlations between intraday and overnight returns, respectively. Each subplot presents the 27 time series of long run correlations between that individual stock and the remaining stocks. The correlations exhibit an obvious upward trend during our sample period of 1998-2016, and are typically high during the 2008 financial crisis.

Figure B.8 in the online appendix plots the eigenvalues of the dynamic covariance matrices, as well as their proportions (the eigenvalues divided by the sum of eigenvalues). The dynamic of eigenvalues reinforces the previous remark that the stock markets experienced high risk in the 2001 9-11 attacks and in the 2008 financial crisis. The largest eigenvalue represents a strong common component, illustrating that a large proportion of the market financial risk can be explained by one single factor. More specifically, the largest eigenvalue proportion increased substantially between 1998 and 2016. The second and third largest eigenvalues still accounted for a considerable proportion of risk in the

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volatile period from 2000 to 2002, but became rather insignificant in the volatile period from 2008 to 2011. The largest intraday eigenvalue proportion reached its peak in 2008, while the largest overnight eigenvalue proportion remained consistently high until 2011. Remarkably, the largest eigenvalue explained nearly 50% of intraday risk in the 2008 financial crisis, and 70% of overnight risk in the August 2011 stock markets fall.

Table B.7 in the online appendix provides estimates of the multivariate model. Compared to the univariate models, the average β_D decreases to 0.8796 from 0.9515, and the average β_N decreases to 0.8926 from 0.9553. Together with the increase of γ_j and ρ_j, this shows that more weights are given to information in the most recent trading days by taking correlations into consideration.

6.3.1 Robust Correlations

One concern is that our initial correlation estimator is based on the Pearson product moment correlation. This Pearson estimator may perform poorly due to the heavy tails of overnight innovations. Therefore, we also try the robust correlation estimator in the initial step, yet the results remain unchanged as shown in Figure B.2 in the online appendix. Figure B.2 plots the largest eigenvalue proportions of the estimated covariance matrices to assess the difference between using robust (in black) and non-robust (in red) correlation estimators in the initial step. We use solid lines for the initial estimators, and dashed lines for the updated estimators. The updated estimators are obtained in the final estimation step, as previously detailed. Despite the large difference of initial estimators, in particular for overnight returns, the updated estimators are roughly similar. Like the eigenvalues, the updated covariances themselves also remain unchanged for different initial estimators.

6.3.2 Idiosyncratic Risk

To investigate the idiosyncratic risk, we estimate the mean equation with a CAPM structure. As reported in Table B.8 in the online appendix, the intraday alpha (α_D) is significant in 6 stocks, and the overnight alpha (α_N) is significant in 15 stocks. The intraday and overnight beta coefficients, β^{DD} and β^{NN}, are very close, varying from 0.5 to 1.5, and all are significantly positive. For several stocks, the beta coefficients associated with previous market returns, β^{DN} and β^{ND}, are also significant with absolute values larger than 0.1. We then investigate the dynamic covariance of the residuals in the CAPM model. The idiosyncratic risk is indeed quite small for most Dow Jones stocks, as implied by Figure (B.10–B.11) in the online appendix.

7 Application to CRSP stocks

In this section, we investigate 10 cap based portfolios with CRSP stocks from July 1992 to December 2015. July 1992 is the first month that CRSP provides opening stock prices. The prices are adjusted for stock splits and dividends with the cumulative factor in CRSP. Stocks with non-active trade status are excluded for that day, as well as stocks with overnight or intraday returns larger than

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50% in absolute value.[5] CRSP sorts all stocks into 10 deciles based on their market capitalization values, and provide the portfolio assignment for each stock each year. We compute three versions of value weighted intraday and overnight returns for 10 cap based deciles, according to the assignment of three market types: NYSE/AMEX/NASDAQ, NASDAQ, and NYSE.

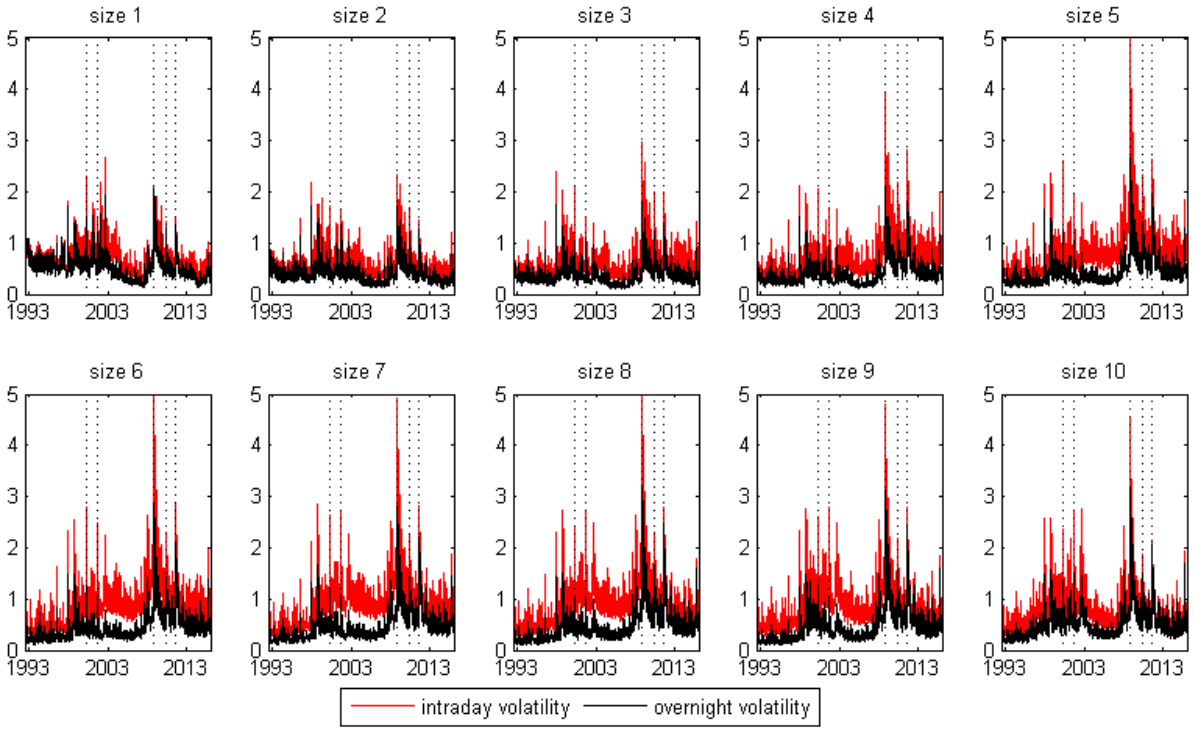
7.1 Preliminary analysis

Table A.1 in Appendix C reports descriptive statistics for overnight and intraday returns of the deciles constructed with NYSE/AMEX/NASDAQ stocks (Panel a). The standard errors are estimated based on the standard deviations of the value-weighted returns across time. The mean intraday and overnight returns are positive and significant, except the mean intraday return in the largest cap decile. Small stocks show higher returns than large stocks, especially during intraday, probably suggesting the existence of size effects. Specifically, the mean intraday (overnight) return is 9.4 bp (6.0 bp) per day in the smallest decile, 9.6 pb (3.2 bp) in the second smallest decile, and only 1.6 bp (2.7 bp) in the largest decile. However, the median values of intraday returns are often larger than 10 bp while their overnight counterparts are overall much smaller around 6 bp. This indicates the right skewness of intraday returns. Panels b and c give the descriptive statistics for NASDAQ deciles and NYSE deciles, respectively. In general, NASDAQ deciles have higher returns than NYSE deciles.

Figure A.4 in Appendix C plots the sample autocorrelation function of intraday and overnight returns for deciles with stocks on NYSE/AMEX/NASDAQ. Both intraday and overnight returns exhibit large and significant positive autocorrelations in small cap deciles. It is worth noting that the autocorrelations of overnight returns decay at a very slow rate, which suggests the existence of long memory. This is consistent with Aboody, Even-Tov, Lehavy, and Trueman (forthcoming). They documented that overnight returns can serve as a measure of firm-specific investor sentiment, partly by showing overnight returns are persistent for periods extending several weeks, and argued that short-term persistence is stronger for harder-to-value firms.

We also notice that the overnight returns in large cap deciles (deciles 5-10) show significant negative first-order autocorrelations, with magnitude around -0.1. For intraday deciles, only the largest cap decile exhibits significantly negative first order autocorrelation, around -0.07. In addition to the autocorrelations, Figure A.5 in Appendix C shows the cross correlations between overnight returns and intraday returns. The correlation between r_t^D and r_t^N on the same day is around 0.1 and significant in most cap deciles, but insignificant in the largest decile. The cross correlation between r_{t-1}^D and r_t^N is significant for all cap deciles, with magnitude even larger than the correlation between r_t^D and r_t^N for decile 1, 2, 3 and 10, which indicates strong positive effects from intraday (overnight) returns to the subsequent overnight (intraday) returns.

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This figure shows the estimated intraday (in red) and overnight (in black) volatilities, $\sqrt{\frac{\nu_j}{\nu_j-2}exp(2\lambda_t^j + 2\sigma^j(\frac{t}{T}))}$, for cap based deciles of NYSE/AMEX/NASDAQ stocks. The five dashed vertical lines from left to right indicate the dates: 10 March 2000 (dot-com bubble), 11 September 2001 (the September 11 attacks), 16 September 2008 (financial crisis), 6 May 2010 (flash crash) and 1 August 2011 (August 2011 stock markets fall), respectively.

Figure 2: Intraday and overnight volatilities: NYSE/AMEX/NASDAQ

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7.2 Overnight and intraday volatilities of CRSP portfolios

We estimate the univariate coupled component GARCH model with the value weighted returns of each size portfolio. Table A.2 of Appendix C reports the estimates and their robust standard errors in the mean equations. Most portfolios exhibit negative and significant δ, indicating overnight returns have positive effects on the subsequent intraday returns. One exception is the largest decile, with insignificant δ. These positive effects seem to be conflicting with the strong reversal effects reported in Berkman, Koch, Tuttle, and Zhang (2012). But note that the reversal effects describe the cross-sectional difference in returns, while our positive effects describe the time-series properties of each return series.

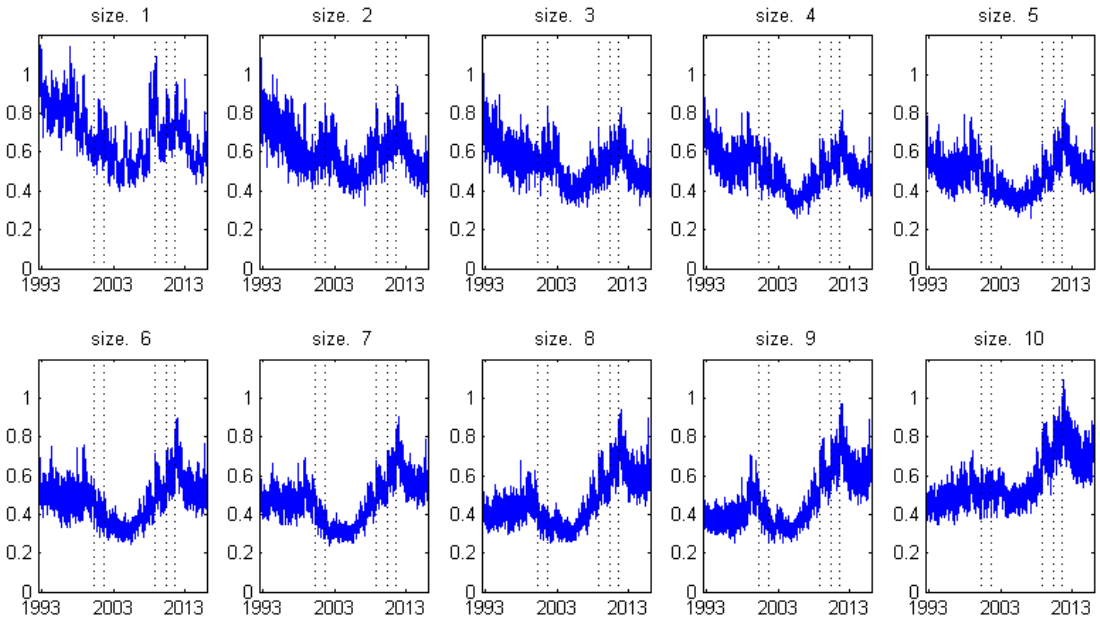
Table A.4 in Appendix C reports the estimates of dynamic parameters for the size portfolios of all NYSE, AMEX and NASDAQ stocks. Parameters β_D and β_N are significantly different from 1, and ρ_D, γ_D, ρ_N and γ_N are significantly positive. The leverage effects are significant, suggesting higher volatility after negative returns. Now the overnight degrees of freedom are larger than 4, and less heavy tailed than that of individual stocks in the previous section.

The main advantage of the coupled component model, relative to the traditional one component version, is that it allows us to investigate the overnight and intraday volatility separately. Figure 2 displays the intraday and overnight volatilities for the size portfolios of all NYSE, AMEX and NASDAQ stocks, $\sqrt{\frac{\nu_j}{\nu_j-2}\exp(2\lambda_t^j + 2\sigma^j(\frac{t}{T}))}$, for $j = D, N$. The five dashed vertical lines from left to right indicate the dates: 10 March 2000 (dot-com bubble), 11 September 2001 (the September 11 attacks), 16 September 2008 (financial crisis), 6 May 2010 (flash crash) and 1 August 2011 (August 2011 stock markets fall). The overnight volatilities (black lines) were typically high during the 2008 financial crisis period for all deciles. Since Asian and European markets are active during the US overnight, we may expect overnight volatilities are largely driven by oversea shocks. Indeed, we find overnight volatilities remained high throughout the period 2009 to 2012, which reflects the European sovereign debt crisis. After that, overnight volatilities became stable again, but still higher than the pre-crisis level. Figure 3 presents the ratio of overnight to intraday volatility of those deciles. The ratio exhibits a downward trend in small cap portfolios (in particular before 2000) and an upward trend in large cap portfolios. Notably, the trend changes monotonically from the smallest cap portfolio to the largest cap portfolio. This indicates that the overnight to intraday volatility ratio is strongly related with size value. We also notice that the ratios were typically high during the 2011 stock markets fall due to the fear of the European sovereign debt crisis. This is consistent with the fact that, during the 2011 crisis, most large shocks stemmed from European markets and occurred in the overnight time of the U.S. market.

Results remain unchanged when we consider NASDAQ stocks or NYSE stocks alone (Figure A.6). However, the slopes are generally flatter for NYSE stocks. It is probably because the small stocks NYSE are much larger than small stocks in NASDAQ. For instance, the average capitalization value of the smallest NYSE decile is similar to that of the decile 3 or decile 4 in NASDAQ stocks. As we can see that the overnight to intraday volatility in decile 1 of NYSE and decile 3 or 4 of NASDAQ

[5]In CRSP, if the closing price is not available on any given trading day, the number in the price field has a negative sign to indicate that it is a bid/ask average and not an actual closing price. We exclude these negative closing price.

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This figure plots the ratio of overnight to intraday volatility for portfolios formed on size. Decile 1 is the portfolio has the smallest capitalization and decile 10 has the largest value. The intraday and overnight volatilities are $\sqrt{\frac{\nu_j}{\nu_j-2}exp(2\lambda_t^j + 2\sigma^j(\frac{t}{T}))}$ for $j = D, N$, respectively.The five dashed vertical lines from left to right indicate the dates: 10 March 2000 (dot-com bubble), 11 September 2001 (the September 11 attacks), 16 September 2008 (financial crisis), 6 May 2010 (flash crash) and 1 August 2011 (August 2011 stock markets fall), respectively.

Figure 3: Ratio of overnight to intraday volatility of size portfolios: NYSE/AMEX/NASDAQ

are indeed quite similar.

With the same approach, we also conduct 10 beta sorted portfolios and 10 standard deviation sorted portfolios. The beta and standard deviation values decrease from decile 1 to decile 10. As plotted in Figure A.7 of Appendix C, nearly all beta sorted and standard deviation sorted deciles exhibit increasing overnight to intraday volatility ratios during the period and were typically high during the European sovereign debt crisis. The only exception is the decile with the largest standard deviation. However, there is no monotonic pattern across deciles.

7.3 Explain the dynamics of overnight to intraday volatility ratio

The empirical results show that the ratio of overnight to intraday volatility for especially large stocks has increased during the last 20 years when accounting for both slowly changing and rapidly changing components. This is contrary to what is often argued with regard to the change in market structure and the predatory practices of certain traders. Portfolios of small stocks on the other hand seem to exhibit a different trend. There are a number of possible explanations for this phenomenon.

Firstly, the trading mechanism has evolved considerably from the early part of the sample and

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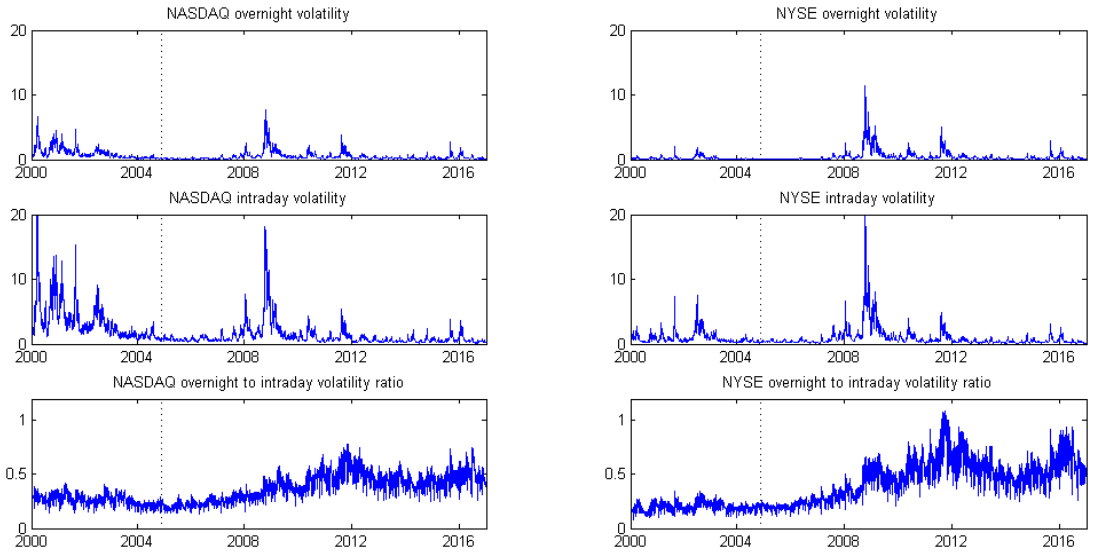
now captures a significant amount of the daily trading volume so that perhaps it is better able to deliver accurate prices than it was. The first change in the trading mechanism we investigate is the minimum tick size change. In the early 1990s, the minimum tick size was largely in 12.5 cents, and NASDAQ dealers colluded implicitly to avoid odd-eighth quotes in order to maintain spreads of at least 25 cents (Christie et al. (1994)). In June 1997, the minimum tick sizes decreased to 6.25 cents with the implement of new SEC rules, and further became a penny after the decimalization in 2001, spreads for most stocks declined (Hasbrouck (2007)). From the opening price of individual stocks, we indeed observe that stocks (even stocks with low prices) often had 25 cents jumps in their opening prices in the 1990s. If stock prices are less than 5 dollars, these 25 cents jumps indicate overnight returns larger than %5, which can lead to big overnight volatility. Moreover, from the cross tabulation analysis of the stock price and size, we observe that small stocks tend to have low prices. This is perhaps the reason why the overnight to intraday volatility ratio decreased during the 1990s.[6] To verify this, we allocate NYSE, AMEX, and NASDAQ stocks (exclude stocks with share prices less than 1 dollar) to ten size groups, using the assignment provided by CRSP. Independently, we sort NYSE, AMEX, and NASDAQ stocks by their spread-to-price values, and split them into 10 groups, using as breakpoints the deciles. The intersections of the sorts produce 100 portfolios. Figure B.5 in the online appendix plots the overnight to intraday volatility ratio for these portfolios. Within the same size group, the slope changes monotonically from the lowest spread-to-price portfolio (almost 0) to the highest spread-to-price portfolio (large negative). [7] We next focus on the post-decimalization period, after February 2000, while most stocks have fairly small spreads.

During 2004, NASDAQ launched opening cross and closing cross to give a single open and close price. To investigate whether these crosses changed the dynamics of the overnight to intraday volatility, we use NYSE as the control group, and estimate the coupled component GARCH model for the value-weighted NASDAQ index and the value-weighted NYSE index. Figure 4 plots the overnight volatility, intraday volatility, and overnight to intraday volatility ratio for both NASDAQ and NYSE. The dashed vertical line indicates the last trading day in October 2004, when NASDAQ had introduced the opening and closing cross. In general, both overnight and intraday volatilities are rather small for both NASDAQ and NYSE in 2004, and the overnight to intraday volatility ratios are similar for NASDAQ and NYSE during 2004. This seems to indicate that the launch of opening

[6]We construct portfolios by size and price, and find low-price groups had more clear downward overnight to intraday volatility ratio during 1990s. After controlling for price, the variation across size become weaker.

[7]Ideally, to compute the price discreteness or bid-ask spreads, we should use the high-frequency intraday returns, but we do not have access to those data. As an alternative, we compute the price discreteness index with the closing ask and bid price provided by CRSP. More specific, we first compute the the minimal closing spread over the year for each stock. These minimal closing spreads mimic the price discreteness fairly well. For instance, the 0.1, 0.5, and 0.9 quantiles of the minimal ask-bid spread across all stocks are respectively 0.125, 0.25 and 0.5 before 1997. These values became 0.0625, 0.125, and 0.375 in 2000, 0.01,0.05 and 0.15 in 2002, and 0.01, 0.01 and 0.07 in 2012. We then standardize the minimal closing spread by the closing price(at the end of the year) to obtain the spread-to-price index. According to this index at year t-1, stocks are equally divided into 10 groups in year t. Together with the 10 size assignments provided by CRSP, we then construct 10 by 10 portfolios. The portfolio returns are from 1994, since the bid and ask prices are not available for most NYSE stocks before 23rd December 1992. Further study is required with more precise spread data.

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This figure plots the overnight volatility, intraday volatility and the overnight to intraday volatility ratio for value-weighted NASDAQ index and value-weighted NYSE index. The dashed vertical line indicates the last trading day in October 2004, when NASDAQ had introduced the opening and closing cross.

Figure 4: NASDAQ and NYSE: Overnight and intraday volatilities

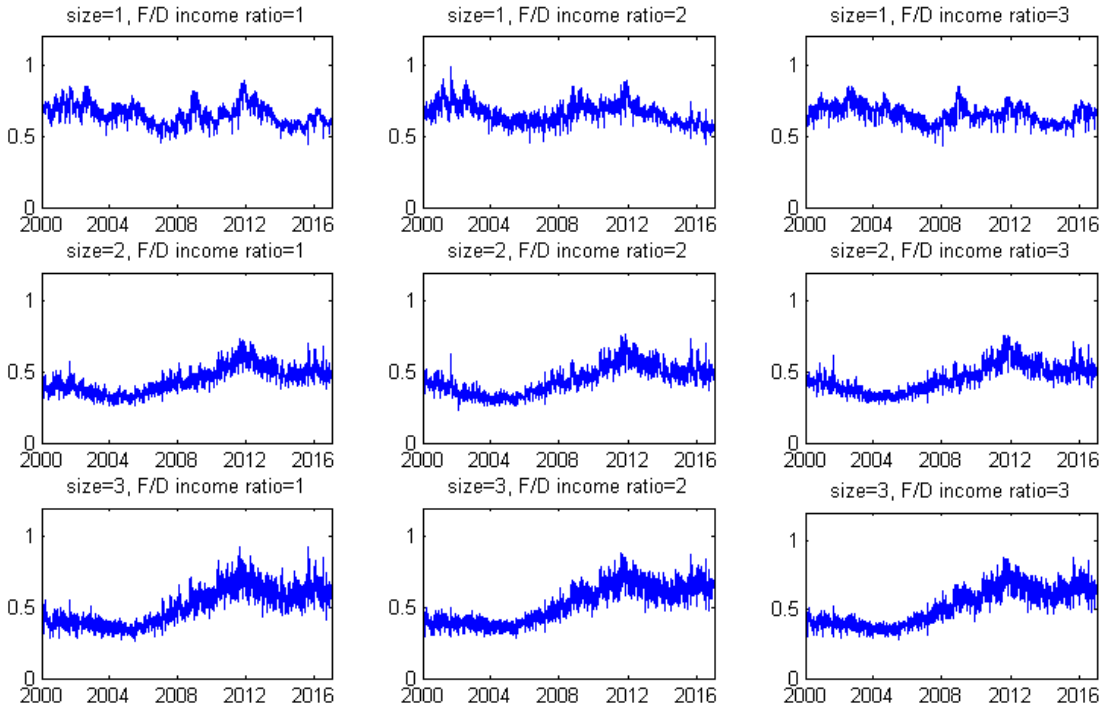
and closing crosses is not a main determinant of the overnight to intraday volatility ratio.

Secondly, we examine whether the variation in international linkage can explain the dynamics of the overnight to intraday volatility ratio. Intuitively, stocks with stronger international linkage should be more sensitive to news about international events that is released while the U.S. market is closed.

We first use the ratio of foreign to domestic income as a proxy of international linkage. Companies with larger proportion of income from foreign countries are supposed to have more exposure to global shocks. We collect the yearly data of pretax foreign income and pretax domestic income from COMPUSTAT, and link them with stock prices obtained from CRSP by index 'CRUSIP'. About 10 percentage of NYSE, AMEX, and NASDAQ stocks in CRSP obtain their income data. At the end of December each year, these stocks (exclude stocks with share prices less than 1 dollar) are allocated to three size groups(small, middle and big), using the assignment provided in CRSP. Specifically, stocks with size assignment 1, 2 or 3 are allocated to the small group, and with assignment 4, 5, 6 or 7 (8,9 or 10) are allocated to the middle(big) group. Independently, we sort NYSE, AMEX, and NASDAQ stocks by the ratio of pretax foreign income to pretax domestic income (F/D income), and split them into 3 groups, using as breakpoints the 30th and 70th percentiles. The intersections of these two sorts produce 9 portfolios. We then estimate the coupled component GARCH model with the equally-weighted returns of these 9 portfolios. [8] Figure 5 shows that the overnight to intraday ratios

[8]Here we use equally-weighted returns instead of value-weighted returns, since the value-weighted returns are largely driven by stocks with big capitalization of each group. Different with other studies, there are only about 60 stocks in

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This figure plots the overnight to intraday volatility ratio for equally-weighted returns of 9 portfolios formed on size and F/D income sorts.

Figure 5: Overnight to intraday volatility ratio: size and F/D income

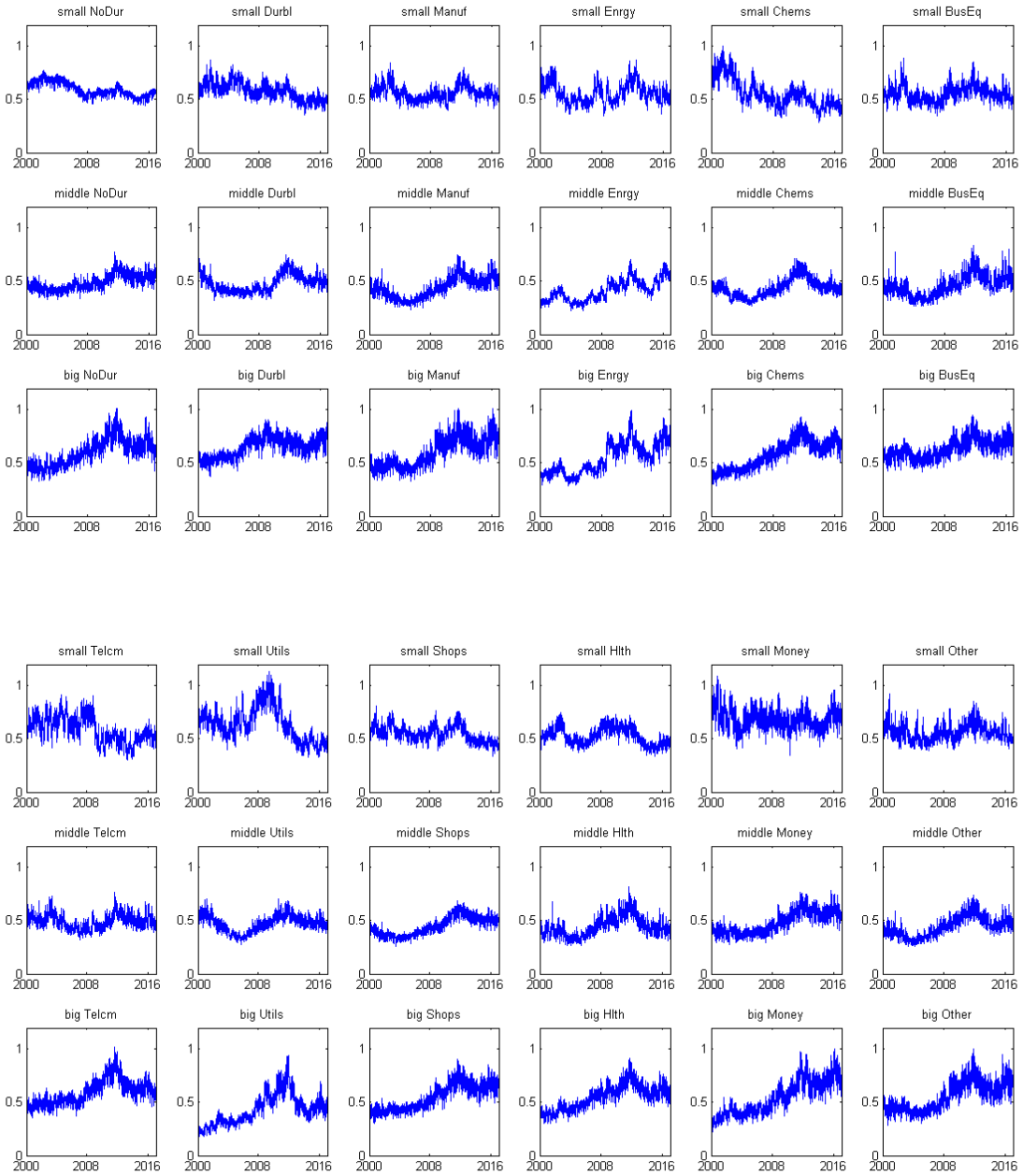
remain unchanged across portfolios with different F/D income, and controlling for F/D income does not weaken the variation across size groups. That is F/D income is not a main factor determining the overnight to intraday volatility ratio.

We further investigate the overnight to intraday volatility ratio across industries since some industries are supposed to have higher international linkage. We use the Fama-French 12 industry classification, and form 12× 3 portfolios by size and industry. The coupled component GARCH model is then estimated for the value-weighted returns of these 36 portfolios. In general, the main variation in overnight to intraday volatility ratio still comes from the size dimension (Figure 6). Although the industry 'Utilities' indeed exhibits smaller overnight to intraday ratio, especially in the early 2000s, but the difference among other industries is small in general. These results are actually consistent with our findings in last section about Dow Jones stocks. Within the Dow Jones there is some variation across firms in terms of the percentage revenue that is earned overseas, from 85% for Intel to 0% for United Health Care, and yet both of these stocks have experienced a similar upward trend in the ratio of overnight to intraday volatility.

We next use the correlation of returns to measure the international linkage. For each stock, we

each group for each year. It sometimes occurs that several stocks have much larger capitalizations than other stocks in the same group, and dominate the value-weighted returns.

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This figure plots the overnight to intraday volatility ratio for portfolios formed on size and industry.

Figure 6: Overnight to intraday volatility ratio: size and industry

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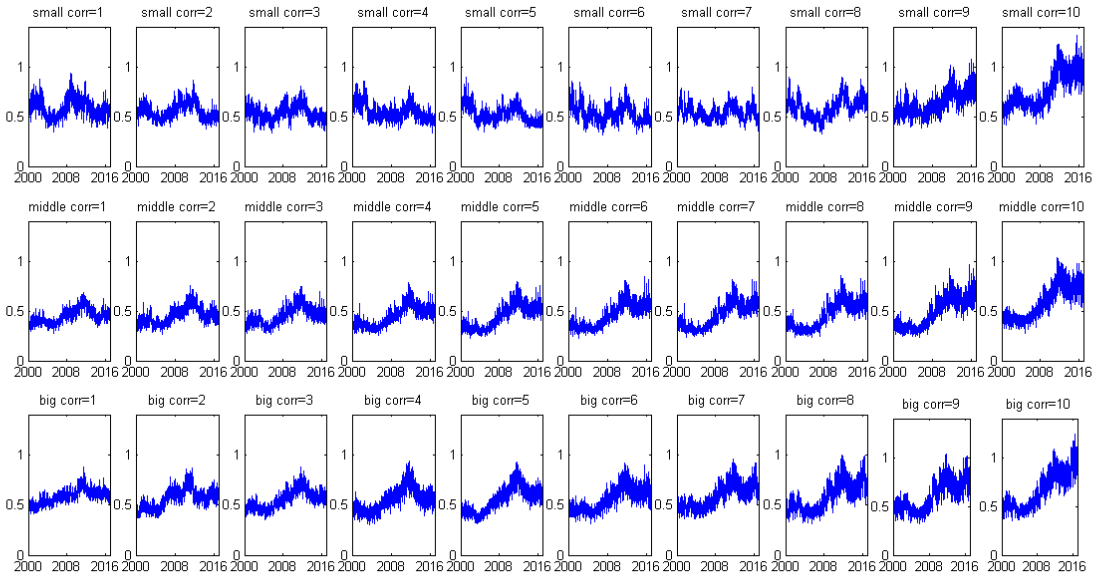
compute the sample correlation of its daily returns and the global excluding U.S. index daily return in each year. The global ex U.S. index is obtained from the Kenneth French data library. We sort all stocks by their correlations in previous year, and split them into 10 groups. Independently, we split stocks into 3 size groups. The intersections of the sorts on correlation and size produce 30 portfolios. Figure 7 plots the overnight to intraday volatility ratio of the value-weighted returns for these 30 portfolios. The slopes of the overnight to intraday ratio changes significantly with international correlations. The overnight to intraday volatility ratios of portfolios with high international correlations portfolios have increased considerably, consistent with the financial integration in recent decades. In contrast, the ratios of portfolios with less international correlations remain quite stable. It is worth to mention that the big stocks with low international correlations now also exhibit flat overnight to intraday volatility ratio. Furthermore, we also observe that high international correlations stocks are more likely also big stocks, consistent with Eun, Huang, and Lai (2008). This can explain the phenomenon that portfolios of big stocks exhibited increasing ratio of overnight to intraday volatility during last 20 years, while portfolios of small stocks did not.

Some other factors perhaps can also explain this phenomenon. For instance, some have argued that the introduction of the EMini contract and its overnight trade ability from 9/9/97 have allowed investors to hedge their overnight positions, which should reduce the riskiness of overnight positions. The liquidity of this contract has slowly grown over the period we study and so this may make a contribution to the changing overnight volatility, except that it should reduce the overnight volatility. In addition, changes in the market structure and technology for trading large firms shares has led to improvements in market quality when the market is open, that is, the electronic trading mechanism delivers less volatility and the market absorbs information faster than it used to. This is partly consistent with the findings of Boehmer, Fong, and Wu (2015) that algorithmic trading has "systematically negative effects on the liquidity of small or low-priced stocks, and AT increases volatility more in those stocks". We leave these for further study.

7.4 Alternative Volatility Measures

As a robustness check for the observed rising trend of most portfolios, we investigate the ratio of VIX to the Rogers and Satchell (1991) volatility (RS volatility). The idea is that the VIX measures one-month ahead volatility, total volatility including presumably intraday and overnight, whereas the aggregated RS volatility only includes intraday volatility. Therefore, the ratio reflects the variability of intraday to overnight to some extent, although it is quite noisy. Figure B.4 in the online appendix presents: (1) the RS volatility on daily Dow Jones stocks, $V_{rs,t}$; (2) one-month ahead RS volatility, $\sqrt{\sum_{i=1}^{22} V_{rs,t+1}^2}$; (3) VIX; and (4) the ratio of VIX to the one month ahead RS volatility. The reported RS volatility is the average RS volatility across the 28 stocks. The ratio of VIX to the one month ahead RS volatility still shows an upward trend during the sample period.

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This figure plots the overnight to intraday volatility ratio for portfolios formed on size and international correlations.

Figure 7: Overnight to intraday volatility ratio: size and international correlation

7.5 Correlations during overnight and intraday: the multivariate model

We next investigate the correlations between overnight and intraday returns with the multivariate model. Figure A.8 in Appendix C presents the long run correlations between intraday and overnight returns in Panel a and Panel b, respectively. Each subplot presents the 9 time series of long run correlations between that decile and the remaining deciles. Both intraday and overnight correlations have increased gradually during the sample period. Especially, the smallest cap decile had rather low correlations with other deciles, around 0.2 (0.3) for overnight (intraday) in 1992, and this number increased considerably to around 0.8 (0.6) for overnight (intraday) in 2015. Despite this substantial increase, when comparing with large stocks, small stocks still co-move less with remaining stocks. We may expect different correlations during trading and non-trading hours. It is possible that short run intraday and overnight correlations are different. But the correlations here are the slowly moving long run correlations that reflect business cycles. We can not observe big difference between these long run intraday and overnight correlations.

8 Conclusion

We have introduced a new coupled component GARCH model for intraday and overnight volatility. This model is able to capture the heavy tails of overnight returns. For each component, we further specify a non-parametric long run smoothly evolving component with a parametric short term fluctuation. The large sample properties of the estimators are provided. For the univariate model we

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show that one can adaptively estimate the parameters of the dynamic process in the presence of the unknown slowly varying trend.

The empirical results show that the ratio of overnight to intraday volatility for especially large stocks has increased during the last 20 years when accounting for both slowly changing and rapidly changing components. This is contrary to what is often argued with regard to the change in market structure and the predatory practices of certain traders. Portfolios of small stocks on the other hand seem to exhibit a different trend. The main explanation for this phenomenon is perhaps the variation in international linkage. Stocks with higher international correlations show considerably higher overnight to intraday volatility ratio, and meanwhile with larger capitalization. The changes of minimal tick size make a contribution to the downward trend of small stocks during 1990s.

We found various other results. First, we found in the multivariate model we find that (slowly moving) correlations between assets have increased during our sample period. In particular, the co-movement of small stocks with the market has increased considerably. We also found that the information in overnight returns is valuable for updating the forecast of the close to close volatility.

In our modelling we have not separated midweek overnight components from weekend components. We may extend the model to allow multiple different components reflecting weekend different from intraweek overnight. There are ten different sessions during a standard week: weekend, Monday day, Monday night, etc. Denote these $j = 1, \ldots, 10$. Let now t reflect week, and define for $j = 1, \ldots, 10$

$$\begin{aligned}\lambda_t^j &= \omega_j(1-\beta_j) + \beta_j\lambda_{t-1}^j + \gamma_j m_{t-1}^j + \gamma_j^*(m_{t-1}^j + 1)\text{sign}(e_{t-1}^j) \\ &+ \sum_{l=1}^{j-1}\rho_{j,l}m_t^l + \sum_{l=1}^{j-1}\rho_{j,l}^*(m_t^l+1)\text{sign}(e_t^l),\end{aligned}$$

where e_t^l, $l = 1, \ldots, 10$ are t-distributed shocks. We may need to impose some structure on $\rho_{j,l}, \rho_{j,l}^*$ to reduce the parameterization. We may also extend the model to allow stocks traded in different time zones, Lin, Engle, and Ito (1994).

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9 Appendix

9.1 Appendix A: Proof of the large sample properties

9.1.1 Proof of Lemma 1

Denote $H^j(s) = \exp(\sigma^j(s))$. We drop the superscript j in what follows and have

$$\begin{aligned} |u_t| &= H(t/T)\,|e_t| = E\,|e_t|\,H(t/T) + H(t/T)\,(|e_t| - E\,|e_t|) \\ \frac{|u_t|}{E\,|e_t|} &= H(t/T) + \frac{H(t/T)}{E\,|e_t|}\,(|e_t| - E\,|e_t|) \\ &=: H(t/T) + \xi_t \end{aligned}$$

where $E\xi_t = 0$. Suppose we know $E\,|e_t|$ .This gives a non-parametric regression function, so we can invoke the Nadaraya-Waston estimator

$$\widetilde{H}(s)^* = \frac{\sum_{t=1}^T K_h(s - t/T)\frac{|u_t|}{E|e_t|}}{\sum_{t=1}^T K_h(s - t/T)}.$$

From Lemma 2, $\{e_t\}$ is a β mixing process with exponential decay, and ξ_t thereby is also a β mixing process with exponential decay. Invoking Theorem 3 in Vogt and Linton (2014), Theorem 4.1 in Vogt (2012) or Kristensen (2009) yields:

$$\sup_{s\in[C_1h,1-C_1h]} \left|\widetilde{H}(s)^* - H_0(s)\right| = O_p\left(\sqrt{\frac{\log T}{Th}} + h^2\right).$$

Denote $\widetilde{\sigma}(s)^* = \log \widetilde{H}(s)^*$. Taylor expansion at $H_0(s)$ gives

$$\widetilde{\sigma}(s)^* = \sigma(s) + \left(\widetilde{H}(s)^* - H(s)\right)\frac{1}{H(s)} - \frac{1}{2}\left(\widetilde{H}(s)^* - H(s)\right)^2 \frac{1}{\bar{H(s)}^2},$$

where $\bar{H}(s)$ is function between $\widetilde{H}(s)^*$ and $H_0(s)$. Therefore,

$$\sup_{s\in[C_1h,1-C_1h]} \left|\widetilde{\sigma}(s)^* - \sigma_0(s)\right| = O_p\left(h^2 + \sqrt{\frac{\log T}{Th}}\right).$$

For $s \in [0,h] \cup [1-h,1]$, we use a boundary kernel to ensure the bias property holds through $[0,1]$.

Until now we have obtained the property for the un-rescaled estimator $\widetilde{\sigma}(s)^*$. Next, we are going to show the convergence rate of the rescaled estimator $\widetilde{\sigma}(s)$. Recall that

$$\widetilde{\sigma}(s) = \widetilde{\sigma}(s) - \frac{1}{T}\sum_{t=1}^T \widetilde{\sigma}(\frac{t}{T}),$$

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and we can rewrite $\widetilde{\sigma}(s)$ as:

$$\widetilde{\sigma}(s) = \widetilde{\sigma}(s)^* - \frac{1}{T}\sum_{t=1}^{T}\widetilde{\sigma}(\frac{t}{T})^*,$$

as $E|e_t|$ in $\widetilde{\sigma}(s)^*$ has vanished due to the rescaling. Plugging this into $\sup_{s\in[C_1h,1-C_1h]}|\widetilde{\sigma}(s)-\sigma_0(s)|$ gives

$$\begin{aligned}
&\sup_{s\in[0,1]}|\widetilde{\sigma}(s)-\sigma_0(s)| \\
&= \sup_{s\in[0,1]}\left|\widetilde{\sigma}(s)^* - \frac{1}{T}\sum_{t=1}^{T}\widetilde{\sigma}(\frac{t}{T})^* - \sigma_0(s)\right| \\
&= \sup_{s\in[0,1]}\left|\widetilde{\sigma}(s)^* - \frac{1}{T}\sum_{t=1}^{T}\widetilde{\sigma}(\frac{t}{T})^* - \sigma_0(s) - \frac{1}{T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T}) + \frac{1}{T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T})\right| \\
&\leq \sup_{s\in[0,1]}\left|\widetilde{\sigma}(s)^* - \sigma_0(s)\right| + \left|\frac{1}{T}\sum_{t=1}^{T}\left(\widetilde{\sigma}(\frac{t}{T})^* - \sigma_0(\frac{t}{T})\right)\right| + \left|\frac{1}{T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T})\right| \\
&= O_p\left(h^2+\sqrt{\frac{\log T}{Th}}\right) + O_p\left(h^2+\sqrt{\frac{\log T}{Th}}\right) + \left|\frac{1}{T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T})\right| \\
&= O_p\left(h^2+\sqrt{\frac{\log T}{Th}}\right) + \left|\frac{1}{T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T})\right|.
\end{aligned}$$

We only have to work out the second term $\left|\frac{1}{T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T})\right|$. According to Theorem 1.3 in Tasaki (2009),

$$\lim_{T\to\infty} T^2\left(\int_0^1\sigma_0(s)ds - \frac{1}{2T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T}) - \frac{1}{2T}\sum_{t=0}^{T-1}\sigma_0(\frac{t}{T})\right) = -\frac{1}{12}\left(\sigma_0'(1)-\sigma_0'(0)\right).$$

Since $\int_0^1\sigma_0(s)ds = 0$ and $\sigma_0'(1)-\sigma_0'(0)$ is bounded by Assumption A4, it follows

$$\begin{aligned}
\left|\frac{1}{T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T})\right| &\leq \left|\frac{1}{2T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T}) + \frac{1}{2T}\sum_{t=0}^{T-1}\sigma_0(\frac{t}{T})\right| + \left|\frac{1}{2T}\sum_{t=1}^{T}\sigma_0(\frac{t}{T}) - \frac{1}{2T}\sum_{t=0}^{T-1}\sigma_0(\frac{t}{T})\right| \\
&= O(T^{-2}) + \frac{1}{2T}\left|\sigma_0(1)-\sigma_0(0)\right| \\
&= O(T^{-1}).
\end{aligned}$$

Therefore, the uniform convergence rate is

$$\begin{aligned}
\sup_{s\in[0,1]}|\widetilde{\sigma}(s)-\sigma_0(s)| &= O_p\left(h^2+\sqrt{\frac{\log T}{Th}}\right) + O(T^{-1}) \\
&= O_p\left(h^2+\sqrt{\frac{\log T}{Th}}\right).
\end{aligned}$$

■

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9.1.2 Proof of Theorem 1

Let $\phi_i = \beta_D$ and θ_k be an element in function $\sigma^D(\cdot)$ (for simplicity, the subscript k is omitted in the following explanation). Recall that $h_t^j = \lambda_t^j + \sigma^j(t/T)$, and the log-likelihood function, without unnecessary constant, can be rewritten as a function of h_t^j

$$l_t^j = -h_t^j - \frac{v_j+1}{2}\ln\left(1+\frac{(u_t^j)^2}{v_j\exp(2h_t^j)}\right) + \ln\Gamma\left(\frac{v_j+1}{2}\right) - \frac{1}{2}\ln v_j - \ln\Gamma\left(\frac{v_j}{2}\right)$$

with the score functions

$$\begin{aligned}\frac{\partial l_t}{\partial\theta} &= \frac{\partial l_t^D}{\partial h_t^D}\frac{\partial h_t^D}{\partial\theta} + \frac{\partial l_t^N}{\partial h_t^D}\frac{\partial h_t^D}{\partial\theta} = m_t^D\frac{\partial h_t^D}{\partial\theta} + m_t^N\frac{\partial h_t^D}{\partial\theta}\\ \frac{\partial l_t}{\partial\beta_D} &= \frac{\partial l_t^D}{\partial h_t^D}\frac{\partial h_t^D}{\partial\beta_D} = m_t^D\frac{\partial h_t^D}{\partial\beta_D} + m_t^N\frac{\partial h_t^D}{\partial\beta_D}.\end{aligned}$$

Recall that $m_t^j = (v_j+1)b_t^j - 1$,with b_t^j independent and identically beta distributed, we have $E\left(m_t^N m_t^D\right) = 0$, $E\left(m_t^j\right)^2$ is time invariant, and $E\left(m_t^j\right)^2 < \infty$. Therefore, we can write

$$\sum_{t=1}^T E\frac{\partial l_t}{\partial\theta}\frac{\partial l_t}{\partial\beta_D} = E\left(m_t^D\right)^2\sum_{t=1}^T\frac{\partial h_t^D}{\partial\theta}\frac{\partial h_t^D}{\partial\beta_D} + E\left(m_t^N\right)^2\sum_{t=1}^T E\frac{\partial h_t^N}{\partial\theta}\frac{\partial h_t^N}{\partial\beta_D}.$$

To prove the theorem, it then suffices to show that

$$\left\|\sum_{t=1}^T E\begin{pmatrix}\frac{\partial h_t^D}{\partial\theta}\\ \frac{\partial h_t^N}{\partial\theta}\end{pmatrix}\begin{pmatrix}\frac{\partial h_t^D}{\partial\beta_D} & \frac{\partial h_t^N}{\partial\beta_D}\end{pmatrix}\right\|_\infty = O(1).$$

By expressing λ_t^j as a function of φ and $\left\{\left(\sigma^j(\frac{t-i}{T}), u_{t-i}^j\right), i\geq 0\right\}$, we can write $\frac{\partial h_t^j}{\partial\theta}$ as

$$\begin{aligned}\begin{pmatrix}\frac{\partial h_t^D}{\partial\theta}\\ \frac{\partial h_t^N}{\partial\theta}\end{pmatrix} &= \sum_{k=0}^T\begin{pmatrix}\frac{\partial h_t^D}{\partial\sigma^D(\frac{t-k}{T})}\frac{\partial\sigma^D(\frac{t-k}{T})}{\partial\theta}\\ \frac{\partial h_t^N}{\partial\sigma^D(\frac{t-k}{T})}\frac{\partial\sigma^D(\frac{t-k}{T})}{\partial\theta}\end{pmatrix}\\ &= \sum_{k=0}^T\begin{pmatrix}\frac{\partial h_t^D}{\partial\sigma^D(\frac{t-k}{T})}\\ \frac{\partial h_t^N}{\partial\sigma^D(\frac{t-k}{T})}\end{pmatrix}\psi_i^D\left(\frac{t-k}{T}\right),\end{aligned}$$

when the limit exists. Thus we obtain,

$$\begin{aligned}&\frac{1}{T}\sum_{t=1}^T E\begin{pmatrix}\frac{\partial h_t^D}{\partial\theta}\\ \frac{\partial h_t^N}{\partial\theta}\end{pmatrix}\begin{pmatrix}\frac{\partial h_t^D}{\partial\beta_D} & \frac{\partial h_t^N}{\partial\beta_D}\end{pmatrix}\\ &= \frac{1}{T}\sum_{k=0}^T\sum_{t=1}^T E\begin{pmatrix}\frac{\partial h_t^D}{\partial\sigma^D(\frac{t-k}{T})}\\ \frac{\partial h_t^N}{\partial\sigma^D(\frac{t-k}{T})}\end{pmatrix}\begin{pmatrix}\frac{\partial h_t^D}{\partial\beta_D} & \frac{\partial h_t^N}{\partial\beta_D}\end{pmatrix}\psi_i^D\left(\frac{t-k}{T}\right)\\ &= \frac{1}{T}\sum_{k=0}^T E\begin{pmatrix}\frac{\partial h_t^D}{\partial\sigma^D(\frac{t-k}{T})}\\ \frac{\partial h_t^N}{\partial\sigma^D(\frac{t-k}{T})}\end{pmatrix}\begin{pmatrix}\frac{\partial h_t^D}{\partial\beta_D} & \frac{\partial h_t^N}{\partial\beta_D}\end{pmatrix}\sum_{t=1}^T\psi_i^D\left(\frac{t-k}{T}\right).\end{aligned}$$

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The second equality follows since $E\begin{pmatrix}\frac{\partial h_t^D}{\partial \sigma^j(\frac{t-k}{T})}\\ \frac{\partial h_t^N}{\partial \sigma^j(\frac{t-k}{T})}\end{pmatrix}\begin{pmatrix}\frac{\partial h_t^D}{\partial \beta_D} & \frac{\partial h_t^N}{\partial \beta_D}\end{pmatrix}$ is invariant across time t by Lemma 4. Taylor expansion of $\sum_{t=1}^T \psi_i^D\left(\frac{t-k}{T}\right)$ around $\sum_{t=1}^T \psi_i^D\left(\frac{t-k}{T}\right)$ gives

$$\begin{aligned}\frac{1}{T}\sum_t \psi_i^D\left(\frac{t-k}{T}\right) &= \frac{1}{T}\sum_t \psi_i^D\left(\frac{t}{T}\right) - \frac{1}{T}\frac{k}{T}\sum_t \psi_i^{D'}\left(\frac{t}{T}\right) + O\left(\frac{k}{T}\right)^2\\ &= O\left(\frac{1}{T}\right) + O\left(\frac{k}{T}\right) + O\left(\frac{k}{T}\right)^2\\ &= O\left(\frac{k}{T}\right).\end{aligned}$$

Hence, it suffices to show

$$\sum_{k=0}^T \left\| k\left(E\begin{pmatrix}\frac{\partial h_t^D}{\partial \sigma^D(\frac{t-k}{T})}\\ \frac{\partial h_t^N}{\partial \sigma^D(\frac{t-k}{T})}\end{pmatrix}\begin{pmatrix}\frac{\partial h_t^D}{\partial \beta_D} & \frac{\partial h_t^N}{\partial \beta_D}\end{pmatrix}\right)\right\|_\infty < \infty,$$

which is obtained by Lemma 3.

The proof with respect to v_D is similar, but the score function is slightly different. The score functions of l_t^D and l_t^N with respect to v_D are

$$\begin{aligned}\frac{\partial l_t^D}{\partial v_D} &= -\frac{1}{2}\ln\left(1+\frac{(u_t^D)^2}{v_D\exp(2h_t^D)}\right) + \frac{\partial}{\partial v_D}\left(\ln\Gamma\left(\frac{v_D+1}{2}\right) - \ln\Gamma\left(\frac{v_D}{2}\right)\right) - \frac{1}{2v_D}\\ &\quad + \frac{v_D+1}{2\left(1+\frac{(u_t^D)^2}{v_D\exp(2h_t^D)}\right)}\frac{(u_t^D)^2}{v_D^2\exp(2h_t^D)}\left(1+2v_D\frac{\partial h_t^D}{\partial v_D}\right) + \frac{\partial h_t^D}{\partial v_D} \qquad (19)\\ \frac{\partial l_t^N}{\partial v_D} &= \frac{v_N+1}{2\left(1+\frac{(u_t^N)^2}{v_N\exp(2h_t^N)}\right)}\frac{(u_t^N)^2}{v_N^2\exp(2h_t^N)}\left(1+2v_N\frac{\partial h_t^N}{\partial v_D}\right) + \frac{\partial h_t^N}{\partial v_D}.\end{aligned}$$

Then we can have

$$\begin{aligned}&\sum_{t=1}^T E\frac{\partial l_t^D}{\partial\theta}\frac{\partial l_t^D}{\partial v_D}\\ &= \sum_{t=1}^T Em_t^D\frac{\partial h_t^D}{\partial\theta}\left[\frac{\partial h_t^D}{\partial v_D} - \frac{1}{2}\ln\left(1+\frac{(u_t^D)^2}{v_D\exp(2h_t^D)}\right)\right]\\ &+ \sum_{t=1}^T Em_t^D\frac{\partial h_t^D}{\partial\theta}\frac{\partial\ln\Gamma\left(\frac{v_D+1}{2}\right) - \frac{1}{2}\ln v_D - \ln\Gamma\left(\frac{v_D}{2}\right)}{\partial v_D}\\ &+ \sum_{t=1}^T Em_t^D\frac{\partial h_t^D}{\partial\theta}\left[\frac{v_D+1}{2\left(1+\frac{(u_t^D)^2}{v_D\exp(2h_t^D)}\right)}\frac{(u_t^D)^2}{v_D^2\exp(2h_t^D)}\left(1+2v_D\frac{\partial h_t^D}{\partial v_D}\right)\right]\\ &= \frac{1}{2}E\left(m_t^D\left(-\ln\left(1+\frac{(\varepsilon_t^D)^2}{v_D}\right) + \frac{v_D+1}{2v_D+(\varepsilon_t^D)^2}\frac{(\varepsilon_t^D)^2}{v_D}\right)\right)\frac{1}{T}\sum_{t=1}^T E\frac{\partial h_t^D}{\partial\theta}\\ &+ E\left(m_t^D\left(1+\frac{(v_D+1)(\varepsilon_t^D)^2}{2v_D+(\varepsilon_t^D)^2}\right)\right)\frac{1}{T}\sum_{t=1}^T E\frac{\partial h_t^D}{\partial v_D}\frac{\partial h_t^D}{\partial\theta}.\end{aligned}$$

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The first term vanishes by Lemma 5. Then we can use the same procedure above to obtain $\sum_{t=1}^{T} E\frac{\partial h_t^D}{\partial v_D}\frac{\partial h_t^D}{\partial \theta} = O(1)$, and to finish the proof for v_D. ■

9.1.3 Proof of Theorem 2

The general strategy is that we first show the estimator obtained by maximizing $\sum l(\phi;\tilde{\theta})$ and $\sum l(\phi;\theta)$ has the same asymptotic distribution, provided $\left\|\tilde{\theta}-\theta_0\right\|$ converges to 0. Then we show $\left\|\tilde{\theta}-\theta_0\right\|$ converges to 0, given the uniform convergence rate for $\tilde{\sigma(s)}$ in Lemma 1. As a result, the asymptotic property of $\tilde{\phi}$ follows as in a parameter model.

Following Severini and Wong (1992), the expansion of $\frac{1}{\sqrt{T}}\sum_{t=1}^{T}\frac{\partial l_t(\phi_{i0},\tilde{\theta})}{\partial\phi}$ at θ_0 gives

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T}\frac{\partial l_t(\phi_0;\tilde{\theta})}{\partial\phi} = \frac{1}{\sqrt{T}}\sum_{t=1}^{T}\frac{\partial l_t(\phi_0;\theta_0)}{\partial\phi} + \frac{1}{\sqrt{T}}\sum_{k}^{\infty}\sum_{t=1}^{T}\frac{\partial^2 l_t(\phi_0;\theta_0)}{\partial\phi\partial\theta_k}(\tilde{\theta}_k-\theta_{k,0}) + o_p(1). \tag{20}$$

According to Theorem 1, we have $E\sum_{t=1}^{T}\frac{\partial^2 l_t(\phi;\theta)}{\partial\phi_i\partial\theta_k} = O(1)$, for each k and i, $k\in\{1,\ldots,\infty\}$ and $i\in\{1,\ldots,14\}$. It follows that

$$\sum_{t=1}^{T}\frac{\partial^2 l_t(\phi_0;\theta_0)}{\partial\phi_i\partial\theta_k} = O_p(\sqrt{T}).$$

Given that the dimension of the sieve space grows slowly and $\tilde{\theta}$ converges to θ, the second term in (20) will be of order $o_p(1)$.

Next we investigate the relationship between the kernel estimator and $\tilde{\theta}$ in a very general setting. Recall that $\sigma(s)=\sum_{j=1}^{\infty}\theta_j\psi_j(s)$ for some orthogonal basis ψ_j. We assume that $\tilde{\sigma}(s)$ is a member of the same normed space as $\sigma(s)$ in which case we can write

$$\tilde{\sigma}(s) = \sum_{j=1}^{\infty}\tilde{\theta}_j\psi_j(s)$$

for some coefficients $\tilde{\theta}_j$, $j=1,2,\ldots$. We have

$$\begin{aligned}\int\left(\tilde{\sigma}(s)-\sigma(s)\right)^2 ds &= \int\left(\sum_{j=1}^{\infty}\left(\tilde{\theta}_j-\theta_j\right)\psi_j(s)\right)^2 ds\\ &= \sum_{j=1}^{\infty}\left(\tilde{\theta}_j-\theta_j\right)^2\int\psi_j^2(s)ds\\ &= \sum_{j=1}^{\infty}\left(\tilde{\theta}_j-\theta_j\right)^2\end{aligned}$$

under the assumption that ψ_j are orthonormal. So given the L_2 rate of convergence of $\tilde{\sigma}$ we have the same convergence rate for the implied coefficients. Hence, given the uniform convergence rate for $\tilde{\sigma}(s)$ in Lemma 1, $\left\|\tilde{\theta}-\theta_0\right\|$ converges to 0, and the estimator obtained by maximizing $\sum l(\phi;\tilde{\theta})$ and $\sum l(\phi;\theta)$ has the same property.

Therefore, the asymptotic property of $\tilde{\phi}$ can be obtained with a similar procedure to Harvey (2013). He gives the consistency and asymptotic normality of the estimator for the parametric beta-t-egarch model. The basic idea is that the first three derivatives of l_t with respect to ϕ (except v_j)

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are linear combinations of $b_t^h(1-b_t)^k$, h, $k=0,1,2,\ldots$, with $b_t = \frac{(1+v)(e_t)^2}{v\exp(2\lambda_t)+(e_t)^2}$. Since b_t is beta distributed, these first three derivatives are all bounded. It is then straightforward to show that the score function satisfies a CLT, and its derivative converges to the information matrix by the ergodic theorem.

Obviously, $\hat{\phi}$ has the same asymptotic property of $\tilde{\phi}$, since $\sum_{t=1}^{T}\frac{\partial l_t(\phi_{i0},\tilde{\theta})}{\partial\phi}$ and $\sum_{t=1}^{T}\frac{\partial l_t(\phi_{i0},\theta)}{\partial\phi}$ have the same asymptotic property. ■

9.1.4 Proof of Theorem 3

Consider the local likelihood function given η_t^j and v_j, i.e., minimize the objective function

$$L_T^j(\sigma^j;s) = \frac{1}{T}\sum_{t=1}^{T}K_h(s-t/T)\left[\sigma^j + \frac{v_j+1}{2}\ln\left(1+\frac{(\eta_t^j\exp(-\sigma^j))^2}{v_j}\right)\right]$$

with respect to ω, for $j=D,N$ separately. The first order and second order derivatives are:

$$\begin{aligned}\frac{\partial L_T^j(\sigma^j;s)}{\partial\sigma^j} &= \frac{1}{T}\sum_{t=1}^{T}K_h(s-t/T)\left[-(v_j+1)b_t^j(\sigma^j)+1\right]\\ \frac{\partial^2 L_T^j(\sigma^j;s)}{\partial\sigma^{j2}} &= 2(v_j+1)\frac{1}{T}\sum_{t=1}^{T}K_h(s-t/T)\left[b_t^j(\sigma^j)\left(1-b_t^j(\sigma^j)\right)\right],\end{aligned} \tag{21}$$

where

$$b_t^j(\sigma^j) = \frac{\frac{(\eta_t^j)^2}{v_j}}{\exp(2\sigma^j)+\frac{(\eta_t^j)^2}{v_j}}.$$

We have

$$\sqrt{Th}\left(\widehat{\sigma^j}(s)-\sigma_0^j(s)\right) = \left[\frac{1}{Th}\frac{\partial^2 L_T^j(\sigma_0^j;s)}{\partial\sigma^{j2}}\right]^{-1}\frac{1}{\sqrt{Th}}\frac{\partial L_T^j(\sigma_0^j;s)}{\partial\sigma^j}+o_p(1),$$

This is asymptotically normal with mean zero and variance (when the t distribution is correct)

$$\text{var}\left[\frac{1}{\sqrt{Th}}\frac{\partial L_T^j(\sigma_0^j;s)}{\partial\sigma^j}\right] = ||K||_2^2 E\left[\left(1-(v_j+1)b_t^j(\sigma_0^j(s))\right)^2\right]_{t/T=s}.$$

This follows because

$$E\left[\left(1-(v_j+1)b_t^j(\sigma_0^j(s))\right)^2\right] = f(t/T)$$

for some smooth function f, and recall $\eta_t^j = \exp(\sigma^j(t/T))\varepsilon_t^j$. It follows that

$$\frac{h^2}{Th}\sum_{t=1}^{T}K_h^2(s-t/T)f(t/T)\to||K||_2^2 f(s),$$

Therefore,

$$\sqrt{Th}\left(\widehat{\sigma^j}(s)-\sigma_0^j(s)\right)\Longrightarrow N\left(0,\frac{||K||_2^2}{E\left[\left(1-(v_j+1)b_t^j\right)^2\right]_{t/T=s}}\right)$$

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Further, since b_t^j is distributed as $beta(\frac{1}{2}, \frac{v_j}{2})$, with

$$E\left[\left(1-(v_j+1)b_t^j\right)^2\right]_{t/T=s} = \frac{2v_j}{(v_j+3)}.$$

It thus follows that

$$\sqrt{Th}\left(\widehat{\sigma^j}(s) - \sigma_0^j(s)\right) \Longrightarrow N\left(0, \sqrt{\frac{(v_j+3)}{2v_j}||K||_2^2}\right).$$

when the t distribution is correct. ■

9.1.5 Other Lemmas

Lemma 2 *If $|\beta_j| < 1$, $j = D, N$, then e_t^j and λ_t^j are strictly stationary and β-mixing with exponential decay.*

Proof. For simplicity, we consider the model without leverage effects

$$\lambda_t^D = \omega_D(1-\beta_D) + \beta_D\lambda_{t-1}^D + \gamma_D m_{t-1}^D + \rho_D m_t^N$$
$$\lambda_t^N = \omega_N(1-\beta_N) + \beta_N\lambda_{t-1}^N + \gamma_N m_{t-1}^N + \rho_N m_{t-1}^D.$$

Let us write it as

$$\begin{pmatrix} \lambda_t^D \\ \lambda_t^N \\ m_t^D \\ m_t^N \end{pmatrix} = \begin{pmatrix} \beta_D & 0 & \gamma_D & 0 \\ 0 & \beta_N & \rho_N & \beta_N \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \lambda_{t-1}^D \\ \lambda_{t-1}^N \\ m_{t-1}^D \\ m_{t-1}^N \end{pmatrix} + \begin{pmatrix} \rho_D m_t^N + \omega_D(1-\beta_D) \\ \omega_N(1-\beta_N) \\ m_t^D \\ m_t^N \end{pmatrix}.$$

Since m_t^N and m_t^D are i.i.d random variables and follow a beta distribution, we can easily find an integer $s \geq 1$ to satisfy

$$E\left|\begin{array}{c} \rho_D m_t^N + \omega_D(1-\beta_D) \\ \omega_N(1-\beta_N) \\ m_t^D \\ m_t^N \end{array}\right|^s < \infty$$

(Condition A_2 in Carrasco and Chen (2002)). The largest eigenvalue of the matrix

$$\begin{vmatrix} \beta_D & 0 & \gamma_D & 0 \\ 0 & \beta_N & \rho_N & \beta_N \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

is smaller than 1 by assumption. Define $X_t = \begin{pmatrix} \lambda_t^D & \lambda_t^N & m_t^D & m_t^N \end{pmatrix}^\intercal$. According to Proposition 2 in Carrasco and Chen (2002), the process X_t is Markov geometrically ergodic and $E|X_t|^s < \infty$. Moreover, if X_t is initialized from the invariant distribution, it is then strictly stationary and β-mixing with exponential decay. The process $\{e_t^j\}$ is a generalized hidden Markov model and stationary β-mixing with a decay rate at least as fast as that of $\{\lambda_t^j\}$ by Proposition 4 in Carrasco and Chen (2002). The extension to the model with leverage effects is straightforward, by defining $X_t = \begin{pmatrix} \lambda_t^D & \lambda_t^N & m_t^D & m_t^N & \text{sign}\left(e_t^D\right) & \text{sign}\left(e_t^N\right) \end{pmatrix}^\intercal$. ■

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Lemma 3 *Under Assumption A1-A4, it holds that*

$$\sum_k k \left\| E\left(\begin{bmatrix} \frac{\partial h_t^D}{\partial \sigma^D (t-k/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D (t-k/T)} \end{bmatrix} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D & \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix} \right) \right\|_\infty < \infty.$$

Proof. By (24) and (25), we have

$$E\begin{pmatrix} \frac{\partial h_{t+1}^D}{\partial \sigma^D (t+k/T)} \\ \frac{\partial h_{t+1}^N}{\partial \sigma^D (t+k/T)} \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_{t+1}^D & \frac{\partial}{\partial \beta_D} h_{t+1}^N \end{pmatrix} = EA_{t+1} \begin{pmatrix} a_t^{DD} \\ a_t^{ND} \end{pmatrix} \begin{pmatrix} \lambda_t^D - \omega_D & 0 \end{pmatrix} A_{t+1}^T; k = 1$$

$$E\begin{pmatrix} \frac{\partial h_{t+1}^D}{\partial \sigma^D (t+k/T)} \\ \frac{\partial h_{t+1}^N}{\partial \sigma^D (t+k/T)} \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_{t+1}^D & \frac{\partial}{\partial \beta_D} h_{t+1}^N \end{pmatrix} = E \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} \lambda_t^D - \omega_D & 0 \end{pmatrix} A_{t+1}^T = 0; k = 0.$$

When $k > 1$, it holds

$$\begin{aligned}
&\operatorname{vec}\begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D (t-k/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D (t-k/T)} \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D & \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix} \\
&= \operatorname{vec} A_t \left(\prod_{i=1}^{k-1} B_{t-i} A_{t-i} \right) \Lambda_{t-k} \begin{pmatrix} \lambda_{t-1}^D - \omega_D & 0 \end{pmatrix} A_t^T \\
&+ \operatorname{vec} A_t \left(\prod_{i=1}^{k-1} B_{t-i} A_{t-i} \right) \Lambda_{t-k} \begin{pmatrix} \lambda_{t-2}^D - \omega_D & 0 \end{pmatrix} A_{t-1}^T B_{t-1}^T A_t^T \\
&+ ... \\
&+ \operatorname{vec} A_t \left(\prod_{i=1}^{k-1} B_{t-i} A_{t-i} \right) \Lambda_{t-k} \begin{pmatrix} \lambda_{t-k+1}^D - \omega_D & 0 \end{pmatrix} A_{t-k+2}^T B_{t-k+2}^T ... A_{t-1}^T B_{t-1}^T A_t^T \\
&= \sum_{j=1}^{k-1} (A_t \otimes A_t) \left(\prod_{i=1}^{j-1} (B_{t-i} \otimes B_{t-i}) (A_{t-i} \otimes A_{t-i}) \right) \operatorname{vec} \left(\left(\prod_{i=j}^{k-1} B_{t-i} A_{t-i} \right) \Lambda_{t-k} \begin{pmatrix} \lambda_{t-j}^D - \omega_D & 0 \end{pmatrix} \right).
\end{aligned}$$

Since $(B_{t-1} \otimes B_{t-1})(A_{t-1} \otimes A_{t-1})$ and $B_t A_t$ are i.i.d, and $EB_t A_t = EB_t EA_t$, we obtain

$$\begin{aligned}
&E\operatorname{vec}\begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D (t-k/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D (t-k/T)} \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D & \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix} \qquad (22) \\
&= \sum_{j=1}^{k-1} E(A_t \otimes A_t) E(B_{t-i} \otimes B_{t-i}) (A_{t-i} \otimes A_{t-i})^{j-1} E\operatorname{vec} \left(\left(\prod_{i=j}^{k-1} B_{t-i} A_{t-i} \right) \Lambda_{t-k} \begin{pmatrix} \lambda_{t-j}^D - \omega_D & 0 \end{pmatrix} \right).
\end{aligned}$$

By (1), we can express λ_{t-1}^D as a function of $\left\{ \left(m_{t-i}^D, m_{t-i+1}^N \right), i > 1 \right\}$. Note that B_t, A_t,and Λ_t are independent of $\left\{ \left(m_s^D, m_s^N \right), s \neq t \right\}$. Therefore, we have

$$\begin{aligned}
&E\left(\left(\prod_{i=j}^{k-1} B_{t-i} A_{t-i} \right) \Lambda_{t-k} \left(\lambda_{t-j}^D - \omega_D \right) \right) \\
&= \gamma_D E \left(\prod_{i=j}^{k-1} B_{t-i} A_{t-i} \right) \Lambda_{t-k} \sum\nolimits_{i=j+1}^{k} \beta_D^{i-1} \left(m_{t-i}^D + \left(m_{t-i}^D + 1 \right) \operatorname{sign}(e_{t-i}^D) \right) \\
&+ \rho_D E \left(\prod_{i=j}^{k-1} B_{t-i} A_{t-i} \right) \Lambda_{t-k} \sum\nolimits_{i=j+1}^{k} \beta_D^{i-1} \left(m_{t-i+1}^N + (m_{t-i+1}^N + 1) \operatorname{sign}(e_{t-i+1}^N) \right),
\end{aligned}$$

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with the first term

$$\begin{aligned}
&\left\| E\left(\prod_{i=j}^{k-1} B_{t-i}A_{t-i}\right)\Lambda_{t-k}\sum\nolimits_{i=j+1}^{k}\beta_D^{i-1}\left(m_{t-i}^D+\left(m_{t-i}^D+1\right)\operatorname{sign}(e_{t-i}^D)\right)\right\|_\infty \\
&\leq \left(\sum\nolimits_{i=j+1}^{k-1}\beta_D^{i-1}\right)\left\|E\left(B_t\left(m_t^D+(m_t^D+1)\operatorname{sign}(e_t^D)\right)A_t\right)\right\|_\infty\left\|EB_tEA_t\right\|_\infty^{k-j-1}\left\|E\Lambda_t\right\|_\infty \\
&+\beta_D^{k-j}\left\|E\Lambda_{t-k}\left(m_{t-k}^D+(m_{t-k}^D+1)\operatorname{sign}(e_{t-k}^D)\right)\right\|_\infty\left\|EB_tEA_t\right\|_\infty^{k-j} \\
&\leq \frac{\beta_D}{1-\beta_D}\left\|E\left(B_t\left(m_t^D+(m_t^D+1)\operatorname{sign}(e_t^D)\right)A_t\right)\right\|_\infty\left\|E\Lambda_t\right\|_\infty\left\|EB_tEA_t\right\|_\infty^{k-j-1} \\
&+\beta_D^{k-j}\left\|E\Lambda_{t-k}\left(m_{t-k}^D+(m_{t-k}^D+1)\operatorname{sign}(e_{t-k}^D)\right)\right\|_\infty\left\|EB_tEA_t\right\|_\infty^{k-j}
\end{aligned}$$

and the second term

$$\begin{aligned}
&\left\| E\left(\prod_{i=j}^{k-1} B_{t-i}A_{t-i}\right)\Lambda_{t-k}\sum\nolimits_{i=j+1}^{k}\beta_D^{i-1}\left(m_{t-i+1}^N+(m_{t-i+1}^N+1)\operatorname{sign}(e_{t-i+1}^N)\right)\right\|_\infty \\
&\leq \frac{\beta_D}{1-\beta_D}\left\|E\left(B_t\left(m_t^N+(m_t^N+1)\operatorname{sign}(e_t^N)\right)A_t\right)\right\|_\infty\left\|E\Lambda_t\right\|_\infty\left\|EB_tEA_t\right\|_\infty^{k-j-1}.
\end{aligned}$$

According to the definition of $\|\|_\infty$,

$$\left\|E\operatorname{vec}\left(\left(\prod_{i=j}^{k-1} B_{t-i}A_{t-i}\right)\Lambda_{t-k}\begin{pmatrix}\lambda_{t-j}^D-\omega_D & 0\end{pmatrix}\right)\right\|_\infty \leq \left\|E\left(\left(\prod_{i=j}^{k-1} B_{t-i}A_{t-i}\right)\Lambda_{t-k}\begin{pmatrix}\lambda_{t-j}^D-\omega_D & 0\end{pmatrix}\right)\right\|_\infty$$

Therefore,

$$\left\|E\operatorname{vec}\left(\left(\prod_{i=j}^{k-1} B_{t-i}A_{t-i}\right)\Lambda_{t-k}\begin{pmatrix}\lambda_{t-j}^D-\omega_D & 0\end{pmatrix}\right)\right\|_\infty \leq c_T\left\|EB_tEA_t\right\|^{k-j-1} \tag{23}$$

with

$$\begin{aligned}
c_T=&\frac{\beta_D}{1-\beta_D}|\gamma_D|\left\|E\left(B_t\left(m_t^D+(m_t^D+1)\operatorname{sign}(e_t^D)\right)A_t\right)\right\|_\infty\left\|E\Lambda_t\right\|_\infty \\
&+\left\|EB_tEA_t\right\|_\infty\left\|E\Lambda_{t-k}\left(m_{t-k}^D+(m_{t-k}^D+1)\operatorname{sign}(e_{t-k}^D)\right)\right\|_\infty \\
&+\frac{\beta_D}{1-\beta_D}|\rho_D|\left\|E\left(B_t\left(m_t^N+(m_t^N+1)\operatorname{sign}(e_t^N)\right)A_t\right)\right\|_\infty\left\|E\Lambda_t\right\|_\infty.
\end{aligned}$$

Substituting (23) into (22) gives

$$\begin{aligned}
&\left\|E\operatorname{vec}\begin{pmatrix}\frac{\partial h_t^D}{\partial\sigma^D(t-k/T)}\\ \frac{\partial h_t^N}{\partial\sigma^D(t-k/T)}\end{pmatrix}\begin{pmatrix}\frac{\partial}{\partial\beta_D}h_t^D & \frac{\partial}{\partial\beta_D}h_t^N\end{pmatrix}\right\|_\infty \\
&\leq\sum_{j=1}^{k-1}\left\|E\left(A_t\otimes A_t\right)\right\|_\infty\left\|E\left(B_{t-i}\otimes B_{t-i}\right)\left(A_{t-i}\otimes A_{t-i}\right)\right\|_\infty^{j-1}c_T\left\|EB_tEA_t\right\|_\infty^{k-j-1} \\
&\leq c_T\left\|E\left(A_t\otimes A_t\right)\right\|_\infty\sum_{j=1}^{k-1}\left\|E\left(B_{t-i}\otimes B_{t-i}\right)\left(A_{t-i}\otimes A_{t-i}\right)\right\|_\infty^{j-1}\left\|EB_tEA_t\right\|_\infty^{k-j-1} \\
&\leq c_T\left\|E\left(A_t\otimes A_t\right)\right\|_\infty\frac{\left\|EB_tEA_t\right\|_\infty^{k-2}}{1-\frac{\left\|E(B_{t-i}\otimes B_{t-i})(A_{t-i}\otimes A_{t-i})\right\|_\infty}{\left\|EB_tEA_t\right\|_\infty}},
\end{aligned}$$

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provided that $\|EB_tEA_t\|_\infty < 1$ and $\|E\left(B_{t-1}A_{t-1} \otimes B_{t-1}A_{t-1}\right)\|_\infty < \|EB_tEA_t\|_\infty$ .It is then straightforward to show

$$\sum_k k \left\| E\text{vec} \begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D(t-k/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D(t-k/T)} \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D & \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix} \right\|_\infty < \infty$$

and thereby

$$\sum_k k \left\| E \begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D(t-k/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D(t-k/T)} \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D & \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix} \right\|_\infty < \infty.$$

■

Lemma 4 *The score functions of h_t^j with respect to β_D, v_D and $\sigma^j(t/T)$ are*

$$\begin{aligned} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D \\ \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix} &= A_t \begin{pmatrix} \lambda_{t-1}^D - \omega_D \\ 0 \end{pmatrix} + A_t B_{t-1} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_{t-1}^D \\ \frac{\partial}{\partial \beta_D} h_{t-1}^N \end{pmatrix} \\ &= \sum_{j=1}^{\infty} A_t \prod_{i=1}^{j-1} B_{t-i}A_{t-i} \begin{pmatrix} \lambda_{t-j}^D - \omega_D \\ 0 \end{pmatrix}. \end{aligned} \tag{24}$$

$$\begin{aligned} \begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D(t-k/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D(t-k/T)} \end{pmatrix} &= A_t B_{t-1} \begin{pmatrix} \frac{\partial h_{t-1}^D}{\partial \sigma^D(t-k/T)} \\ \frac{\partial h_{t-1}^N}{\partial \sigma^D(t-k/T)} \end{pmatrix} \\ &= A_t \left(\prod_{i=1}^{k-1} B_{t-i}A_{t-i} \right) \Lambda_{t-k}, k > 1 \\ \begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D(t/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D(t/T)} \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}; \text{ and } \begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D(t-1/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D(t-1/T)} \end{pmatrix} = A_t \begin{pmatrix} a_{t-1}^{DD} \\ a_{t-1}^{ND} \end{pmatrix}, \end{aligned} \tag{25}$$

with $\Lambda_t = \begin{pmatrix} a_t^{DD} \\ a_t^{ND} \end{pmatrix}$ .If the top-Lyapunov exponent of the sequence of A_tB_{t-1} is strictly negative, $\begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D \\ \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix}$, $\begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D(t-k/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D(t-k/T)} \end{pmatrix}$ and $\begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D(t-k/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D(t-k/T)} \end{pmatrix} \begin{pmatrix} \frac{\partial h_t^D}{\partial \beta_D} & \frac{\partial h_t^N}{\partial \beta_D} \end{pmatrix}$ are strictly stationary.

Proof. Since $h_t^j = \lambda_t^j + \sigma^j(t/T)$, we can write h_t^j in a recursive formula as

$$\begin{aligned} h_t^D &= \sigma^D(t/T) - \beta_D \sigma^D(\frac{t-1}{T}) + \omega_D(1-\beta_D) + \beta_D h_{t-1}^D + \gamma_D m_{t-1}^D \\ &\quad + \rho_D m_t^N + \gamma_D^*(m_{t-1}^D + 1)\text{sign}(u_{t-1}^D) + \rho_D^*(m_t^N + 1)\text{sign}(u_t^N) \end{aligned} \tag{26}$$

$$\begin{aligned} h_t^N &= \sigma^N(t/T) - \beta_N \sigma^N(\frac{t-1}{T}) + \omega_N(1-\beta_N) + \beta_N h_{t-1}^N + \gamma_N m_{t-1}^N \\ &\quad + \rho_N m_{t-1}^D + \rho_N^*(m_{t-1}^D + 1)\text{sign}(u_{t-1}^D) + \gamma_N^*(m_{t-1}^N + 1)\text{sign}(u_{t-1}^N). \end{aligned} \tag{27}$$

and m_t^D and m_t^N can be expressed as

$$\begin{aligned} m_t^D &= \frac{(1+v_D)(u_t^D)^2 \exp(-2h_t^D)}{v_D + (u_t^D)^2 \exp(-2h_t^D)} - 1, \quad v_D > 0 \\ m_t^N &= \frac{(1+v_N)(u_t^N)^2 \exp(-2h_t^N)}{v_N + (u_t^N)^2 \exp(-2h_t^N)} - 1, \quad v_N > 0. \end{aligned}$$

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Taking the first order derivative of equation (26) and (27) with respect to β_D gives

$$\begin{aligned}\frac{\partial h_t^D}{\partial \beta_D} &= -\sigma^D(\frac{t-1}{T}) - \omega_D + h_{t-1}^D + \beta_D \frac{\partial}{\partial \beta_D} h_{t-1}^D + \frac{\partial}{\partial \beta_D}\gamma_D m_{t-1}^D + \frac{\partial}{\partial \beta_D}\rho_D m_t^N \\ &\quad + \frac{\partial}{\partial \beta_D}\gamma_D^*(m_{t-1}^D + 1)\text{sign}(u_{t-1}^D) + \frac{\partial}{\partial \beta_D}\rho_D^*(m_t^N + 1)\text{sign}(u_t^N) \qquad (28) \\ \frac{\partial h_t^N}{\partial \beta_D} &= \beta_N \frac{\partial}{\partial \beta_D} h_{t-1}^N + \frac{\partial}{\partial \beta_D}\gamma_N m_{t-1}^N + \frac{\partial}{\partial \beta_D}\rho_N m_{t-1}^D \\ &\quad + \frac{\partial}{\partial \beta_D}\rho_N^*(m_{t-1}^D + 1)\text{sign}(u_{t-1}^D) + \frac{\partial}{\partial \beta_D}\gamma_N^*(m_{t-1}^N + 1)\text{sign}(u_{t-1}^N) \qquad (29)\end{aligned}$$

and the derivatives of m_{t-1}^D and m_{t-1}^N are

$$\begin{aligned}\frac{\partial}{\partial \beta_D} m_{t-1}^D &= \frac{\partial m_{t-1}^D}{\partial h_{t-1}^D}\frac{\partial}{\partial \beta_D} h_{t-1}^D = -2\left(v_D + 1\right) b_{t-1}^D \left(1 - b_{t-1}^D\right) \frac{\partial}{\partial \beta_D} h_{t-1}^D \\ \frac{\partial}{\partial \beta_D} m_{t-1}^N &= \frac{\partial m_{t-1}^N}{\partial h_{t-1}^N}\frac{\partial}{\partial \beta_D} h_{t-1}^N = -2\left(v_N + 1\right) b_{t-1}^N \left(1 - b_{t-1}^N\right) \frac{\partial}{\partial \beta_D} h_{t-1}^N.\end{aligned}$$

Substituting them back into (28) and (29) gives

$$\begin{aligned}\frac{\partial h_t^D}{\partial \beta_D} &= \lambda_{t-1}^D - \omega_D + \left(\beta_D + a_{t-1}^{DD}\right)\frac{\partial}{\partial \phi} h_{t-1}^D + a_t^{DN}\frac{\partial}{\partial \phi} h_t^N \\ \frac{\partial h_t^N}{\partial \beta_D} &= 0 + \left(\beta_N + a_{t-1}^{NN}\right)\frac{\partial}{\partial \phi} h_{t-1}^N + a_{t-1}^{ND}\frac{\partial}{\partial \phi} h_{t-1}^D\end{aligned}$$

with the matrix form

$$\begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D \\ \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix} = A_t \begin{pmatrix} \lambda_{t-1}^D - \omega_D \\ 0 \end{pmatrix} + A_t B_{t-1} \begin{pmatrix} \frac{\partial}{\partial \beta_D} h_{t-1}^D \\ \frac{\partial}{\partial \beta_D} h_{t-1}^N \end{pmatrix}.$$

Note that $A_t B_{t-1}$ and $A_t \begin{pmatrix} \lambda_{t-1}^D - \omega_D \\ 0 \end{pmatrix}$ are strictly stationary and ergodic, by Theorem 4.27 in Douc, Moulines, and Stoffer (2014), when the top-Lyapunov exponent of the sequence of $A_t B_{t-1}$ is strictly negative, $\begin{pmatrix} \frac{\partial}{\partial \beta_D} h_t^D \\ \frac{\partial}{\partial \beta_D} h_t^N \end{pmatrix}$ converges and is strictly stationary.

Likewise, taking the first order derivative of h_t^j with respect to $\sigma^D\left(\frac{t-k}{T}\right)$ yields

$$\begin{aligned}\frac{\partial h_t^D}{\partial \sigma^D((t-k)/T)} &= \left(\beta_D + a_{t-1}^{DD}\right)\frac{\partial h_{t-1}^D}{\partial \sigma^D((t-k)/T)} + a_t^{DN}\frac{\partial h_t^N}{\partial \sigma^D((t-k)/T)},\ k > 1 \\ \frac{\partial h_t^D}{\partial \sigma^D(t/T)} &= 1,\ \frac{\partial h_t^D}{\partial \sigma^D((t-1)/T)} = a_{t-1}^{DD} + a_t^{DN} a_{t-1}^{ND}\end{aligned}$$

$$\begin{aligned}\frac{\partial h_t^N}{\partial \sigma^D((t-k)/T)} &= \left(\beta_N + a_{t-1}^{NN}\right)\frac{\partial h_{t-1}^N}{\partial \sigma^D((t-k)/T)} + a_{t-1}^{ND}\frac{\partial h_{t-1}^D}{\partial \sigma^D((t-k)/T)},\ k > 1 \\ \frac{\partial h_t^N}{\partial \sigma^D(t/T)} &= 0,\ \frac{\partial h_t^N}{\partial \sigma^D((t-1)/T)} = a_{t-1}^{ND},\end{aligned}$$

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and (25) follows. Similarly, $\begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D((t-k)/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D((t-k)/T)} \end{pmatrix}$ is strictly stationary across time t.

Finally, we can write

$$\begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D((t-k)/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D((t-k)T)} \\ \frac{\partial h_t^D}{\partial \beta_D} \\ \frac{\partial h_t^N}{\partial \beta_D} \end{pmatrix} = \begin{pmatrix} A_t B_{t-1} & 0 \\ 0 & A_t B_{t-1} \end{pmatrix} \begin{pmatrix} \frac{\partial h_{t-1}^D}{\partial \sigma^D((t-k)/T)} \\ \frac{\partial h_{t-1}^N}{\partial \sigma^D((t-k)/T)} \\ \frac{\partial h_{t-1}^D}{\partial \beta_D} \\ \frac{\partial h_{t-1}^N}{\partial \beta_D} \end{pmatrix} + \begin{pmatrix} A_t \begin{pmatrix} \lambda_{t-1}^D - \omega_D \\ 0 \end{pmatrix} \\ 0 \end{pmatrix}.$$

Both $\begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D((t-k)/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D((t-k)/T)} \\ \frac{\partial h_t^D}{\partial \beta_D} \\ \frac{\partial h_t^N}{\partial \beta_D} \end{pmatrix}$ and $\begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^D((t-k)/T)} \\ \frac{\partial h_t^N}{\partial \sigma^D((t-k)/T)} \end{pmatrix} \begin{pmatrix} \frac{\partial h_t^D}{\partial \beta_D} & \frac{\partial h_t^N}{\partial \beta_D} \end{pmatrix}$ are strictly stationary, since the top-Lyapunov exponent of the sequence $\begin{pmatrix} A_t B_{t-1} & 0 \\ 0 & A_t B_{t-1} \end{pmatrix}$, same as that of $A_t B_{t-1}$, is strictly negative by assumption. ■

Lemma 5 *When Assumption A1-A4 holds, we have* $\frac{1}{T}\sum_{t=1}^{T} E\frac{\partial h_t^D}{\partial \theta} = 0$.

Proof. Similar to the proof of Theorem 1, we only need to show $\sum_{t=1}^{T} k \left\| E \begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^N((t-k)/T)} \\ \frac{\partial h_t^N}{\partial \sigma^N((t-k)T)} \end{pmatrix} \right\|_\infty < \infty$. Note that $E \begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^N((t-k)/T)} \\ \frac{\partial h_t^N}{\partial \sigma^N((t-k)/T)} \end{pmatrix} = EA_t B_{t-1} A_{t-1} B_{t-2} .. A_{t-k+2} B_{t-k+1} A_{t-k+1} \Lambda_{t-k} = A (BA)^{k-1} \Lambda$, when $k > 1$. Obviously, $\sum_{t=1}^{T} k \left\| E \begin{pmatrix} \frac{\partial h_t^D}{\partial \sigma^N((t-k)/T)} \\ \frac{\partial h_t^N}{\partial \sigma^N((t-k)/T)} \end{pmatrix} \right\|_\infty < \infty$. ■

9.2 Appendix B: Derivatives in the multivariate model

We now give the first-order and second-order derivatives of the global log-likelihood function in the multivariate model, given λ_t and v. Without subscripts j and ignoring some unnecessary parts, the log-likelihood function is

$$l_t = \log|\Theta| - \sum_{i=1}^{n} \left(\frac{v_i + 1}{2} \ln \left(1 + \frac{(\iota_i^\intercal \mathrm{diag}\left(\exp(-\lambda_t)\right) \Theta u_t)^2}{v_i} \right) \right).$$

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Then

$$\begin{aligned}
\mathrm{d}l_t &= \mathrm{d}\log|\Theta| - \sum_{i=1}^{n}\frac{(v_i+1)\left(\iota_i^{\intercal}\mathrm{diag}\left(\exp(-\lambda_t)\right)\Theta u_t\right)\exp(-\lambda_{it})}{v_i+\left(\iota_i^{\intercal}\mathrm{diag}\left(\exp(-\lambda_t)\right)\Theta u_t\right)^2}\mathrm{tr}u_t\iota_i^{\intercal}\mathrm{d}\Theta \\
&= \mathrm{tr}\left(\Theta^{-1}\mathrm{d}\Theta\right) - \mathrm{tr}\left(\sum_{i=1}^{n}\frac{(v_i+1)\exp(-2\lambda_{it})\iota_i^{\intercal}\Theta u_t}{v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2}u_t\iota_i^{\intercal}\mathrm{d}\Theta\right) \\
&= \mathrm{tr}\left[\left(\Theta^{-1} - \sum_{i=1}^{n}\frac{(v_i+1)\exp(-2\lambda_{it})\iota_i^{\intercal}\Theta u_t}{v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2}u_t\iota_i^{\intercal}\right)\mathrm{d}\Theta\right] \\
&= \left[\mathrm{vec}\left(\Theta^{-1} - \sum_{i=1}^{n}\iota_i u_t^{\intercal}\frac{(v_i+1)\exp(-2\lambda_{it})\iota_i^{\intercal}\Theta u_t}{v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2}\right)\right]^{\intercal}\mathrm{dvec}\Theta \\
&= \left[\mathrm{vec}\left(\Theta^{-1} - \sum_{i=1}^{n}\iota_i u_t^{\intercal}\frac{(v_i+1)\exp(-2\lambda_{it})\iota_i^{\intercal}\Theta u_t}{v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2}\right)\right]^{\intercal}D_n\mathrm{dvech}\Theta, \qquad (30)
\end{aligned}$$

where D_n is the duplication matrix so that $\mathrm{vec}\Theta = D_n\mathrm{vech}\Theta$. Therefore, the first order derivative of the global log-likelihood function is

$$\begin{aligned}
\frac{\partial L_T(\Theta;\lambda_t,s)}{\partial\mathrm{vech}\Theta} = &-\frac{1}{T}D_n^{\intercal}\mathrm{vec}\sum_{i=1}^{n}\left(\iota_i\sum_{t=1}^{T}\left(K_h(s-t/T)u_t^{\intercal}\frac{(v_i+1)\exp(-2\lambda_{it})\iota_i^{\intercal}\Theta u_t}{v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2}\right)\right) \\
&+ D_n^{\intercal}\mathrm{vec}\left(\Theta^{-1}\right). \qquad (31)
\end{aligned}$$

To compute the Hessian matrix, we evaluate the differential of the Jacobian matrix in (30)

$$\begin{aligned}
&\mathrm{dvec}D_n^{\intercal}\left(\Theta^{-1} - \sum_{i=1}^{n}\iota_i u_t^{\intercal}\frac{(v_i+1)\exp(-2\lambda_{it})\iota_i^{\intercal}\Theta u_t}{v_i+\left(\exp(-2\lambda_{it})\iota_i^{\intercal}\Theta u_t\right)^2}\right)D_n \\
&= D_n^{\intercal}\mathrm{dvec}\Theta^{-1} - D_n^{\intercal}\mathrm{vec}\sum_{i=1}^{n}\left(\mathrm{d}\frac{(v_i+1)\exp(-2\lambda_{it})\iota_i^{\intercal}\Theta u_t}{v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2}\right)\iota_i u_t^{\intercal} \\
&= D_n^{\intercal}\mathrm{dvec}\Theta^{-1} - D_n^{\intercal}\sum_{i=1}^{n}\frac{\left(v_i-\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2\right)(v_i+1)\exp(-2\lambda_{it})}{\left(v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2\right)^2}\mathrm{vec}\left(\iota_i\iota_i^{\intercal}\mathrm{d}\Theta u_t u_t^{\intercal}\right) \\
&= -D_n^{\intercal}\left(\Theta^{-1}\otimes\Theta^{-1}\right)D_n\mathrm{dvech}\Theta \\
&- D_n^{\intercal}\sum_{i=1}^{n}\frac{\left(v_i-\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2\right)(v_i+1)\exp(-2\lambda_{it})}{\left(v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2\right)^2}\left(u_t u_t^{\intercal}\right)\otimes\left(\iota_i\iota_i^{\intercal}\right)D_n\mathrm{dvech}\Theta.
\end{aligned}$$

The Hessian matrix of the global log-likelihood function is thus

$$\begin{aligned}
&\frac{\partial^2 L_T(\Theta;\lambda_t,s)}{\partial\mathrm{vech}\Theta\partial\left(\mathrm{vech}\Theta\right)^{\intercal}} \\
&= -D_n^{\intercal}\left(\sum_{i=1}^{n}\left(\sum_{t=1}^{T}\frac{K_h(s-t/T)\left(v_i-\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2\right)(v_i+1)}{T\left(v_i+\exp(-2\lambda_{it})\left(\iota_i^{\intercal}\Theta u_t\right)^2\right)^2\exp(2\lambda_{it})}u_t u_t^{\intercal}\right)\otimes\left(\iota_i\iota_i^{\intercal}\right)\right)D_n \\
&- D_n^{\intercal}\left(\Theta^{-1}\otimes\Theta^{-1}\right)D_n \qquad (32)
\end{aligned}$$

■

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9.3 Appendix C: tables and figures

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Panel a: size portfolios of NYSE/AMEX/NASDAQ stocks

	mean		s.e.		median			
decile	day(%)	night(%)	day(%)	night(%)	day(%)	night(%)	T stat.	avg. no.
1-small	0.0942	0.0606	0.0107	0.0079	0.1184	0.0642	2.5290	493.3080
2	0.0964	0.0323	0.0093	0.0060	0.1380	0.0389	5.8126	571.2961
3	0.0793	0.0308	0.0096	0.0055	0.1436	0.0371	4.3742	604.6714
4	0.0601	0.0392	0.0111	0.0061	0.1318	0.0456	1.6503	630.0765
5	0.0484	0.0414	0.0130	0.0067	0.1233	0.0546	0.4786	643.4898
6	0.0426	0.0388	0.0137	0.0070	0.1161	0.0552	0.2447	655.4365
7	0.0401	0.0376	0.0141	0.0072	0.1051	0.0492	0.1546	664.1175
8	0.0416	0.0253	0.0143	0.0073	0.1101	0.0409	1.0122	673.6501
9	0.0324	0.0324	0.0136	0.0073	0.1049	0.0487	0.0003	683.3976
10-large	0.0155	0.0270	0.0126	0.0081	0.0543	0.0441	-0.7725	699.5996

Panel b: size portfolios of NASDAQ stocks

	mean		s.e.		median			
size	day(%)	night(%)	day(%)	night(%)	day(%)	night(%)	T stat.	avg. no.
1-small	0.1056	0.0623	0.0135	0.0104	0.1110	0.0599	2.5372	262.8418
2	0.0902	0.0497	0.0106	0.0073	0.1254	0.0542	3.1557	305.5648
3	0.1060	0.0238	0.0102	0.0065	0.1531	0.0294	6.7834	321.6926
4	0.0670	0.0461	0.0107	0.0064	0.1325	0.0548	1.6795	335.6151
5	0.0645	0.0436	0.0130	0.0070	0.1401	0.0546	1.4182	346.0700
6	0.0565	0.0552	0.0157	0.0080	0.1373	0.0670	0.0753	351.3748
7	0.0471	0.0478	0.0159	0.0081	0.1294	0.0690	-0.0388	358.4989
8	0.0382	0.0575	0.0162	0.0082	0.1069	0.0765	-1.0609	361.1587
9	0.0341	0.0439	0.0169	0.0086	0.1124	0.0569	-0.5148	368.2421
10-large	0.0028	0.0672	0.0184	0.0108	0.0686	0.0862	-3.0218	378.3676

Panel c: size portfolios of NYSE stocks

	mean		s.e.		median			
decile	day(%)	night(%)	day(%)	night(%)	day(%)	night(%)	T stat.	avg. no.
1-small	0.0793	0.0063	0.0108	0.0056	0.1021	0.0067	6.0203	219.2011
2	0.0474	0.0014	0.0100	0.0053	0.0866	0.0044	4.0480	231.0752
3	0.0443	0.0088	0.0110	0.0057	0.0919	0.0134	2.8591	232.3123
4	0.0383	0.0145	0.0122	0.0062	0.0883	0.0136	1.7341	232.0882
5	0.0428	0.0117	0.0134	0.0068	0.0947	0.0128	2.0683	232.8432
6	0.0413	0.0134	0.0135	0.0069	0.0927	0.0152	1.8429	233.4402
7	0.0355	0.0142	0.0131	0.0068	0.0871	0.0237	1.4465	235.8511
8	0.0322	0.0190	0.0126	0.0068	0.0855	0.0325	0.9232	236.4532
9	0.0255	0.0242	0.0127	0.0073	0.0698	0.0352	0.0895	238.8781
10-large	0.0179	0.0164	0.0120	0.0079	0.0455	0.0342	0.1040	240.4777

This table gives the summary statistics for the intraday and overnight returns of 10 portfolios formed on size: mean, median, standard error of the mean, T statistics, and the average number of stocks. The T statistics are used to test the null hypothesis that the mean value of overnight returns equals the mean value of intraday returns.

Table A.1: Descriptive statistics for intraday and overnight returns: portfolios formed on size

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In particular:

$$\begin{aligned}
\text{var}(r_t^N|\mathcal{F}_{t-1}) &= \frac{v_N}{v_N-2}\exp(2\lambda_t^N + 2\sigma^N(t/T)) \\
\text{var}(r_t^D|\mathcal{F}_{t-1}) &= \delta^2\frac{v_N}{v_N-2}\exp(2\lambda_t^N + 2\sigma^N(t/T)) + \frac{v_D}{v_D-2}\exp(2\sigma^D(t/T))E\left[\exp(2\lambda_t^D)|\mathcal{F}_{t-1}\right] \\
\text{var}\left(r_t^D + r_t^N|\mathcal{F}_{t-1}\right) &= \frac{v_N(1-\delta)^2}{v_N-2}\exp(2\lambda_t^N + 2\sigma^N(t/T)) + \frac{v_D}{v_D-2}\exp(2\sigma^D(t/T))E\left[\exp(2\lambda_t^D)|\mathcal{F}_{t-1}\right].
\end{aligned}$$

In the online appendix we show that

$$\begin{aligned}
E\left[\exp(2\lambda_t^D)|\mathcal{F}_{t-1}\right] = &\frac{1}{2}\exp(-2\rho_D)\Lambda_t \ {}_{\times 1}F_1(1/2, 1/2+v_N/2, (2\rho_D+2\rho_D^*)(v_N+1)) \\
&+ \frac{1}{2}\exp(-2\rho_D)\Lambda_t \ {}_{\times 1}F_1(1/2, 1/2+v_N/2, (2\rho_D-2\rho_D^*)(v_N+1)),
\end{aligned} \tag{33}$$

where $\Lambda_t = \exp(2\omega_D(1-\beta_D)+2\beta_D\lambda_{t-1}^D+2\gamma_D m_{t-1}^D+2\gamma_D^*(m_{t-1}^D+1)\text{sign}(e_{t-1}^D)$ and ${}_1F_1$ is the Kummer's function

$${}_1F_1(\alpha,\beta,c) = 1 + \sum_{k=0}^{\infty}\left(\prod_{r=1}^{k-1}\frac{\alpha+r}{\beta+r}\right)\frac{c^k}{k!}, \quad \alpha,\beta>0.$$

From this we obtain an explicit formula for $\text{var}(r_t^D|\mathcal{F}_{t-1})$ and $\text{var}\left(r_t^D + r_t^N|\mathcal{F}_{t-1}\right)$, which can be used for forecasting. In the online appendix we also give explicit expressions for the unconditional moments.

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Panel A: size portfolios of NYSE/AMEX/NASDAQ stocks							
decile	δ	μ_D	μ_N	Π_{11}	Π_{12}	Π_{21}	Π_{22}
1-small	-0.0306	0.0722	0.0457	0.2304	0.0028	0.0592	0.1504
	(0.0282)	(0.0113)	(0.0081)	(0.0247)	(0.0267)	(0.0170)	(0.0224)
2	-0.1375	0.0722	0.0183	0.2526	-0.0045	0.1002	0.1359
	(0.0391)	(0.0097)	(0.0063)	(0.0239)	(0.0340)	(0.0151)	(0.0251)
3	-0.2024	0.0643	0.0230	0.2038	-0.0365	0.0838	0.0368
	(0.0514)	(0.0101)	(0.0060)	(0.0247)	(0.0406)	(0.0151)	(0.0268)
4	-0.2375	0.0557	0.0362	0.0827	-0.0115	0.0604	-0.0171
	(0.0495)	(0.0117)	(0.0065)	(0.0260)	(0.0477)	(0.0157)	(0.0281)
5	-0.2247	0.0478	0.0424	0.0205	-0.0070	0.0411	-0.0719
	(0.0573)	(0.0136)	(0.0070)	(0.0299)	(0.0550)	(0.0158)	(0.0284)
6	-0.1841	0.0428	0.0403	0.0079	-0.0111	0.0305	-0.0726
	(0.0570)	(0.0142)	(0.0072)	(0.0280)	(0.0562)	(0.0153)	(0.0290)
7	-0.1877	0.0399	0.0397	0.0158	-0.0101	0.0351	-0.0920
	(0.0537)	(0.0145)	(0.0074)	(0.0261)	(0.0524)	(0.0143)	(0.0281)
8	-0.1887	0.0411	0.0266	0.0111	0.0021	0.0327	-0.1051
	(0.0573)	(0.0147)	(0.0074)	(0.0274)	(0.0551)	(0.0148)	(0.0290)
9	-0.1673	0.0312	0.0346	0.0375	0.0018	0.0467	-0.1117
	(0.0547)	(0.0140)	(0.0074)	(0.0257)	(0.0520)	(0.0143)	(0.0280)
10-large	-0.0482	0.0164	0.0289	-0.0685	0.0094	0.0498	-0.1012
	(0.0459)	(0.0128)	(0.0081)	(0.0247)	(0.0440)	(0.0170)	(0.0286)
Panel b: size portfolios of NYSE stocks							
decile	δ	μ_D	μ_N	Π_{11}	Π_{12}	Π_{21}	Π_{22}
1-small	-0.3385	0.0699	0.0027	0.1242	-0.0412	0.0483	-0.0361
	(0.0734)	(0.0114)	(0.0062)	(0.0393)	(0.0804)	(0.0225)	(0.0391)
2	-0.2984	0.0454	-0.0016	0.0441	-0.0351	0.0644	-0.0779
	(0.0895)	(0.0105)	(0.0056)	(0.0366)	(0.0832)	(0.0257)	(0.0531)
3	-0.2602	0.0434	0.0069	0.0253	-0.0260	0.0630	-0.0941
	(0.0875)	(0.0114)	(0.0059)	(0.0360)	(0.0856)	(0.0253)	(0.0526)
4	-0.2736	0.0380	0.0144	0.0098	0.0004	0.0383	-0.0920
	(0.0691)	(0.0126)	(0.0064)	(0.0312)	(0.0696)	(0.0193)	(0.0399)
5	-0.2336	0.0432	0.0113	-0.0044	-0.0122	0.0377	-0.1049
	(0.0700)	(0.0138)	(0.0069)	(0.0330)	(0.0684)	(0.0187)	(0.0380)
6	-0.2123	0.0413	0.0138	-0.0033	0.0157	0.0282	-0.1199
	(0.0699)	(0.0138)	(0.0070)	(0.0326)	(0.0689)	(0.0182)	(0.0359)
7	-0.2305	0.0347	0.0147	0.0225	0.0063	0.0335	-0.1198
	(0.0644)	(0.0134)	(0.0069)	(0.0300)	(0.0629)	(0.0164)	(0.0332)
8	-0.1973	0.0315	0.0200	0.0291	-0.0051	0.0373	-0.1165
	(0.0616)	(0.0129)	(0.0069)	(0.0286)	(0.0589)	(0.0166)	(0.0318)
9	-0.1558	0.0257	0.0258	0.0060	-0.0099	0.0412	-0.1082
	(0.0587)	(0.0130)	(0.0074)	(0.0275)	(0.0553)	(0.0171)	(0.0337)
10-large	-0.0505	0.0193	0.0171	-0.0755	0.0063	0.0475	-0.0945
	(0.0462)	(0.0122)	(0.0080)	(0.0253)	(0.0454)	(0.0188)	(0.0300)
Panel c: size portfolios of NASDAQ stocks							
decile	δ	μ_D	μ_N	Π_{11}	Π_{12}	Π_{21}	Π_{22}
1-small	0.1181	0.0824	0.0526	0.2004	0.0284	0.0086	0.1359
	(0.0240)	(0.0134)	(0.0106)	(0.0194)	(0.0264)	(0.0155)	(0.0190)
2	-0.0246	0.0698	0.0349	0.2347	-0.0129	0.0792	0.1539
	(0.0295)	(0.0107)	(0.0075)	(0.0213)	(0.0267)	(0.0142)	(0.0207)
3	-0.0578	0.0792	0.0126	0.2536	-0.0080	0.0749	0.1377
	(0.0384)	(0.0106)	(0.0068)	(0.0236)	(0.0330)	(0.0143)	(0.0253)
4	-0.1190	0.0563	0.0383	0.1818	-0.0323	0.0796	0.0554
	(0.0448)	(0.0111)	(0.0067)	(0.0227)	(0.0345)	(0.0139)	(0.0231)
5	-0.1717	0.0613	0.0420	0.0631	-0.0174	0.0364	-0.0190
	(0.0451)	(0.0137)	(0.0073)	(0.0257)	(0.0433)	(0.0136)	(0.0231)
6	-0.1544	0.0572	0.0566	0.0127	-0.0250	0.0238	-0.0492
	(0.0496)	(0.0164)	(0.0083)	(0.0278)	(0.0473)	(0.0132)	(0.0233)
7	-0.1409	0.0480	0.0501	-0.0018	-0.0150	0.0149	-0.0628
	(0.0516)	(0.0164)	(0.0083)	(0.0252)	(0.0475)	(0.0127)	(0.0238)
8	-0.1346	0.0377	0.0608	0.0200	-0.0025	0.0247	-0.0754
	(0.0469)	(0.0167)	(0.0085)	(0.0228)	(0.0436)	(0.0117)	(0.0229)
9	-0.1006	0.0335	0.0467	0.0383	-0.0129	0.0374	-0.0923
	(0.0486)	(0.0173)	(0.0087)	(0.0225)	(0.0445)	(0.0115)	(0.0224)
10-large	0.0152	0.0009	0.0737	-0.0545	0.0342	0.0541	-0.0988
	(0.0414)	(0.0187)	(0.0110)	(0.0234)	(0.0367)	(0.0122)	(0.0223)

Estimates of mean equation parameters for 10 portfolios formed on size , with standard errors in parentheses. The mean equation is specified as $\begin{pmatrix} 1 & \delta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r_t^D \\ r_t^N \end{pmatrix} = \begin{pmatrix} \mu_D \\ \mu_N \end{pmatrix} + \begin{pmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{21} & \Pi_{22} \end{pmatrix} \begin{pmatrix} r_{t-1}^D \\ r_{t-1}^N \end{pmatrix} + \begin{pmatrix} u_t^D \\ u_t^N \end{pmatrix}$.

Table A.2: Estimates of the mean equations

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	decile1	decile2	decile3	decile4	decile5	decile6	decile7	decile8	decile9	decile10
cap nyse/amex/nasdaq	0.0230	0.0676	0.1252	0.2298	0.3747	0.6418	1.1669	2.1138	4.8682	38.9714
cap nyse	0.0993	0.2274	0.3800	0.6326	1.0605	1.7561	2.5854	4.9514	10.3084	56.4557
cap nasdaq	0.0169	0.0424	0.0786	0.1262	0.2099	0.3456	0.6023	1.0475	2.1272	22.7218
beta	1.5301	1.2385	1.0712	0.9612	0.8637	0.7639	0.6643	0.5264	0.3531	0.2501
std. dev.	0.0612	0.0391	0.0315	0.0271	0.0236	0.0212	0.0185	0.0157	0.0129	0.0091

The first three rows show the average capitalization value (in million) of size portfolios of NYSE/AMEX/NASDAQ, NYSE, and NASDAQ stocks, respectively. Since the capitalization value varies across years, we present the average value in 2015 instead of the whole period. The fourth (last) row shows the average beta (standard deviation) value of portfolios formed on beta (standard deviation).

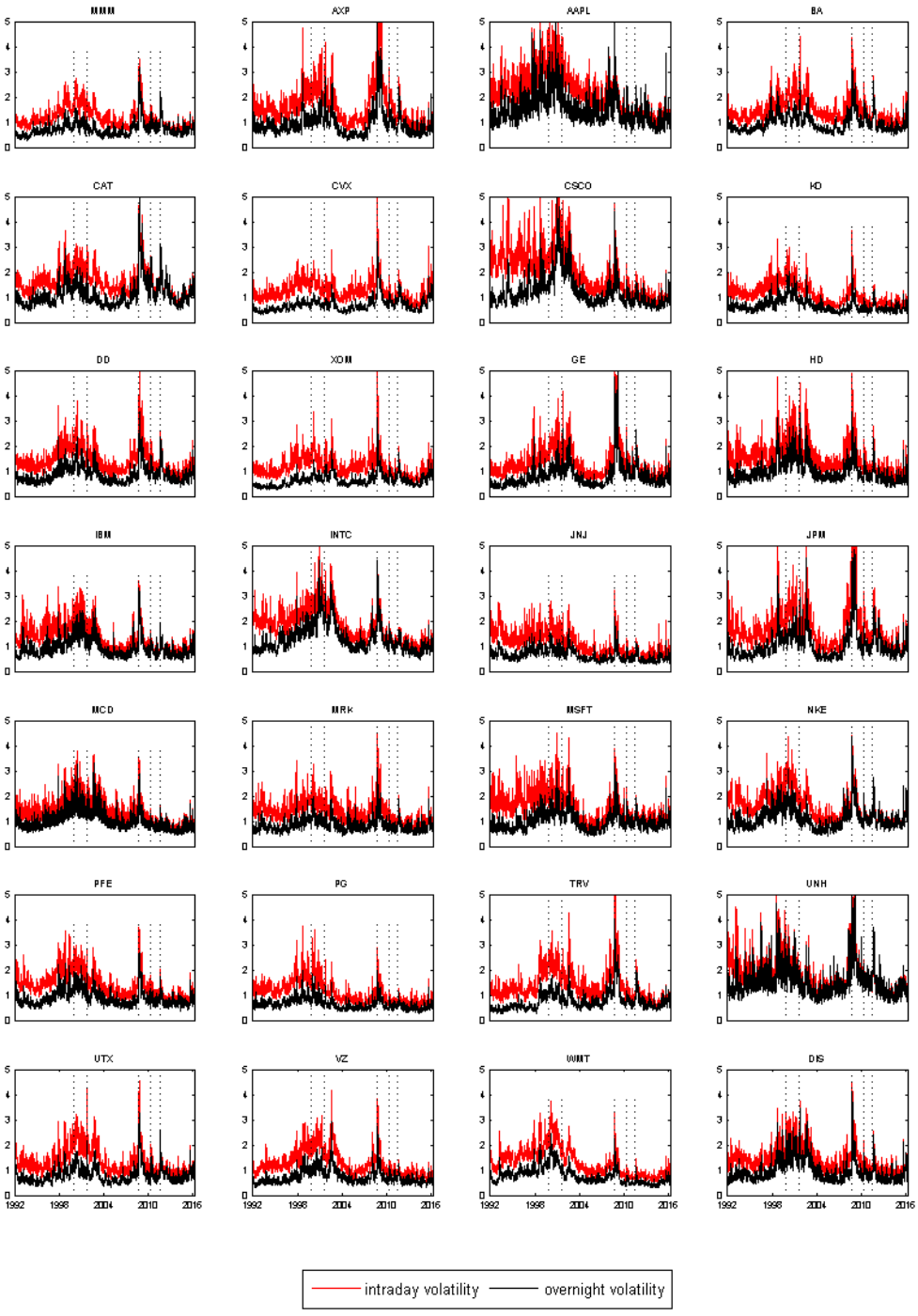
Table A.3: Average capitalization, beta and standard deviation values

	decile1	decile2	decile3	decile4	decile5	decile6	decile7	decile8	decile9	decile10
β_D	0.9508	0.9277	0.9276	0.9261	0.9215	0.9060	0.9302	0.9344	0.9418	0.9499
	(0.0077)	(0.0106)	(0.0101)	(0.0100)	(0.0103)	(0.0117)	(0.0087)	(0.0077)	(0.0068)	(0.0056)
γ_D	0.0485	0.0568	0.0619	0.0605	0.0562	0.0528	0.0426	0.0414	0.0372	0.0277
	(0.0049)	(0.0051)	(0.0054)	(0.0055)	(0.0051)	(0.0048)	(0.0042)	(0.0042)	(0.0041)	(0.0035)
ρ_D	0.0332	0.0503	0.0402	0.0413	0.0312	0.0348	0.0280	0.0281	0.0349	0.0322
	(0.0044)	(0.0054)	(0.0053)	(0.0056)	(0.0055)	(0.0055)	(0.0050)	(0.0049)	(0.0048)	(0.0043)
γ_D^*	-0.0093	-0.0097	-0.0162	-0.0208	-0.0223	-0.0268	-0.0260	-0.0272	-0.0319	-0.0363
	(0.0026)	(0.0029)	(0.0030)	(0.0032)	(0.0032)	(0.0035)	(0.0031)	(0.0030)	(0.0030)	(0.0031)
ρ_D^*	-0.0020	-0.0102	-0.0168	-0.0171	-0.0216	-0.0219	-0.0232	-0.0274	-0.0277	-0.0282
	(0.0023)	(0.0029)	(0.0031)	(0.0031)	(0.0033)	(0.0035)	(0.0031)	(0.0032)	(0.0031)	(0.0031)
ν_D	10.7398	9.7630	9.1173	7.5708	9.0529	11.0662	11.3517	11.1961	10.4456	9.5211
	(1.2586)	(1.1054)	(0.9099)	(0.6376)	(0.8902)	(1.2855)	(1.3326)	(1.2980)	(1.1378)	(1.0101)
ω_D	-0.4778	-0.6712	-0.6608	-0.5677	-0.4206	-0.3241	-0.2966	-0.3039	-0.3781	-0.4399
	(0.0297)	(0.0273)	(0.0268)	(0.0261)	(0.0240)	(0.0216)	(0.0224)	(0.0228)	(0.0239)	(0.0232)
β_N	0.9575	0.9355	0.9396	0.9447	0.9342	0.9308	0.9417	0.9394	0.9480	0.9531
	(0.0060)	(0.0083)	(0.0073)	(0.0069)	(0.0080)	(0.0083)	(0.0074)	(0.0075)	(0.0065)	(0.0058)
γ_N	0.0415	0.0555	0.0537	0.0527	0.0504	0.0502	0.0442	0.0480	0.0491	0.0363
	(0.0046)	(0.0052)	(0.0052)	(0.0052)	(0.0055)	(0.0055)	(0.0053)	(0.0056)	(0.0056)	(0.0048)
ρ_N	0.0522	0.0582	0.0538	0.0557	0.0545	0.0515	0.0433	0.0501	0.0471	0.0359
	(0.0047)	(0.0052)	(0.0050)	(0.0052)	(0.0053)	(0.0052)	(0.0049)	(0.0052)	(0.0048)	(0.0043)
γ_N^*	-0.0125	-0.0195	-0.0199	-0.0160	-0.0222	-0.0236	-0.0222	-0.0260	-0.0254	-0.0330
	(0.0024)	(0.0030)	(0.0029)	(0.0027)	(0.0031)	(0.0033)	(0.0030)	(0.0034)	(0.0032)	(0.0034)
ρ_N^*	-0.0056	-0.0106	-0.0176	-0.0177	-0.0211	-0.0227	-0.0268	-0.0275	-0.0291	-0.0311
	(0.0026)	(0.0030)	(0.0030)	(0.0029)	(0.0031)	(0.0032)	(0.0031)	(0.0033)	(0.0034)	(0.0032)
ν_N	7.1157	6.6378	6.1400	5.1110	4.9166	4.8815	4.6815	4.3985	4.3945	5.1849
	(0.5842)	(0.4938)	(0.4245)	(0.3100)	(0.2870)	(0.2850)	(0.2682)	(0.2437)	(0.2389)	(0.3222)
ω_N	-0.9078	-1.2087	-1.3144	-1.2991	-1.2501	-1.2194	-1.2254	-1.2657	-1.3134	-1.0869
	(0.0340)	(0.0293)	(0.0294)	(0.0309)	(0.0274)	(0.0263)	(0.0268)	(0.0279)	(0.0305)	(0.0272)

Estimates of the dynamic parameters for size portfolios with standard errors in parentheses.

Table A.4: Estimates of the dynamic parameters for size portfolios: NYSE/AMEX/NASDAQ

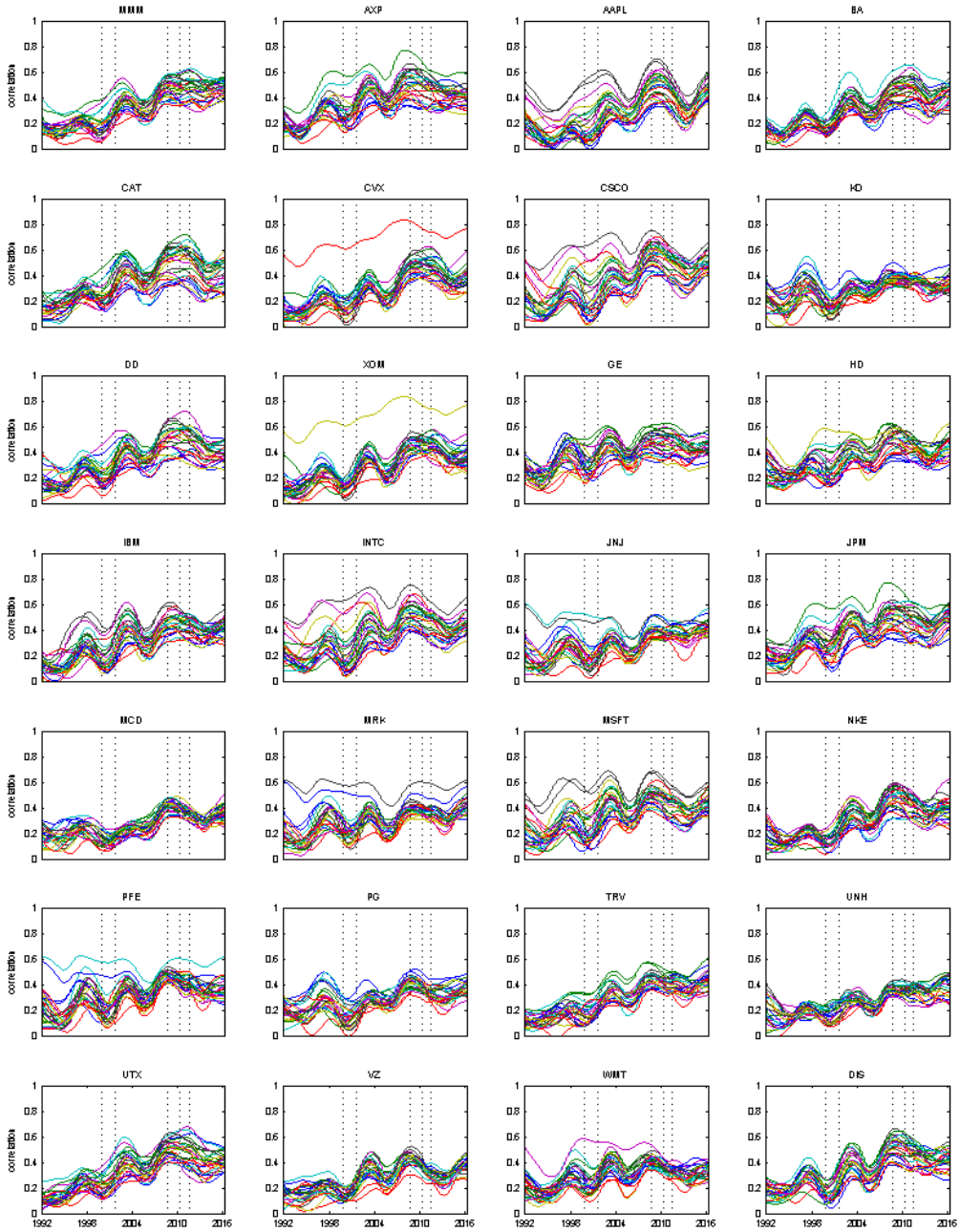
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This figure shows the estimated intraday (in red) and overnight (in black) volatilities, $\sqrt{\frac{\nu_j}{\nu_j-2}exp(2\lambda_t^j + 2\sigma^j(\frac{t}{T}))}$, based on the univariate coupled component model, with one subplot for each stock. The five dashed vertical lines from left to right represent the dates: 10 March 2000 (dot-com bubble), 17 September 2001 (the September 11 attacks), 16 September 2008 (financial crisis), 6 May 2010 (flash crash) and 1 August 2011 (August 2011 stock markets fall), respectively.

Figure A.1: Intraday and overnight volatilities of Dow Jones stocks

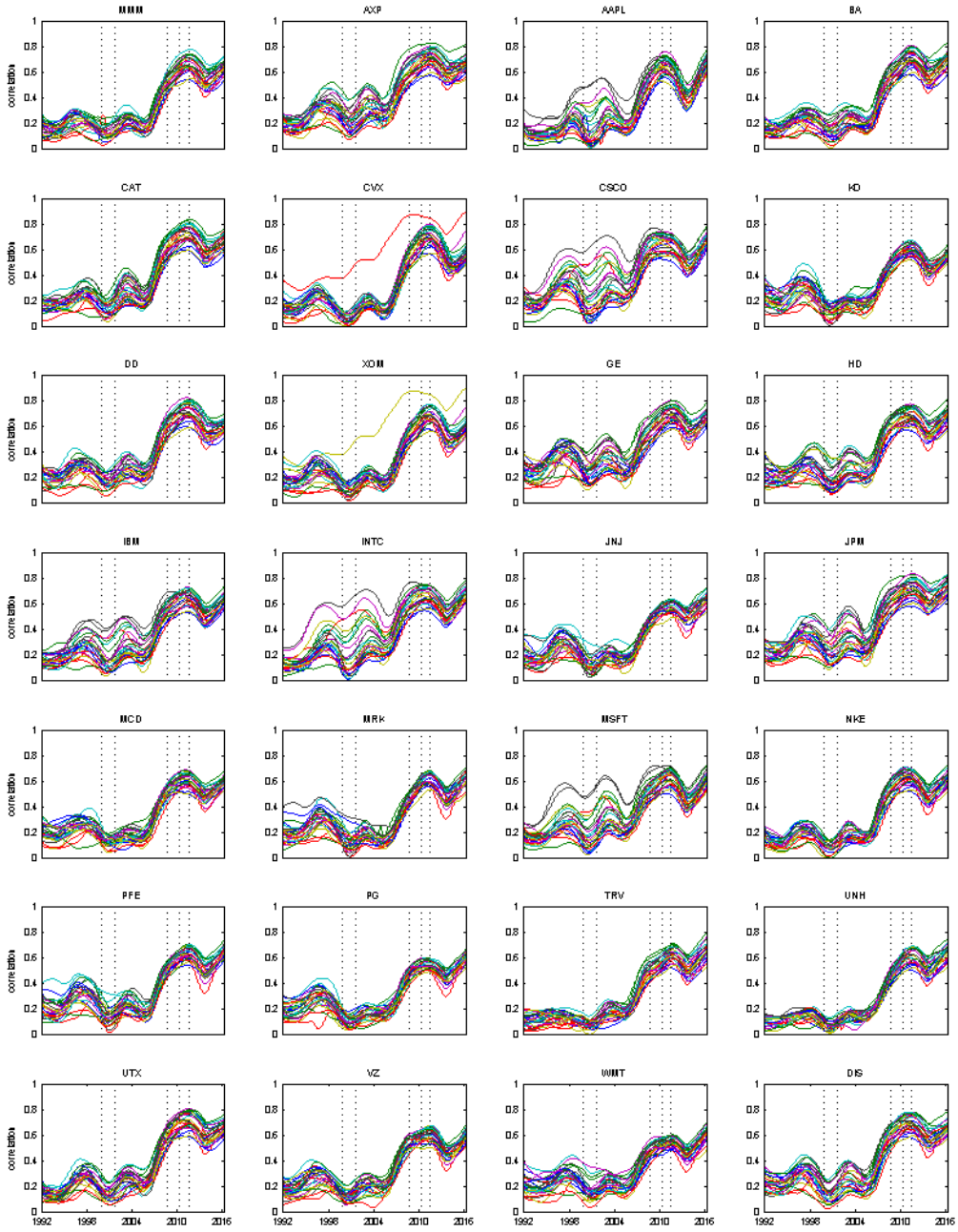
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Each panel presents the long run intraday correlations between that individual stock and the remaining stocks, as implied by the multivariate coupled component model. The five dashed vertical lines from left to right represent the dates: 10 March 2000 (dot-com bubble), 17 September 2001 (the September 11 attacks), 16 September 2008 (financial crisis), 6 May 2010 (flash crash) and 1 August 2011 (August 2011 stock markets fall), respectively.

Figure A.2: Long run intraday correlations between Dow Jones stocks

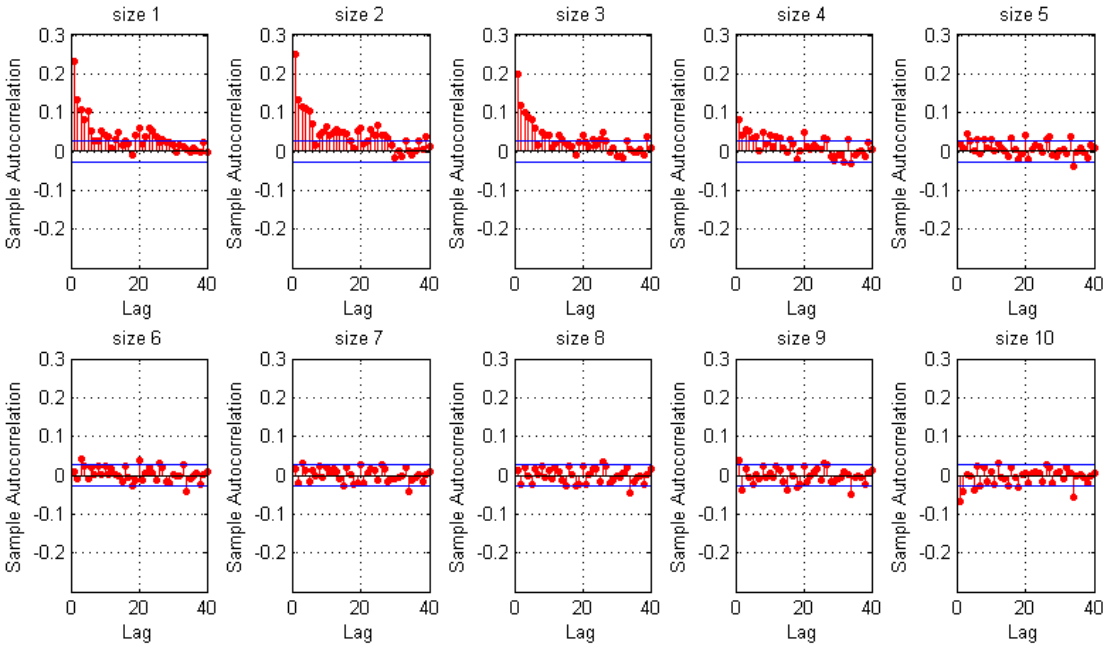
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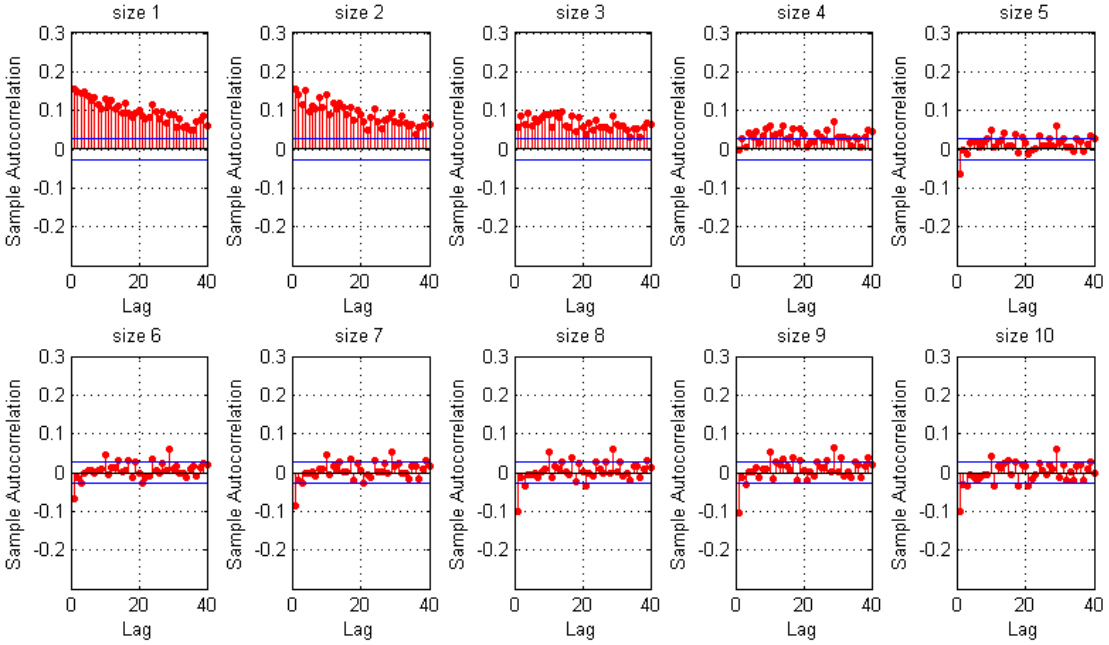
Each panel presents the long run overnight correlations between that stock and the remaining stocks, as implied by the multivariate coupled component model. The five dashed vertical lines from left to right indicate the dates: 10 March 2000 (dot-com bubble), 11 September 2001 (the September 11 attacks), 16 September 2008 (financial crisis), 6 May 2010 (flash crash) and 1 August 2011 (August 2011 stock markets fall), respectively.

Figure A.3: Long run overnight correlations between Dow Jones stocks

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(a) intraday

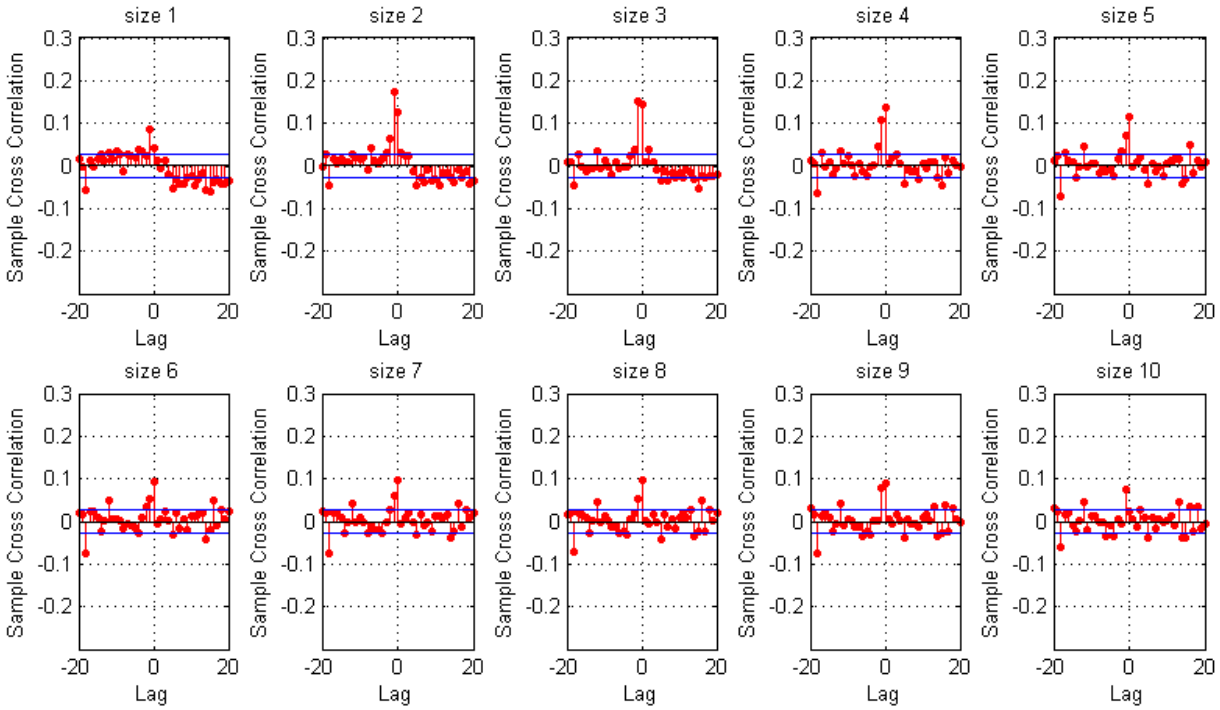


(b) overnight

This figure plots the autocorrelations of intraday (overnight) returns for cap based portfolios on NYSE/AMEX/NASDAQ stocks in Panel a (Panel b).

Figure A.4: Autocorrelations of intraday and overnight returns

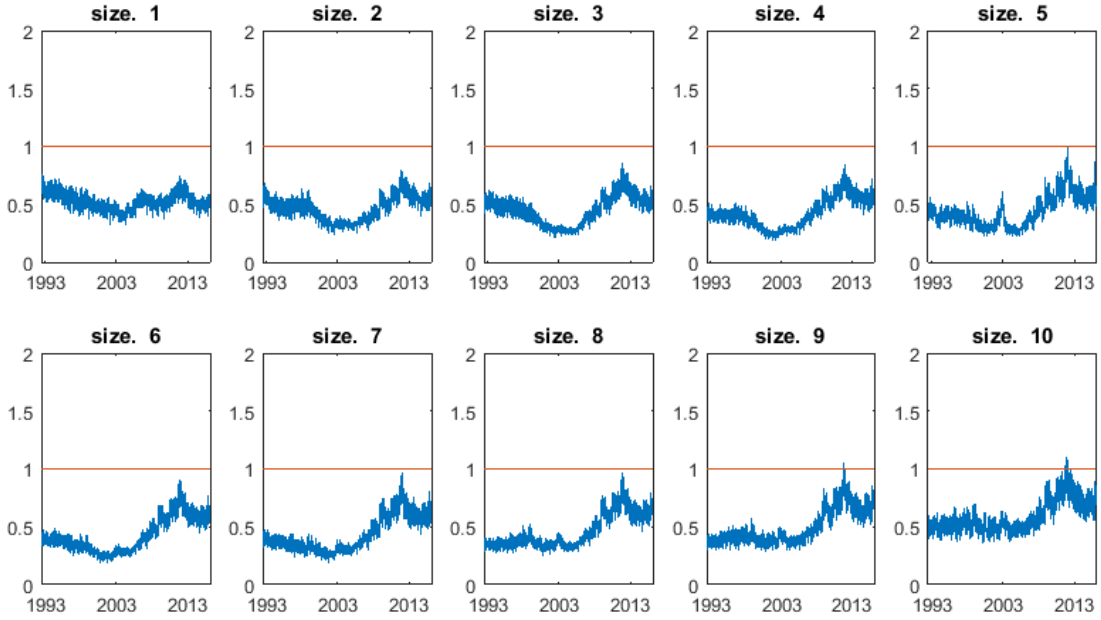
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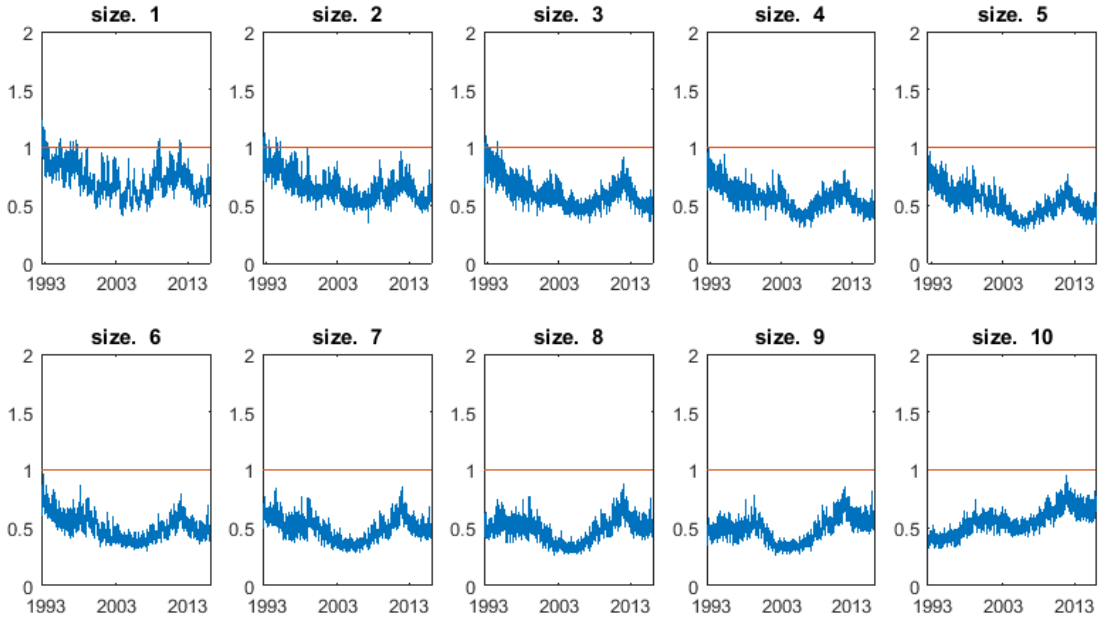
Cross correlations between overnight and intraday returns, r_t^N and r_{t+Lag}^D, for the cap based portfolios in NYSE/AMEX/NASDAQ

Figure A.5: Cross correlations between overnight and intraday returns: NYSE/AMEX/NASDAQ

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(a) NYSE

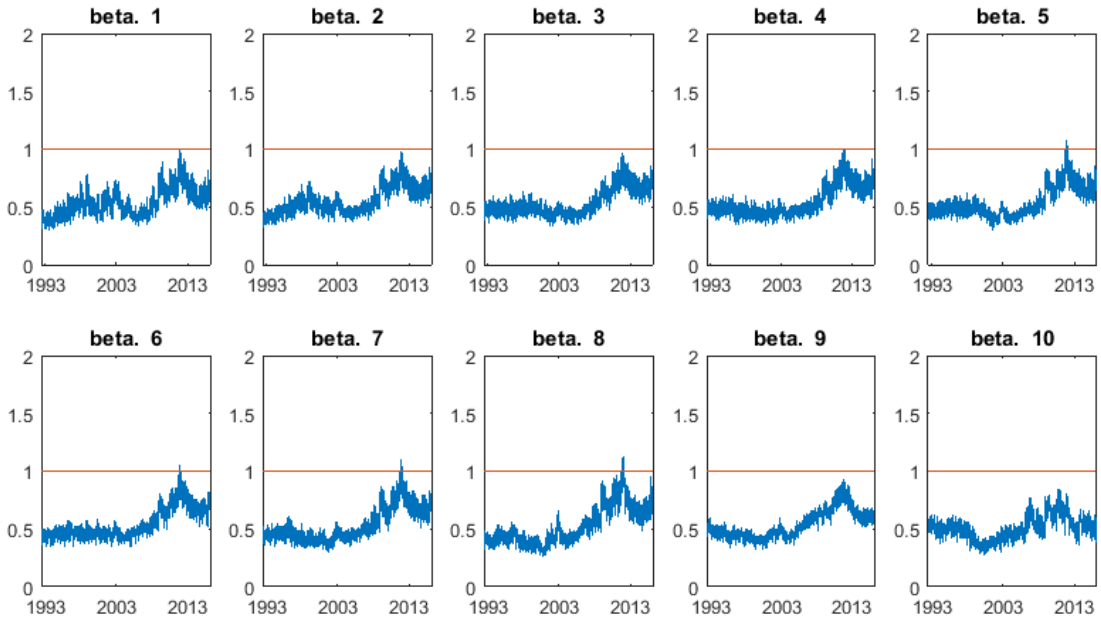


(b) NASDAQ

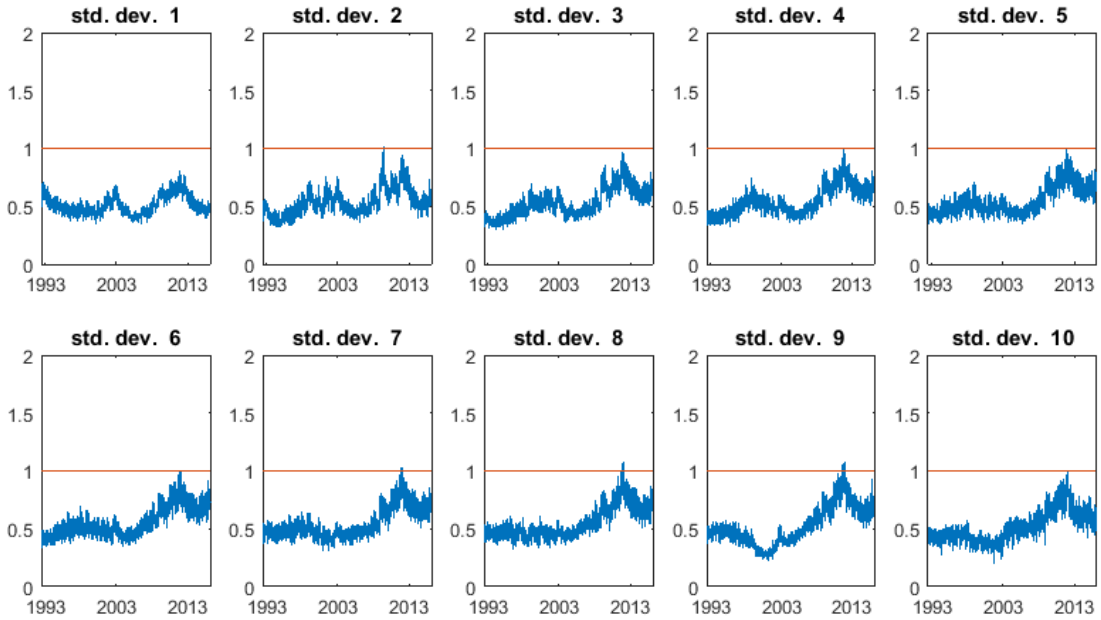
Panel a (Panel b) plots the ratio of overnight to intraday volatility for cap based portfolios for NYSE (NASDAQ) stocks: decile 1 with the smallest capitalization value and decile 10 with the largest capitalization value. Intraday and overnight volatilities are defined as $\sqrt{\frac{\nu_j}{\nu_j-2}exp(2\lambda_t^j + 2\sigma^j(\frac{t}{T}))}$, for $j = D, N$, respectively.

Figure A.6: Ratio of overnight to intraday volatility for cap based portfolios: NYSE and NASDAQ

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(a) beta based deciles

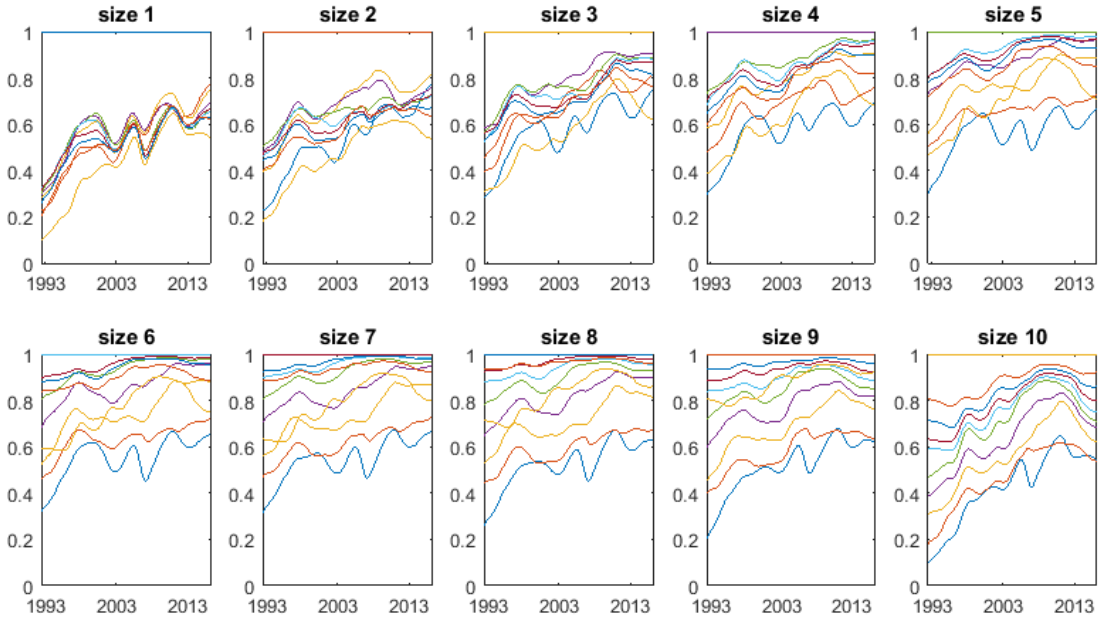


(b) standard deviation based deciles

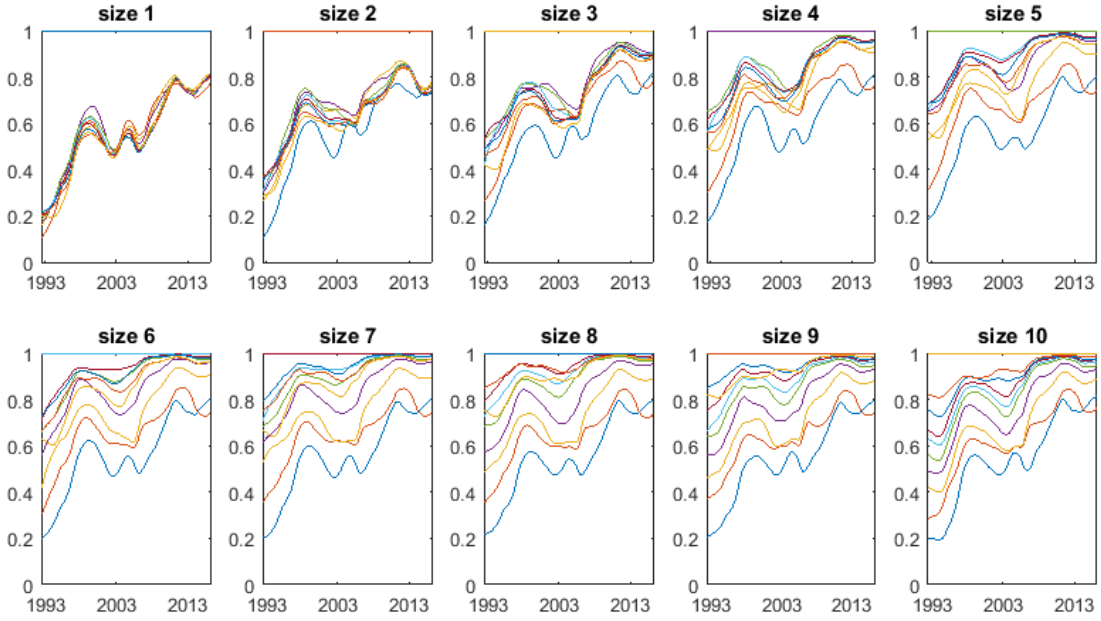
Panel a plots the ratio of overnight to intraday volatility for portfolios formed on beta. Decile 1 has the largest beta, around 1.53 on average, while decile 10 has the smallest beta, around 0.25 on average. Panel b plots the ratio for standard deviation sorted deciles. Decile 1 has the largest standard deviation, around 0.06 on average, and decile 10 has the smallest standard deviation, around 0.009 on average. Intraday and overnight volatilities are defined as $\sqrt{\frac{\nu_j}{\nu_j-2}exp(2\lambda_t^j + 2\sigma^j(\frac{t}{T}))}$, for $j = D, N$, respectively.

Figure A.7: Ratio of overnight to intraday volatility for portfolios formed on beta or standard deviation: NYSE/AMEX/ NASDAQ

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(a) intraday correlations



(b) overnight correlations

Each panel presents the long-run intraday correlations between that size decile and the remaining size deciles, as implied by the multivariate coupled component model.

Figure A.8: Long run correlations between cap based deciles

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